

PERCOLATION IN ACYLINDRICALLY HYPERBOLIC GROUPS

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ABSTRACT. Let G be an acylindrically hyperbolic group. We prove that Bernoulli bond percolation on every Cayley graph of G has a nonuniqueness phase, in which there are infinitely many infinite clusters. This generalizes Hutchcroft's result for Gromov hyperbolic graphs to relatively hyperbolic groups, mapping class groups and rank-1 CAT(0) groups, for example.

Keywords. acylindrical hyperbolicity, hyperbolic group, percolation

MSC classes: 20F67, 57K20, 60K35

1. INTRODUCTION

In geometric group theory, groups are often studied as geometric objects. Some groups resemble $\mathbb{Z}^d$ while some others resemble free groups and surface groups. Between these extremes, there is a range of degrees of hyperbolicity. Here, the notion of hyperbolicity can either be phrased internally using the Dehn function, small cancellation or the prevalence of Morse elements, or by using the group action on hyperbolic spaces, e.g., word hyperbolicity, relative hyperbolicity, hierarchical hyperbolicity and acylindrical hyperbolicity. For an overview of this aspect, we refer to M. Bestvina's survey [Bes23]. There have been efforts to study these notions of hyperbolicity by means of stochastic processes such as random walks and Markov chains ([Sis18], [MT18], [MS20], [GS21]).

On the other hand, groups naturally arise in probability theory as sources of many homogeneous graphs with vertex-transitive automorphism group. In this paper, we study percolation in groups. It was classically studied for Euclidean lattices $\mathbb{Z}^d$ in relation to physical situations where liquid passes through a porous medium. I. Benjamini and O. Schramm considered its generalization to Cayley graphs of groups and sketched the general landscape of the expected phenomena [BS96]. See also papers by I. Benjamini, R. Lyons, Y. Peres and O. Schramm ([BLPS99b], [BLPS99a]) regarding the critical percolation in general groups.

Given a connected, locally finite (simplicial) graph Γ , *Bernoulli bond percolation* on Γ is defined by assigning independent Bernoulli random variables with expectation p to the edges. Edges whose Bernoulli random variable takes value 0 are deleted, and those with Bernoulli random variable taking

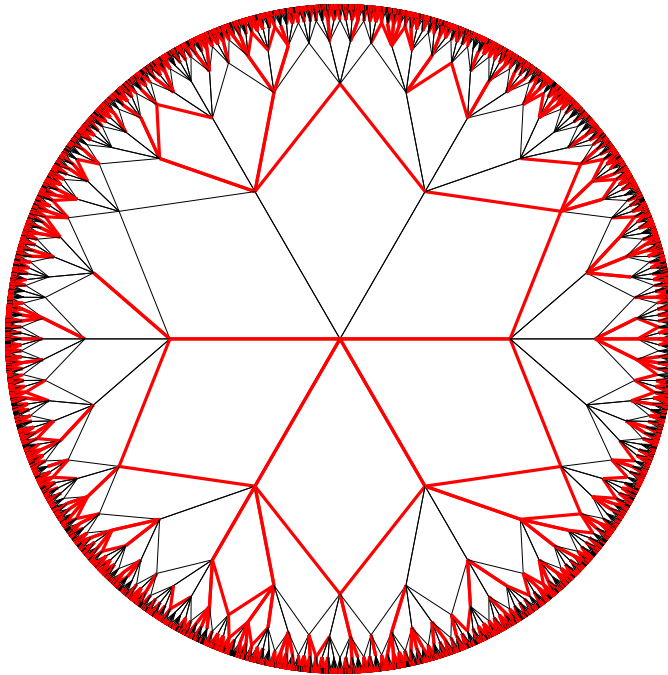


FIGURE 1. Percolation in a planar hyperbolic graph.

value 1 are retained. We can then ask how the connected components, i.e., clusters, of the resulting random subgraph $\Gamma[p]$ are shaped. To this end, we define two parameters, called the *critical parameter*

$$p_c = p_c(\Gamma) := \inf \{p \in [0, 1] : \Gamma[p] \text{ contains an infinite cluster almost surely}\}$$

and the *uniqueness threshold*

$$p_u = p_u(\Gamma) := \inf \{p \in [0, 1] : \Gamma[p] \text{ contains a unique infinite cluster almost surely}\}.$$

See Section 2.1 for further basics of percolation theory.

Benjamini and Schramm posed several conjectures regarding percolation in the Cayley graphs of groups beyond $\mathbb{Z}^d$. Among them is the following:

Conjecture 1.1 ([BS96, Conjecture 6]). *A connected, locally finite, quasi-transitive graph Γ is nonamenable if and only if $p_c(\Gamma) < p_u(\Gamma)$.*

See [HJ06] for an overview of this conjecture.

Let us list some facts for context. Given a connected graph Γ , we have $0 \leq p_c \leq p_u \leq 1$ by definition. It is a fact that for each $p > p_c$, the random graph $\Gamma[p]$ almost surely has an infinite cluster. When Γ is quasi-transitive in addition, for each $p > p_u$ the random graph $\Gamma[p]$ almost surely has a unique infinite cluster. This is due to O. Häggström and Y. Peres [HP99] for unimodular cases, and due to R. H. Schonmann [Sch99] in general (see also [HPS99]). Lastly, C. M. Newman and L. S. Schulman proved in [NS81] for

quasi-transitive Γ that, for each $p \in (0, 1)$, there exists $N_\infty(p) \in \{0, 1, +\infty\}$ such that the number of infinite clusters in $\Gamma[p]$ is almost surely $N_\infty(p)$.

Hence for quasi-transitive graphs, $N_\infty(p) = 0$ almost surely for $p < p_c$, $N_\infty(p) = +\infty$ almost surely for $p_c < p < p_u$ and $N_\infty(p) = 1$ almost surely for $p > p_u$. In particular, if $p_c < p_u$ then there exists (uncountably many) p such that $\Gamma[p]$ has infinitely many infinite clusters almost surely.

Now, for *nonamenable* quasi-transitive graphs, it is known that $N_\infty(p_c) = 0$ almost surely. This is due to I. Benjamini, R. Lyons, Y. Peres and O. Schramm [BLPS99b, Theorem 1.1] and is generalized by T. Hutchcroft to graphs with exponential growth [Hut16, Theorem 1.2]. Hence, for a non-amenable quasi-transitive graph, $p_c < p_u$ if and only if $N_\infty(p) = +\infty$ for some (uncountably many) p 's.

Let us go back to the conjecture. The equality $p_c(\mathbb{Z}^d) = p_u(\mathbb{Z}^d)$ was first established in a deep theorem by M. Aizenman, H. Kesten and C. M. Newman [AKN87], and a simpler method was later obtained by R. M. Burton and M. Keane [BK89]. A. Gandolfi, M. S. Keane and C. M. Newman observed in [GKN92] that Burton and Keane's method generalizes to amenable graphs. Hence, the only nontrivial direction is the “only if” direction. A significant breakthrough was made by T. Hutchcroft, who showed the conjecture for nonamenable quasi-transitive graphs that admit an action by a non-unimodular group [Hut20b]. We note that the first example of a quasi-transitive graph Γ for which all three cases— $N_\infty = 0$, $N_\infty = \infty$ and $N_\infty = 1$ —take place was the direct product of trees and $\mathbb{Z}$, which admits a non-unimodular automorphism group [GN90].

Hence, the remaining case is nonamenable graphs with unimodular automorphism groups. The most natural examples in this category are Cayley graphs of nonamenable groups.

We thus focus on Cayley graphs of nonamenable groups. Let G be a group with a finite generating set S . The Cayley graph $\Gamma(G, S)$ consists of the vertex set G and the edge set $\{\overline{vw} : v^{-1}w \in S \cup S^{-1}\}$. The group G naturally acts by graph automorphisms and the action is vertex-transitive, i.e., for each $v, w \in V(\Gamma) = G$ there exists $g \in G$ such that $gv = w$. The action is also free, i.e., each vertex has trivial stabilizer.

Strong evidence for Conjecture 1.1 is given by I. Pak and T. Smirnova-Nagnibeda [PSN00], who proved that every nonamenable group has a Cayley graph for which $p_c < p_u$. Furthermore, A. Nachmias and Y. Peres proved that Cayley graphs of nonamenable groups with high girth relative to the spectral radius satisfy Conjecture 1.1 [NP12]. One can then ask if there are groups for which *every* Cayley graph satisfies Conjecture 1.1.

In this regard, Benjamini and Schramm proved that every transitive, non-amenable, planar graph with one end has a nonuniqueness phase [BS01]. This generalizes S. Lalley's earlier work about Fuchsian groups [Lal98]. The nonuniqueness was proved by D. Gaboriau [Gab05] for graphs admitting nonconstant harmonic Dirichlet functions and by R. Lyons ([Lyo00], [Lyo13]) for every Cayley graph of groups admitting an action with cost > 1 .

Beyond planar graphs, Hutchcroft showed that the nonuniqueness phase exists for every Cayley graph of nonamenable word hyperbolic groups [Hut19]. In fact, Hutchcroft proved the result for more general quasi-transitive non-amenable Gromov hyperbolic graphs, by using the Bonk–Schramm embedding of such graphs into a real hyperbolic space $\mathbb{H}^d$ [BS00]. See also J. Czapkowski’s independent work on certain groups acting on $\mathbb{H}^3$ [Cza24].

The main point of this paper is to generalize Hutchcroft’s result to other nonamenable groups. Namely, we have:

Theorem A (Corollary 7.11). *Let G be an acylindrically hyperbolic group and let Γ be a Cayley graph of G with respect to a finite generating set. Then we have $p_c(\Gamma) < p_u(\Gamma)$; in particular, there exist uncountably many $p \in (0, 1)$ such that $\Gamma[p]$ has infinitely many infinite clusters.*

Acylindrically hyperbolic groups encompass word hyperbolic groups, relatively hyperbolic groups and many other groups that act on a Gromov hyperbolic space. These groups exhibit interesting dynamical and group-theoretical properties ([BF02], [Ham08], [DGO17]), as well as probabilistic behavior ([Sis18], [MS20], [Cho25]), that are shared with word hyperbolic groups. We list some examples of acylindrically hyperbolic groups:

- (non-elementary) relatively hyperbolic groups;
- non-elementary Kleinian groups (possibly with $\mathbb{Z}^d$ subgroups);
- free products of nontrivial groups;
- the mapping class group of a finite-type hyperbolic surface;
- the outer automorphism group $\text{Out}(F_n)$ of a non-abelian free group F_n [BF14];
- the automorphism group $\text{Aut}(G)$ of a non-elementary hyperbolic group G ([Gen19], [GH21]);
- rank-1 CAT(0) groups such as irreducible CAT(0) cubical groups [BF09], and
- many Artin groups and 3-manifold groups ([HMS24], [MO15]).

We refer readers to D. Osin’s survey [Osi16] for more details.

Fix a vertex $v \in G$. We denote by $\chi_p(v)$ the expected size (the number of vertices) of the cluster of v . This does not depend on the choice of v in the case of Cayley graphs, so we will drop v and write χ_p . The facts that $\chi_p < +\infty$ for $p < p_c$ and $\chi_p \gtrsim (p_c - p)^{-1}$ are due to M. Aizenman and C. M. Newman [AN84, Proposition 3.1]. In the course of the proof, we show that $\chi_p \lesssim (p_c - p)^{-1}$, following Hutchcroft’s criterion. This is one example of the so-called *mean-field critical behavior* of percolation in acylindrically hyperbolic groups. More consequences follow from the following fact:

Theorem B. *Let G and Γ be as in Theorem A. Then at $p_c = p_c(\Gamma)$, we have $\nabla_{p_c}(\Gamma) < +\infty$.*

See Section 2.2 for the definition of the triangle diagram ∇_{p_c} . This notion was introduced by Aizenman and Newman to study various mean-field

critical behaviors. We summarize some consequences of the finitude of ∇_{p_c} in Theorem 2.5.

In fact, the triangle condition $\nabla_{p_c} < +\infty$ is derived from yet another fact, namely, the L^2 -boundedness of an operator associated with the transition probability. In [Hut19, Section 2], Hutchcroft considered a quantity $p_{2 \rightarrow 2}(\Gamma)$ for a connected, locally finite, quasi-transitive graph Γ . This quantity has the property that $p_{2 \rightarrow 2} \leq p_u$ and $\nabla_p < +\infty$ for each $p < p_{2 \rightarrow 2}$. Hence, once $p_c < p_{2 \rightarrow 2}$ holds, both the triangle condition $\nabla_{p_c} < +\infty$ and the inequality $p_c < p_u$ follow:

$$[L^2\text{-boundedness} : p_c(\Gamma) < p_{2 \rightarrow 2}(\Gamma)] \Rightarrow [\nabla_{p_c}(\Gamma) < +\infty] \wedge [p_c(\Gamma) < p_u(\Gamma)].$$

We will not explain the L^2 -boundedness in detail. Nonetheless, the L^2 -boundedness and the triangle condition guarantee several mean-field critical behaviors. We summarize them in Section 2.2. We refer to [Hut19] and [Hut20a] for a deeper story.

1.1. The hitchhiker's guide to the nonuniqueness. This paper is concerned with probabilistic phenomena on geometric objects that entail hyperbolicity. The probabilistic ingredients are already given by Hutchcroft [Hut19]. Namely, the gap $p_c < p_u$ and the triangle diagram bound $\nabla_{p_c} < +\infty$ follow from Equation 2.7 and 2.8. Theorem 2.10 and 2.18 provide a way to guarantee these equations. The rest of the paper will focus on why acylindrically hyperbolic groups satisfy the assumptions of Theorem 2.10 and 2.18.

Thus, for readers familiar with the theory of acylindrically hyperbolic groups, the quickest way to read this paper is as follows:

- (1) read Definition 2.12 and 2.14,
- (2) read Theorem 2.10 and Theorem 2.18,
- (3) read Section 5 and study Proposition 5.8,
- (4) read Section 6 and study Proposition 6.1, and
- (5) read Section 7 and combine Proposition 7.5, 7.8 and 7.9.

Nevertheless, we recommend that readers read Section 2 through Section 7. This is because:

- Section 2 presents basic percolation theory and an overview of Hutchcroft's theory. This helps understand Theorem 2.10 and Theorem 2.18.
- Section 2.3 describes in detail the intuition behind our strategy.
- Section 3 provides necessary hyperbolic geometry.
- Section 4 gives a different proof of Hutchcroft's hyperbolic magic lemma without using Benjamini–Schramm's Euclidean magic lemma or Bonk–Schramm's embedding theorem. This is not needed for general acylindrically hyperbolic groups but is an important prototype containing essential ideas.

Moreover, when restricted to *groups acting properly on a Gromov hyperbolic space*, Proposition 4.7 and 6.1 are sufficient for the main theorem; see Corollary 6.6. Hence, readers interested in relatively

hyperbolic groups (such as non-elementary Kleinian groups and free products of groups) may read Section 4 and Section 6 only.

This paper is mostly self-contained but hides two secret ingredients. First, Hutchcroft’s approach is eventually based on the analysis of the transfer operator. We invite readers to [Hut19, Section 2] for details of operator analysis. Second, the argument for barriers (Proposition 6.1 and 7.9) can be explained by the fact that acylindrically hyperbolic groups act on a quasi-tree that enjoys the bottleneck property, which arises from Bestvina–Bromberg–Fujiwara’s construction ([BBF15], [BBFS19]).

1.2. Discussion. Our result in this article can be seen as a positive response to the Benjamini–Schramm conjecture in the realm of acylindrically hyperbolic groups. This motivates a variety of problems that relate percolation theory to geometric group theory. Below, we list a selection of problems that reflect our own preferences.

In Theorem A, we prove that a nonuniqueness phase exists for every Cayley graph of a certain class of groups. However, such an argument cannot be directly generalized to general finitely generated groups.

Question 1.2. *Is the existence of a nonuniqueness phase a group invariant? That is to say, for a finitely generated group G , if there exists a finite generating set S such that there exists a nonuniqueness phase for $\text{Cay}(G, S)$, does the same phenomenon happen for every Cayley graph of G ?*

If this is the case, Pak and Smirnova-Nagnibeda’s theorem will settle Benjamini–Schramm’s conjecture.

As we mentioned, many groups are known to be acylindrically hyperbolic. On the other hand, there are nonamenable groups that are not acylindrically hyperbolic. In particular, Theorem A does not deal with *direct products*. Given a nonamenable group, a cheap recipe to create a new nonamenable group is to take a direct product with another group. Our method does not handle such groups, including $F_2 \times \mathbb{Z}$. Hence, an innocent question follows:

Question 1.3. *Does every Cayley graph of $F_2 \times \mathbb{Z}$ have a nonuniqueness phase?*

We now elaborate on different viewpoints on this question.

A group G is called *free-by-cyclic* if the following short exact sequence

$$1 \rightarrow F \rightarrow G \rightarrow \mathbb{Z} \rightarrow 1$$

holds for some finite-rank free group F . In this case, G is isomorphic to the semidirect product $F \rtimes_{\phi} \langle t \rangle$ determined by an automorphism $\phi \in \text{Aut}(F)$. The easiest example is $F \times \mathbb{Z}$, given by the trivial automorphism. Genevois and Horbez [GH21] and Ghosh [Gho23] show that a free-by-cyclic group $G = F \rtimes_{\phi} \langle t \rangle$ is acylindrically hyperbolic if and only if ϕ is of infinite order. Currently, there is no strategy to prove the nonuniqueness of infinite clusters when ϕ is of finite order. We note that generalizing the study of Grimmett and Newman [GN90] on $T_k \times \mathbb{Z}^d$ for a regular tree T_k with valency $k \gg 1$,

Hutchcroft [Hut20b] showed that Benjamini–Schramm’s conjecture holds for all quasi-transitive graphs that admit a nonunimodular automorphism group, which includes $T_k \times \mathbb{Z}^d$ for $k \geq 3$.

Question 1.4. *Does the Benjamini–Schramm conjecture hold for all free-by-cyclic groups?*

If $\Gamma = (V, E)$ is a simple graph and $\mathcal{G} = (G_v)_{v \in V}$ is a system of groups labelled by vertices of Γ , the *graph product* $\Gamma\mathcal{G}$ of $\mathcal{G}$ is defined by

$$\Gamma\mathcal{G} := (*_{v \in V} G_v) / \langle\langle [g, h] \mid g \in G_u, h \in G_v, \{u, v\} \in E \rangle\rangle.$$

In particular, if all groups in the system $\mathcal{G}$ are infinite cyclic, the corresponding graph product is known as a right-angled Artin group. A. Minasyan and D. Osin [MO15, Corollary 2.13] show that $\Gamma\mathcal{G}$ is acylindrically hyperbolic in many cases. Precisely, $\Gamma\mathcal{G}$ satisfies one of the following:

- (1) Γ is a join of two nontrivial subgraphs, that is, $\Gamma\mathcal{G}$ is a direct product of nontrivial groups.
- (2) $\Gamma\mathcal{G}$ is virtually cyclic.
- (3) $\Gamma\mathcal{G}$ is acylindrically hyperbolic.

The first case includes direct products of free groups for which the Benjamini–Schramm conjecture is still open.

Question 1.5. *Does the Benjamini–Schramm conjecture hold for all graph products of groups?*

We are also interested in Benjamini–Schramm’s conjecture in the class of fundamental groups of (connected compact) 3-dimensional manifolds, briefly, 3-manifold groups. Minasyan–Osin [MO15, Theorem 2.8] show that most 3-manifold groups are acylindrically hyperbolic. In detail, they show that every 3-manifold whose fundamental group is neither polycyclic nor acylindrically hyperbolic is an extension of an acylindrically hyperbolic group by an infinite cyclic group. What we want to know is the following.

Question 1.6. *Does the Benjamini–Schramm conjecture hold for all 3-manifold groups? In particular, for every Seifert fibered space M , does the Benjamini–Schramm conjecture hold for $\pi_1(M)$?*

In contrast to general $\text{CAT}(0)$ spaces, rank rigidity is known for $\text{CAT}(0)$ cube complexes by P.-E. Caprace and M. Sageev [CS11]. In particular, all irreducible, non-virtually cyclic $\text{CAT}(0)$ cubical groups are covered by Theorem A. Thus, the nonuniqueness problem for percolation in $\text{CAT}(0)$ cubical groups will be completely understood once it is understood for metric products. The groups $F_2 \times \mathbb{Z}$ and $F_2 \times F_2$ serve as barometers for this question. We conclude this section with the following question:

Question 1.7. *Does the Benjamini–Schramm conjecture hold for $\text{CAT}(0)$ cubical groups, or more generally, quasi-transitive median graphs? In particular, for a $\text{CAT}(0)$ cubical group G , does $p_c(\text{Cay}(G, S)) = p_u(\text{Cay}(G, S))$ hold for every finite generating set S of G if and only if G is virtually $\mathbb{Z}^d$?*

Disclosure of AI use. The first version of this paper was posted on arXiv in August 2025; no AI was used in its preparation. While revising the paper in September 2026, Choi proofread the paper by feeding the manuscript into a publicly available LLM and prompting “list up all the typos, grammar errors and unnatural expressions”, “scrutinize everything”, “do it again.” Then the authors read the output, which helped find typos and mismatched constants. Every change to the paper, such as the restructuring of proofs and the addition/deleteion of lemmas and definitions, were conceived and made by the authors. Naturally, the two authors are responsible for all the maths written here.

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2. PRELIMINARIES I: PROBABILITY THEORY

We assume that readers are familiar with finitely generated groups and simplicial graphs. When a group G is given a generating set S , we define the word metric d_S as

$$d_S(g, h) := \inf \{n \geq 0 : \exists s_1, \dots, s_n \in S \cup S^{-1} [h = gs_1 \cdots s_n]\},$$

$$\|g\|_S := d_S(id, g), \quad B_S(R) := \{g \in G : \|g\|_S \leq R\}.$$

A d_S -path $(g_1, \dots, g_n)$ is a combinatorial path on the Cayley graph $Cay(G, S)$, that is, a sequence that satisfies $g_i^{-1}g_{i+1} \in S \cup S^{-1}$ for each i .

2.1. Basics of percolation theory. This subsection is intended as a quick introduction to percolation theory. We refer to [Gri89] for further details. Readers who are familiar with percolation theory or want to keep it as a blackbox can skip this subsection.

Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a connected simplicial graph, with $\mathcal{V}$. Throughout, Γ has uniformly bounded valency. Let $0 \leq p \leq 1$. On the product space $\Omega = \{0, 1\}^{\mathcal{E}}$ indexed by edges of Γ , we put the product of Bernoulli measures with expectation p . That is, $\mathbb{P}(\omega \in \Omega : \omega(e) = 1) = p$ for each e , and $\omega(e)$ and $\omega(e')$ are independent for $e \neq e'$. Each $\omega \in \Omega$ gives rise to a graph $\Gamma(\omega)$, which is the subgraph of Γ after removing those edges e with $\omega(e) = 0$.

It is convenient to use the notation:

$$v \leftrightarrow_{\omega} w \Leftrightarrow \text{“}v \text{ and } w \text{ are connected in } \Gamma(\omega)\text{”}$$

for $v, w \in \mathcal{V}$. Given $v \in \mathcal{V}$, we define the cluster

$$C_\omega(v) := \{w \in \mathcal{V} : w \leftrightarrow_\omega v\}.$$

Then by Fubini's theorem, we have

$$\mathbb{E}_p(\#C_\omega(v)) = \sum_{w \in \mathcal{V}} \mathbb{P}_p(v \leftrightarrow_\omega w).$$

By convention, we will write

$$v \leftrightarrow_\omega \infty \Leftrightarrow \#C_\omega(v) = +\infty.$$

(Recall that we focus on locally finite graphs.)

Note that the space $\Omega = \{0, 1\}^\mathcal{E}$ and the random graph $\Gamma(\omega) \subseteq \Gamma$ for $\omega \in \Omega$ are defined without reference to p . The parameter p affects the underlying probability measure only. To express the role of p more explicitly, we denote the random graph by $\Gamma[p]$ and the underlying measure by $\mathbb{P}_p$.

One can ask if the expected size of clusters increases as p increases, or if the expected size of clusters depends on the choice of the root vertex. These questions can be answered using the following tools.

The space Ω is given a natural order: for $\omega, \omega' \in \Omega$ we write $\omega \leq \omega'$ if $\omega(e) \leq \omega'(e)$ for each $e \in \mathcal{E}$. We say that an event $A \subseteq \Omega$ is *increasing* if

$$\forall \omega, \omega' \in \Omega [\omega \in A \wedge \omega \leq \omega' \Rightarrow \omega' \in A].$$

Standard examples of increasing events include $\{\omega : v \leftrightarrow_\omega w\}$ for given $v, w \in \mathcal{V}$, or $\{\omega : v \leftrightarrow_\omega \infty\}$ for a given $v \in \mathcal{V}$.

Fact 2.1. *Let $A \subseteq \Omega$ be an increasing event. Then $\mathbb{P}_p(A) \leq \mathbb{P}_{p'}(A)$ holds for each $0 \leq p \leq p' \leq 1$.*

We now state the Harris–FKG inequality, which was first described by T. E. Harris [Har60] and later generalized by C. M. Fortuin, P. W. Kasteleyn and J. Ginibre [FKG71]:

Proposition 2.2 (Harris–FKG). *Let $A, B \subseteq \Omega$ be increasing events. Then*

$$\mathbb{P}_p(A \cap B) \geq \mathbb{P}_p(A) \cdot \mathbb{P}_p(B)$$

holds for each p .

The Harris–FKG inequality can be used, for example, to show that the average size of clusters does not depend on the choice of the root vertex.

The Harris–FKG inequality looks like a generalization of the equality that holds for independent events. One can ask if the reverse inequality also holds in certain circumstances. The BK inequality partially answers this.

Given an increasing event A and $\omega \in A$, there can be a set $W \subseteq \{e \in \mathcal{E} : \omega(e) = 1\} \subseteq \mathcal{E}$ such that $1_W \subseteq A$ holds, i.e.,

$$\forall \omega' \in \Omega [\forall e \in W [\omega'(e) = 1] \Rightarrow \omega' \in A].$$

In this situation, we call W a *witness* for A in ω . If $W' \subseteq W$ are both witnesses for A in ω , we say that W' is a *sub-witness* of W .

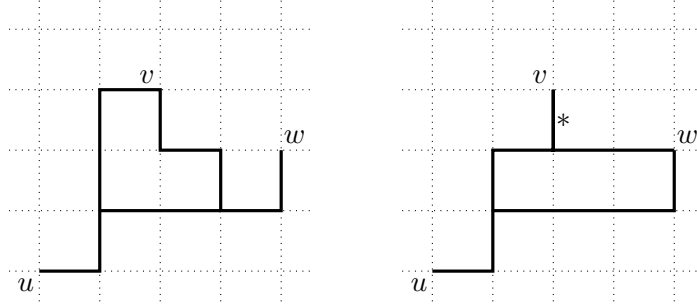


FIGURE 2. Two configurations from percolation in $\mathbb{Z}^2$. In the left configuration, the paths $u \rightarrow \uparrow \uparrow \rightarrow v$ and $v \downarrow \rightarrow \downarrow \rightarrow w$ are disjoint witnesses for $A := \{u \leftrightarrow v\}$ and $B := \{v \leftrightarrow w\}$, respectively. In the right configuration, there are several witnesses for the events A and B in ω , but none of them are disjoint; they all contain the edge $(*)$.

For example, let $v, w \in \mathcal{V}$, let $\omega \in \Omega$ and suppose that there exists a path $(e_1, \dots, e_n)$ in $\Gamma(\omega)$ connecting v to w . Then this path becomes a witness for $A := \{v \leftrightarrow w\}$ in ω : $\{e_1, \dots, e_n\}$ are all given value 1 by ω , and for $\omega' \in \Omega$, $\omega'(e_i) = 1$ for each i implies $\omega' \in A$.

Now, for two increasing events $A, B \subseteq \Omega$, we define another event $A \circ B \subseteq \Omega$, called the *disjoint occurrence* of A and B , as follows:

$$A \circ B := \left\{ \omega \in \Omega : \begin{array}{l} \omega \in A \cap B, \exists \text{ witness } W \subseteq \mathcal{E} \text{ for } A \text{ in } \omega \text{ and} \\ \exists \text{ witness } W' \subseteq \mathcal{E} \text{ for } B \text{ in } \omega \text{ such that } W \cap W' = \emptyset \end{array} \right\}.$$

Figure 2 shows two different configurations in $A \cap B$, where $A := \{u \leftrightarrow v\}$ and $B := \{v \leftrightarrow w\}$. The left one lies in $A \circ B$ and the right one does not.

We can now state the BK inequality, which is due to J. van den Berg and H. Kesten [vdBK85].

Proposition 2.3 (BK inequality). *Let $A, B \subseteq \Omega$ be increasing events for which every witness has a finite sub-witness. Then for each p we have*

$$\mathbb{P}_p(A \circ B) \leq \mathbb{P}_p(A) \cdot \mathbb{P}_p(B).$$

Often, we want more precise information about the growth of $\mathbb{P}_p(A)$ for a given event A . For this it is beneficial to have a formula for derivatives of $\mathbb{P}_p(A)$. Russo's formula serves this purpose.

Given an increasing event $A \subseteq \Omega$ and $\omega \in \Omega$, we say that $e \in \mathcal{E}$ is *pivotal* for A if ω enters A after turning e on, and ω is excluded from A by turning e off. More formally, e is pivotal for A (given ω) if $\omega^e \in A$ and $\omega_e \notin A$ for

$$\omega^e(f) := \begin{cases} 1 & f = e \\ \omega(f) & f \in \mathcal{E} \setminus \{e\} \end{cases}, \quad \omega_e(f) := \begin{cases} 0 & f = e \\ \omega(f) & f \in \mathcal{E} \setminus \{e\} \end{cases}.$$

We now record Russo's formula [Rus81]:

Proposition 2.4. *Let $A \subseteq \Omega$ be an increasing event. Then*

$$\left(\frac{d}{dp}\right)_+ \mathbb{P}_p(A) \geq \sum_{e \in \mathcal{E}} \mathbb{P}_p(e \text{ is pivotal for } A) = \frac{1}{1-p} \sum_{e \in \mathcal{E}} \mathbb{P}_p(e \text{ is closed and pivotal for } A).$$

Here, $(d/dp)_+$ denotes the lower right Dini derivative.

We now focus on percolation in groups. For this purpose, let G be a finitely generated group and let Γ be its Cayley graph. We then consider Bernoulli percolation on Γ . In this setting, $\mathbb{P}_p$ is G -ergodic: every G -invariant event occurs with probability 0 or 1.

We now define the *critical parameter* for Γ :

$$p_c = p_c(\Gamma) := \inf\{p \in [0, 1] : \mathbb{P}_p(C(id) \text{ is infinite}) > 0\}.$$

Then all clusters are almost surely finite for $p < p_c$. By ergodicity of $\mathbb{P}_p$ under the G -action, there is almost surely an infinite cluster for $p > p_c$.

We then define the *uniqueness threshold* for Γ :

$$p_u = p_u(\Gamma) := \inf\{p \in [0, 1] : \mathbb{P}_p(\text{there is a unique infinite cluster}) > 0\}.$$

For every value of p , the number $N_\infty(p)$ of infinite clusters is almost surely constant and is among $\{0, 1, +\infty\}$. Furthermore, $N_\infty(p) = +\infty$ holds throughout $p_c < p < p_u$, and $N_\infty(p) = 1$ holds throughout $p > p_u$. What happens at $p = p_c$ and $p = p_u$ is mysterious: we refer readers to [HP24] for a summary of history and recent progress.

2.2. Overview of Hutchcroft's strategy. We now explain Hutchcroft's theory in [Hut19]. Throughout, Γ will be a Cayley graph of a finitely generated group G . This graph is connected, has uniformly bounded valency and is vertex-transitive.

Recall that for a given parameter $0 \leq p \leq 1$ we defined the random graph $\Gamma[p]$ by randomly deleting edges from Γ . We define the two-point function

$$\tau_p(g, h) := \mathbb{P}_p(g \leftrightarrow h) = \mathbb{P}_p(\exists \text{ path connecting } g \text{ and } h \text{ in } \Gamma[p]).$$

We abbreviate $\tau_p(id, g)$ by $\tau_p(g)$. Then $\tau_p(g, h) = \tau_p(g^{-1}h)$ for each $g, h \in G$. We now introduce the *triangle diagram*

$$\nabla_p := \sup_{g \in G} \sum_{h, k \in G} \tau_p(g, h) \tau_p(h, k) \tau_p(k, g).$$

We call the expected size of the identity cluster the *susceptibility*:

$$\chi_p := \mathbb{E}_p[\#C(id)] = \sum_{g \in G} \tau_p(g).$$

It is a fact that $\chi_p < +\infty$ for $0 \leq p < p_c$ and $\lim_{p \nearrow p_c} \chi_p = +\infty$.

Let us now define

$$\iota_p := 1 - \sup \left\{ \frac{\sum_{g, h \in A} \tau_p(g, h)}{\chi_p \cdot \#A} : A \subseteq G \text{ finite} \right\}.$$

A naive counting shows that $\iota_p \geq 0$ always holds. It is however nontrivial to show that ι_p gets closer to 1 as $p \nearrow p_c$, which is one of our main goals.

We can now state:

Theorem 2.5 ([Hut19, Proposition 2.7]). *Let Γ be a Cayley graph of a finitely generated group. If*

$$(2.1) \quad \liminf_{p \nearrow p_c} \frac{p_c - p}{1 - p} \chi_p \sqrt{1 - \iota_p^2} = 0,$$

then $p_c(\Gamma) < p_u(\Gamma)$ and $\nabla_{p_c}(\Gamma) < +\infty$.

Moreover, we have the following mean-field critical behaviors:

$$(2.2) \quad \chi_p := \mathbb{E}_p(\#C(id)) \asymp (p_c - p)^{-1} \quad p \nearrow p_c,$$

$$(2.3) \quad \mathbb{E}_p \left[(\#C(id))^{k+1} \right] / \mathbb{E}_p \left[(\#C(id))^k \right] \asymp (p_c - p)^{-2} \quad k \geq 1, p \nearrow p_c,$$

$$(2.4) \quad \mathbb{P}_p(\#C(id) = +\infty) \asymp p - p_c \quad p \searrow p_c,$$

$$(2.5) \quad \mathbb{P}_{p_c}(\#C(id) \geq n) \asymp n^{-1/2} \quad n \nearrow +\infty,$$

$$(2.6) \quad \mathbb{P}_{p_c}(\text{intrinsic radius of } C(id) \geq n) \asymp n^{-1} \quad n \nearrow +\infty.$$

To see how the triangle condition implies the mean-field critical behaviors, see [Hut20b, Section 7] and [Hut22a] and the references therein. In fact, there are some remarkably precise estimates of the implied constants for $\asymp$ in the above display: see [Hut22b] for example.

In fact, Hutchcroft derives the triangle condition from Equation 2.1 by means of the so-called L^2 -boundedness. We refer the interested readers to [Hut19, Section 2] for the derivation, and only record some consequences of the L^2 -boundedness.

Theorem 2.6 ([Hut19, Proposition 2.3], [Hut20a], [Hut22b], [Hut24]). *Let Γ be a Cayley graph of a finitely generated group. If*

$$\liminf_{p \nearrow p_c} \frac{p_c - p}{1 - p} \chi_p \sqrt{1 - \iota_p^2} = 0,$$

then we have the following:

- (1) *For $p \simeq p_c$, the two-point function $\tau_p(id, g)$ decays exponentially in $\|g\|_S$.*
- (2) *At the critical percolation, the intrinsic and the extrinsic radii exhibit similar critical behaviors. More precisely, we have*

$$\mathbb{P}_{p_c}(\sup\{r : B_S(id, r) \subseteq C(id)\} \geq n) \asymp n^{-1} \asymp \mathbb{P}_{p_c}(\text{diam}_S(C(id)) \geq n).$$

Furthermore, the implied constant for $\asymp$ in the above display depends exponentially on $|p - p_c|$.

- (3) *For $p \simeq p_c$, the intrinsic distance in $\Gamma[p]$ is not hugely distorted from the word metric in Γ : huge distortions are exponentially unlikely.*
- (4) *For $p \simeq p_c$ with $p > p_c$, the volume growth of the ball in $C(id)$ is purely exponential.*

We now turn to the strategy for Equation 2.1. It will follow from

$$(2.7) \quad \limsup_{p \nearrow p_c} (p_c - p) \chi_p < +\infty,$$

$$(2.8) \quad \limsup_{p \nearrow p_c} \left\{ \frac{\sum_{g,h \in A} \tau_p(g,h)}{\chi_p \cdot \#A} : A \subseteq G \text{ finite} \right\} = 0.$$

In order to show Equation 2.7, Hutchcroft proved the following for Gromov hyperbolic graphs that admit a vertex-transitive action by a unimodular group. We restrict ourselves to the case of Cayley graphs.

Proposition 2.7 (Supporting Hyperplane Theorem, [Hut19, Corollary 4.3]). *Let G be a non-elementary word hyperbolic group with a finite generating set S . Then there exists $r > 0$ such that the following holds.*

For each finite set $A \subseteq G$ there exists $A' \subseteq A$ with $\#A' \geq \#A/2$ such that for each $u \in A'$, there exists $v \in G$ with $d_S(u, v) \leq r$ such that $H_G(u, v)$ is a proper discrete halfspace with $A \subseteq H_G(u, v)$.

This is accompanied by:

Proposition 2.8. *Let G be a non-elementary word hyperbolic group with a finite generating set S . Then there exists $R > 0$ such that, for each $u, v \in G$ that give rise to a proper discrete halfspace $H_G(u, v)$, there exists $g \in G$ with $\|g\|_S \leq 2d_S(u, v) + R$ such that $H_G(u, v)$ and $gH_G(u, v)$ are disjoint.*

The precise shape of $H_G(u, v)$ is not relevant here. We only need that $H_G(u, v)$ is large enough to contain A , but also small enough such that some reasonably close translates of $H_G(u, v)$ do not overlap. Let us put them in more abstract language:

Definition 2.9. *Let G be a group with a finite generating set S and let $A, B \subseteq G$ be subsets with elements $a \in A, b \in B$. We say that (A, a) and (B, b) are decoupled if*

- (1) *no edge of $\text{Cay}(G, S)$ connects A to B , and*
- (2) *there exists a d_S -path $P = P_1 \circ P_2$ connecting a to b , where P_1 does not meet B and P_2 does not meet A .*

Theorem 2.10 ([Hut19, Section 5.1]). *Let $\Gamma = \text{Cay}(G, S)$ be the Cayley graph of a group G with a finite generating set S . Let $\mathcal{H}$ be a finite collection of subsets of G containing id and let $F \subseteq G$ be a finite subset. Suppose that every finite set $A \subseteq G$ satisfies*

$$\# \left\{ a \in A : \exists H_a \in \mathcal{H}, h_a \in F \left[\begin{array}{c} A \subseteq aH_a, \\ (H_a, \text{id}) \text{ and } (h_a H_a, h_a) \text{ are decoupled} \end{array} \right] \right\} \geq \#A/2.$$

Then Equation 2.7 holds for Γ .

This was proven in [Hut19, Section 5.1] for proper discrete halfspaces in G . We present Hutchcroft's proof in Appendix A for completeness.

The proof of Equation 2.8 is more involved. Using Benjamini and Schramm's magic lemma for Euclidean spaces, Hutchcroft proved a magic lemma for real hyperbolic space $\mathbb{H}^d$:

Proposition 2.11 (hyperbolic magic lemma, [Hut19, Proposition 4.1]). *Let X be a closed convex subset of $\mathbb{H}^d$ and let Y be a coarsely dense and uniformly locally finite subset of X . Then for every $\epsilon > 0$ there exists a constant $N(\epsilon)$ such that for every finite set $A \subseteq Y$ there exists a subset $A' \subseteq A$ with the following properties:*

- (1) $\#A' \geq (1 - \epsilon)\#A$.
- (2) For each $v \in A'$, there exists a pair of halfspaces $H_1, H_2 \subseteq \mathbb{H}^d$ such that $d_{\mathbb{H}^d}(v, H_1 \cup H_2) \geq \epsilon^{-1}$ and $\#(A \setminus (H_1 \cup H_2)) \leq N(\epsilon)$.

The precise shape of halfspaces H_i in Proposition 2.11 is again not important, but we will have to impose some “smallness” of H_i in terms of ϵ . More precisely, we need that $\mathbb{E}_p \# \{g \in C(id) : gx_0 \in H_i\} \lesssim \epsilon \chi_p$. Let us introduce some terminology.

Definition 2.12. *Let G be a group with a finite generating set S . For subsets $A, B, C \subseteq G$, we say that B is a d_S -barrier between A and C if every d_S -path $(g_0, g_1, \dots, g_n)$ in G starting at A (i.e., $g_0 \in A$) and ending at C (i.e., $g_n \in C$) intersects B (i.e., $\exists i[g_i \in B]$).*

We record Hutchcroft’s observation about barriers:

Lemma 2.13 ([Hut19, Proof of Lemma 5.4]). *Let G be a group with a finite generating set S . Let $A, B \subseteq G$ be such that B is a d_S -barrier between id and A . Then*

$$\mathbb{E}_p \#(C(id) \cap A) \leq \mathbb{E}_p \#(C(id) \cap B) \cdot \chi_p$$

for each $0 \leq p < p_c$.

Proof. For each $a \in A$ and $b \in B$ we define $E_b := \{\omega : id \leftrightarrow b\}$ and $F_{b,a} := \{\omega : b \leftrightarrow a\}$. Then

$$\begin{aligned} \sum_{a \in A, b \in B} \mathbb{P}_p(E_b \circ F_{b,a}) &\leq \sum_{a \in A, b \in B} \mathbb{P}_p(E_b) \mathbb{P}_p(F_{b,a}) = \sum_{b \in B} \mathbb{P}_p(E_b) \cdot \sum_{a \in A} \mathbb{P}_p(F_{b,a}) \\ &\leq \sum_{b \in B} \mathbb{P}_p(E_b) \cdot \mathbb{E}_p \# C(b) \\ &= \sum_{b \in B} \mathbb{P}_p(E_b) \cdot \chi_p = \mathbb{E}_p \#(C(id) \cap B) \cdot \chi_p. \end{aligned}$$

Meanwhile, for each $a \in A$ we claim that

$$\bigcup_{b \in B} (E_b \circ F_{b,a}) = \{\omega : id \leftrightarrow a\}.$$

The inclusion “ $\subseteq$ ” is clear. Now for “ $\supseteq$ ”, let $\omega \in \Omega$ be a configuration such that $id \leftrightarrow a$. Take a shortest path P in $\Gamma(\omega)$ connecting id and a , which does not revisit any vertex. Since B is a d_S -barrier between id and A , P visits a vertex $b \in B$. Then the subpaths of P between id and b , and between b and a , are disjoint (finite) witnesses for E_b and $F_{b,a}$, respectively. Hence, $\omega \in E_b \circ F_{b,a}$ as desired.

In conclusion, we have

$$\begin{aligned} \mathbb{E}_p \#(C(id) \cap A) &= \sum_{a \in A} \mathbb{P}_p\{id \leftrightarrow a\} \\ &\leq \sum_{a \in A, b \in B} \mathbb{P}_p(E_b \circ F_{b,a}) \leq \mathbb{E}_p \#(C(id) \cap B) \cdot \chi_p. \quad \square \end{aligned}$$

Having Lemma 2.13 in hand, it is desirable to construct a barrier B between the origin and a halfspace such that the “capacity” $\mathbb{E}_p \#(C(id) \cap B)$ of B is uniformly small for all $0 \leq p < p_c$. For example, B should not be the entire G ; the susceptibility $\chi_p = \mathbb{E}_p \#C(id)$ tends to infinity as $p \nearrow p_c$. Likewise, B should not contain an arbitrarily large d_S -metric ball. A geometric intuition is that if B is a *codimension 1* subset of the ambient set, then the portion of the cluster in B is finite because the cluster tends to escape B before growing large in it. The following notion captures this phenomenon.

Definition 2.14. *Let G be a group with a finite generating set S . We say that a set $B \subseteq G$ is r -roughly branching if there exists $B' \subseteq G$ such that:*

- (1) *B is contained in the r -neighborhood of B' in the word metric d_S .*
- (2) *For every $k \geq 1$, if $g_1, \dots, g_k$ and $h_1, \dots, h_k$ are distinct sequences of elements of B' , then $g_1 \cdots g_k \neq h_1 \cdots h_k$.*

Lemma 2.15 ([Hut19, Lemma 5.5]). *Let G be a group with a finite generating set S . Then for each $r > 0$ there exists M such that for every r -roughly branching subset $B \subseteq G$ and for every $0 \leq p \leq p_c$ we have $\mathbb{E}_p \#(C(id) \cap B) \leq M$.*

We sketch the proof for a 0-roughly branching set B ; see [Hut19] for a full proof. By the Harris–FKG inequality, $\tau_p(gh) \geq \tau_p(g) \cdot \tau_p(h)$ for each $g, h \in G$. Hence, for $g_1, \dots, g_k \in B$, we have $\tau_p(g_1 \cdots g_k) \geq \tau_p(g_1) \cdots \tau_p(g_k)$. Meanwhile, the k -fold product map from B^k to G : $(g_1, \dots, g_k) \mapsto g_1 \cdots g_k$ is injective by the assumption. This implies

$$\chi_p \geq \sum_{g \in B^k} \tau_p(g) = \sum_{g_1, \dots, g_k \in B} \tau_p(g_1 \cdots g_k) \geq \prod_{i=1}^k \left(\sum_{g_i \in B} \tau_p(g_i) \right).$$

For a given $0 \leq p < p_c$, this is true regardless of k . Note that $\chi_p < +\infty$. This forces $\sum_{g \in B} \tau_p(g) \leq 1$ for each $0 \leq p < p_c$. Since $p \mapsto \sum_{g \in B} \tau_p(g)$ is a lower semicontinuous function on $[0, 1]$ (cf. [Hut16, Lemma 5]), the same bound holds for $p = p_c$ as well.

We state the “smallness” of halfspaces in $\mathbb{H}^d$ in terms of nested barriers.

Proposition 2.16 ([Hut19, Lemma 5.6]). *Let $X \ni x_0$ be a closed convex set of $\mathbb{H}^d$ and suppose that $G \leq \text{Isom}(X)$ acts properly and coboundedly on X . Let S be a finite generating set of G . Then there exist $r, R > 0$ such that for each halfspace $H \subseteq \mathbb{H}^d$, there exists an r -roughly branching subset*

$$B = B_1 \sqcup B_2 \sqcup \dots \sqcup B_{\lfloor d(x_0, H)/R \rfloor} \subseteq G$$

such that B_i is a d_S -barrier between id and $\{g : gx_0 \in H\}$ for each $i = 1, \dots, \lfloor d(x_0, H)/R \rfloor$.

In the above, the capacity of B is uniformly bounded in p and H ; hence, there exists B_i such that $\mathbb{E}_p \#(C(id) \cap B_i) \lesssim 1/d(x_0, H)$. By Lemma 2.13 we conclude

Corollary 2.17 ([Hut19, Lemma 5.4]). *Let $X \subseteq \mathbb{H}^d$ and $G \leq \text{Isom}(X)$ be as in Proposition 2.16. Then there exists $K > 0$ such that for each halfspace $H \subseteq \mathbb{H}^d$ we have*

$$\mathbb{E}_p \#(C(id) \cap \{g \in G : gx_0 \in H\}) \leq \frac{K}{d(x_0, H)} \chi_p$$

for each $0 \leq p < p_c$.

Proposition 2.11 and Corollary 2.17 describe all we need for halfspaces. Let us now state an abstract version:

Theorem 2.18 ([Hut19, Proof of Proposition 5.2]). *Let $\Gamma = \text{Cay}(G, S)$ be the Cayley graph of a finitely generated group G . Suppose that there exists $r > 1$, and for each $D, E > 0$ there exist*

$$S_D = \sqcup_{i=1}^{\infty} S_{D,i} \subseteq G, \quad \mathcal{G}_{D,E} \subseteq G$$

and a collection $\mathcal{H}_D$ of subsets of G such that

- (1) S_D is r' -roughly branching for some $r' = r'_D$,
- (2) for each $\mathcal{H} \in \mathcal{H}_D$ there exists an r -roughly branching subset $B = B_1 \sqcup \dots \sqcup B_{D/r} \subseteq G$ such that B_i is a d_S -barrier between id and $\mathcal{H}$ for $i = 1, \dots, D/r$;
- (3) for each $D, E > 0$, $\sqcup_{i \geq E} S_{D,i}$ is a d_S -barrier between id and $\mathcal{G}_{D,E}$.

Suppose that for each $\epsilon > 0$ and $D, E > 0$, there exists a constant $N = N(\epsilon, D, E)$ such that for every finite set $A \subseteq G$ there exists $A' \subseteq A$ satisfying:

- (1) $\#A' \geq (1 - \epsilon)\#A$;
- (2) For each $a \in A'$ there exist $\mathcal{H}_1, \mathcal{H}_2 \in \mathcal{H}_D$ such that

$$\#(A \setminus a \cdot (\mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{G}_{D,E})) \leq N.$$

Then Equation 2.8 holds for Γ .

In Hutchcroft's original formulation for Gromov hyperbolic graphs, the sets S_D and $\mathcal{G}_{D,E}$ are not needed. It is not hard to adapt Hutchcroft's proof to the current version; we include it for completeness.

Proof. By Lemma 2.15, for each D we have $\sum_{g \in S_D} \tau_{p_c}(g) < +\infty$. Furthermore, note that $\sqcup_{i=1}^E S_{D,i}$ exhausts S_D as E increases. Hence, for each $D > 0$ and $\epsilon > 0$ there exists $E = E(D, \epsilon) > 0$ such that $\sum_{g \in \sqcup_{i \geq E} S_{D,i}} \tau_{p_c}(g) \leq \epsilon$. Then by Lemma 2.13, we have $\sum_{g \in \mathcal{G}_{D,E}} \tau_p(g) \leq \epsilon \cdot \chi_p$ for each $0 < p < p_c$.

Now, let $M = M(r)$ for r as in Lemma 2.15. Then by Assumption (2) and Lemma 2.13, we have $\sum_{g \in \mathcal{H}} \tau_p(g) \leq rM\chi_p/D$ for each $0 < p < p_c$ and for each $\mathcal{H} \in \mathcal{H}_D$.

Let us now fix $\epsilon > 0$. We take $D > rM/\epsilon$, and then $E = E(D, \epsilon)$. Now let $N = N(\epsilon, D, E)$. Lastly, recall that $\lim_{p \nearrow p_c} \chi_p = +\infty$; there exists p_0 such that $\chi_p \geq N/\epsilon$ for $p_0 < p < p_c$.

We now claim that

$$\frac{\sum_{g,h \in A} \tau_p(g, h)}{\#A} \leq 5\epsilon\chi_p \quad (\forall \text{ finite } A \subseteq G, \forall p_0 < p < p_c).$$

To show this, let $A \subseteq G$ be a finite set and let $p_0 < p < p_c$. Let $A' \subseteq A$ be as in the statement for ϵ, D, E . Then we have

$$\begin{aligned} \sum_{g,h \in A} \tau_p(g, h) &\leq \sum_{g \in A \setminus A', h \in G} \tau_p(g, h) + \sum_{g \in A', h \in A \setminus g \cdot (\mathcal{H}_1(g) \cup \mathcal{H}_2(g) \cup \mathcal{G}_{D,E})} \tau_p(g, h) \\ &\quad + \sum_{g \in A', k \in \mathcal{H}_1(g) \cup \mathcal{H}_2(g) \cup \mathcal{G}_{D,E}} \tau_p(id, k) \\ &\leq \epsilon(\#A) \cdot \chi_p + (\#A') \cdot N + (\#A') \cdot \left(\frac{rM}{D} \chi_p + \frac{rM}{D} \chi_p + \epsilon\chi_p \right) \\ &\leq 5\epsilon(\#A)\chi_p. \end{aligned}$$

Since ϵ is arbitrary, we conclude that Equation 2.8 holds. $\square$

Combining the aforementioned facts about word hyperbolic groups and convex subsets of $\mathbb{H}^d$, together with the Bonk–Schramm embedding theorem, Hutchcroft showed that word hyperbolic groups satisfy Equation 2.7 and 2.8.

2.3. Intuition and examples. We now explain how to generalize Hutchcroft's results to relatively hyperbolic groups and acylindrically hyperbolic groups. Our ultimate goals are Theorem 2.10 and Theorem 2.18, which will allow us to apply Theorem 2.5. Theorem 2.10 is a combination of Proposition 2.7 and 2.8, and Theorem 2.18 is a combination of Proposition 2.11 and 2.16. Among them, Proposition 2.8 generalizes to non-elementary group actions on δ -hyperbolic spaces without much difficulty. So we focus on generalizing Proposition 2.7 and 2.11 first, and then discuss Proposition 2.16.

Our primary example will be the free group $G := F_2 \simeq \langle a, b \rangle$ of rank 2. We will consider its two Cayley graphs:

$$\Gamma_{std} := \text{Cay}(F_2, \{a, b\}), \quad \Gamma := \text{Cay}(F_2, \{a^i : i \in \mathbb{Z}\} \cup \{b\}).$$

Here, Γ_{std} is a 4-regular tree. One should regard it as a δ -hyperbolic space on which (relatively) hyperbolic groups act properly. Meanwhile, Γ is a locally infinite quasi-tree (i.e., quasi-isometric to a tree). One should regard it as a δ -hyperbolic space that acylindrically hyperbolic groups act on.

Let us first discuss Γ_{std} . For $x, y \in G$, let us define standard halfspaces

$$\mathcal{H}_{std}(x, y) := \{g \in G : d_{std}(g, x) \leq d_{std}(g, y)\}.$$

One standard halfspace is drawn in Figure 3. The following logic shows that, given the collection $\mathcal{H} := \{\mathcal{H}_{std}(id, g) : g \in G\}$ of halfspaces, Proposition

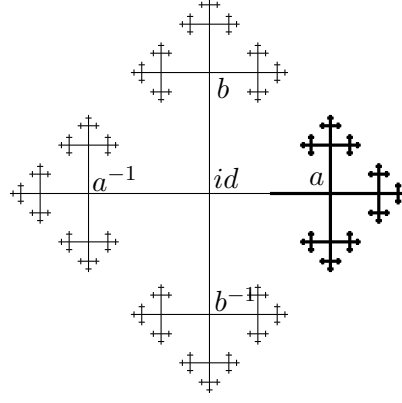


FIGURE 3. The graph Γ_{std} with some vertices labelled and a halfspace highlighted.

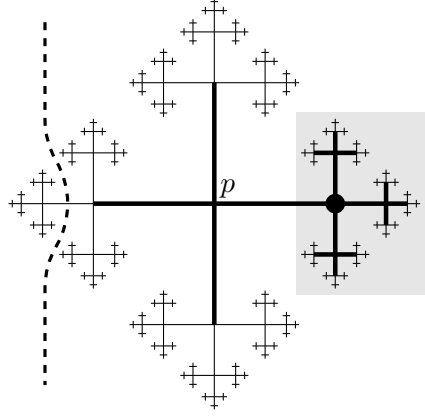


FIGURE 4. A metric ball A in Γ_{std} and its center. The point p lies in its thickened boundary $\partial_2 A$. The dashed line is a hyperplane at distance ≤ 2 from p that bounds a halfspace containing A . The shaded area is a halfspace at distance ≤ 1 from p containing all but 4 points of A .

2.7 holds for *convex subsets* A . Since Γ_{std} is nonamenable, there exists $R > 0$ such that at least 50% of A is concentrated in the boundary

$$\partial_R A := \{u \in A : d_{std}(u, A^c) \leq R\}.$$

Now for each $u \in \partial_R A$, there exists a hyperplane P such that $d_{std}(u, P) < R$ and such that P separates u and a point of A^c . Since A is a convex set containing u , P bounds a halfspace that contains the entire A . This is the desired conclusion of Proposition 2.7. A similar argument implies the conclusion of Proposition 2.11 for *strictly convex subsets* A such as metric balls in Γ_{std} . See Figure 4.

Let us now consider a non-convex subset such as

$$A := B_{std}(20) \setminus B_{std}(10) = \{g \in G : \|g\|_{std} \in [11, 20]\}.$$

Then $\partial A = \partial B_{std}(11) \cup \partial B_{std}(20)$ takes up more than half of A . However, elements u of $\partial B_{std}(11)$, the *inward* boundary, do not satisfy the desired properties: there is no $v \in G$ such that $d(u, v) = 1$ and such that $\mathcal{H}_{std}(u, v)$ contains the entire A . Also, there are no $v, w \in G$ such that $d(u, v) = d(u, w) = 1$ and such that $\mathcal{H}_{std}(v, u) \cup \mathcal{H}_{std}(w, u)$ contain all but (uniformly) finitely many elements of A .

Hence, the point is to prove that most of A is concentrated in $\partial B_{std}(20)$, the *outward* boundary. Let us give a heuristic for this statement. Note that for each $u \in A \setminus \partial B_{std}(20)$, there exist three “children” $u_1, u_2, u_3 \in A$, i.e., $d_{std}(u_i, u) = 1$ and $\|u_i\|_{std} = \|u\|_{std} + 1$. Hence, for $i = 0, \dots, 20$ we have

$$\#\{g \in A : \|g\|_{std} = i, g \notin \partial B_{std}(20)\} \leq \frac{1}{3} \#\{g \in A : \|g\|_{std} = i + 1\}.$$

By using this fact inductively, one can show that the outward boundary $\partial B_{std}(20)$ takes up at least 50% of A .

We will prove Proposition 2.11 for arbitrary subsets of Γ_{std} and, more generally, groups acting properly on a δ -hyperbolic space X in Section 4.

Let us now discuss Proposition 2.16 for Γ_{std} . In the real hyperbolic space $\mathbb{H}^d$, a halfspace H at distance D from the basepoint is barred by D hyperplanes $P_1, \dots, P_D$ such that $d(P_i, P_{i+1}) = 1$. Moreover, from the fact that hyperplanes have positive codimension in $\mathbb{H}^d$, one can show that the union $\cup_i P_i$ of these hyperplanes is roughly branching.

How about halfspaces in Γ_{std} ? Let us introduce one notation: for each $g \in G$, let $I(g)$ denote the initial power of b appearing in the word g . For example, $I(b^3ab) := 3$, $I(b^4a^{-2}b^{-10}) := 4$, etc.

In between id and a halfspace $H := \mathcal{H}_{std}(b^8, id) = \{g \in G : I(g) \geq 4\}$, we can place 3 disjoint sets $B'_i := \{g : I(g) = i\}$. Then each of B'_1, B'_2, B'_3 is a d_{std} -barrier between id and H . However, the B'_i 's are too large and should not be considered “codimension-1” subsets. Indeed, the B'_i 's are not roughly branching and contain arbitrarily large d_{std} -metric balls. This leads to the fact that $\mathbb{E}_p \#(C(id) \cap B'_i)$ is not uniformly bounded in $p \in (0, p_c)$. In short, the B'_i 's are not suitable for the conclusion of Proposition 2.16.

We can remedy this by considering “horizontal” barriers $B_i := \{b^i a^k : k \in \mathbb{Z}\}$ that are “thinner” than B'_i 's. We claim that B_1 is still a d_{std} -barrier between id and H . To see this, suppose to the contrary that a d_{std} -path $P = (id = g_0, g_1, \dots, g_N \in H)$ avoids B_1 . Then $I(g_i)$ grows from 0 to 4 as i changes from 0 to N . Hence, there is a moment $i(1)$ where $I(g_{i(1)}) = 1$, i.e., $g_{i(1)} = bw$ for some w not starting with $b^{\pm 1}$. Since P avoids B_1 , the subword w contains the letter b . That is, $g_{i(1)} = ba^k b^\epsilon \cdot v$ in its reduced form for some $k \in \mathbb{Z} \setminus \{0\}$, $\epsilon \in \{1, -1\}$ and a subword v .

We claim that the letter b^ϵ after ba^k cannot be erased along the path, i.e., g_i starts with $ba^k b^\epsilon$ for every $i \geq i(1)$. If b^ϵ is to be erased at some step

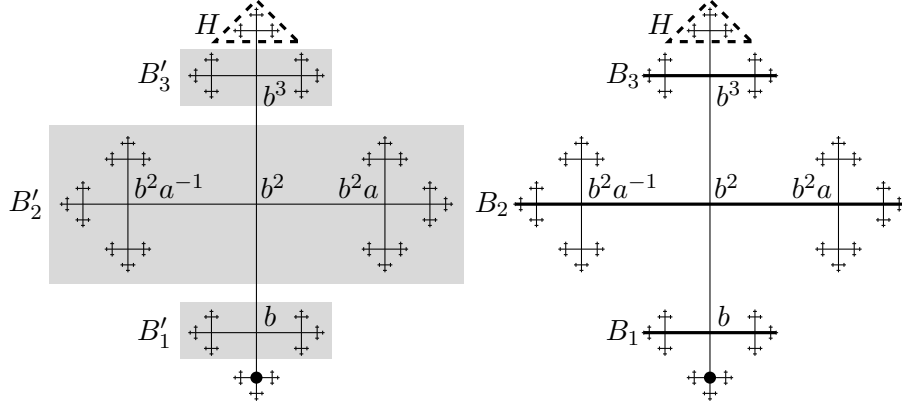


FIGURE 5. Two drawings of Γ_{std} . The vertex id is marked with a black dot.

$i \geq i(1)$, the only possibility is $g_i = ba^k b^e$ and $g_{i+1} = ba^k \in B_1$, contradicting the assumption. But if every g_i starts with $ba^k b^e$, including at $i = N$, then g_N cannot land in $H = \mathcal{H}_{std}(b^8, id)$. This is a contradiction.

Moreover, $\cup_i B_i$ is contained in the 1-neighborhood of $\{b^k a^l : k \geq 1, l \neq 0\}$, which freely generates a free semigroup. This is clear from the fact that

$$b^{i(1)} a^{k(1)} \dots b^{i(N)} a^{k(N)} = b^{j(1)} a^{m(1)} \dots b^{j(N)} a^{m(N)} \Rightarrow i(l) = j(l), k(l) = m(l)$$

for each $1 \leq l \leq N$. This idea will be developed rigorously in Section 6.

We now turn to Γ depicted in Figure 6. Let us define

$$\mathcal{H}_\Gamma(x, y) := \{g \in G : d_\Gamma(g, x) \leq d_\Gamma(g, y)\}, \quad \mathcal{H} := \{\mathcal{H}_\Gamma(id, g) : g \in G\}.$$

Now, fix a large D and let

$$A' := \{a^i : 0 \leq i \leq 6^D\}.$$

Let A'' be a set of 60 elements, and let $A := A' \cup A''$. Hence, A' takes up at least 99% of A . Note that A' is contained in a ball of radius 1 in Γ . Hence, the collection

$$\mathcal{C} := \{u \cdot \mathcal{H}_\Gamma(id, b^2)^c : u \in A'\}$$

consists of halfspaces at d_Γ -distance at most 3 from id . If Γ were a graph of degree 4, then we could have picked an A'' that meets every halfspace in $\mathcal{C}$.

But this is not the case, as Γ has infinite valency. Rather, the halfspaces in $\mathcal{C}$ are pairwise disjoint. Hence, although some halfspaces in $\mathcal{C}$ might intersect the finite set A'' , such halfspaces are very few. Moreover, every halfspace in $\mathcal{C}$ is disjoint from A' . We conclude that $u \cdot \mathcal{H}_\Gamma(id, b^2)$ contains A for most choices of $u \in A' \subseteq A$. In other words, the conclusion of Proposition 2.7 holds for A and $\mathcal{H}$. We expand this idea for general WPD actions on δ -hyperbolic spaces in Section 5.

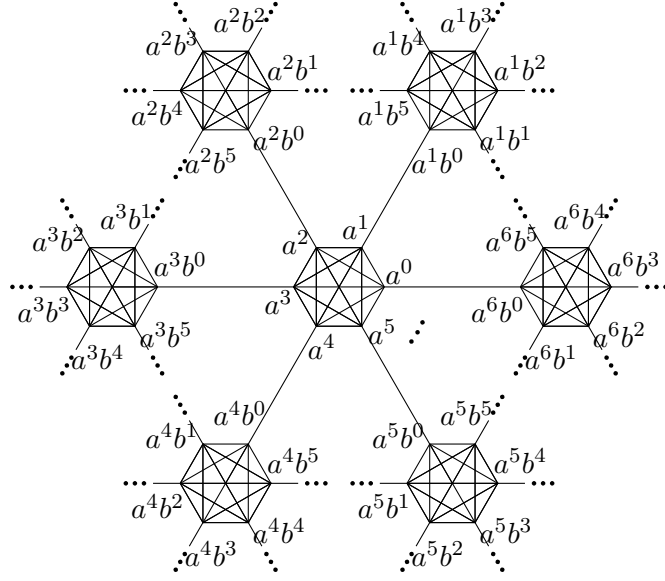


FIGURE 6. Drawing of (part of) Γ . Each a -orbit forms an infinite clique, and the entire graph is quasi-isometric to a (countably) ∞ -regular tree.

Meanwhile, generalizing Proposition 2.11 and Proposition 2.16 for WPD actions is much harder. This is why we need auxiliary objects S_D and $\mathcal{G}_{D,E}$ in Theorem 2.18. Let us only describe the intuition behind these objects.

Note that G is acting acylindrically on a δ -hyperbolic space Γ . Here, the $\langle b \rangle$ -orbit (denoted by $Ax(b)$) is isometrically embedded in Γ , whereas the orbit of a is distorted. With this in mind, we informally define

$$S_D := \left\{ g \in G : \begin{array}{l} \text{In } \Gamma, [id, g]_\Gamma \text{ fellow travels with} \\ \text{each translate of } Ax(b) \text{ for length } \leq D \end{array} \right\}.$$

In other words, S_D is the set of words that do not contain the subword $b^{\pm D}$. This set becomes a roughly branching set (Proposition 7.8). Viewed in Γ_{std} , elements of S_D are those words that are reached from id by moving in an “almost horizontal” direction. These resemble the horizontal hyperplanes B_i ’s we defined above (but not exactly the same).

Next, we informally define

$$\mathcal{G}_{D,E} := \left\{ g \in G : \begin{array}{l} \text{For any } h \in G \text{ such that } \|h\|_{std} \leq E, \\ [id, g]_\Gamma \text{ does not fellow travel with} \\ h \cdot Ax(b) \text{ for length } \geq D \end{array} \right\}.$$

This is the collection of words that do not fellow travel with $Ax(b)$ “in the beginning”. These resemble the halfspace $H = \mathcal{H}_{std}(b^{2E}, id)$ earlier: to reach H , one has to first move in the “almost horizontal direction” for length

E , but is allowed to move freely afterwards. Intuitively, $\mathcal{G}_{D,E}$ is barred by $S_D \cap \{g : \|g\|_S \geq E/2\}$. We formally prove this in Proposition 7.9.

3. PRELIMINARIES II: HYPERBOLIC GEOMETRY

Given three real numbers A, B, C , we write $A =_C B$ if $|A - B| < C$.

A *geodesic* in a metric space (X, d_X) is an isometric embedding $\gamma : I \rightarrow X$ of a closed connected subset $I \subseteq \mathbb{R}$ into X . We will frequently refer to the image of γ as γ . Throughout, every metric space (X, d_X) is assumed to be *geodesic*, i.e., every pair of points is connected by a geodesic segment. We will however not assume that (X, d_X) is locally compact or complete.

Let $\gamma : [0, L] \rightarrow X$ be a geodesic and let $A_1, \dots, A_n \subseteq X$. We say that γ *passes through* $(A_1, \dots, A_n)$ *in order* if there exist $0 \leq t_1 \leq \dots \leq t_n \leq L$ such that $\gamma(t_i) \in A_i$ for each i .

Let (X, d_X) be a geodesic metric space. Given a set $A \subseteq X$, we define its R -neighborhood

$$\mathcal{N}_R(A) := \{x \in X : \exists a \in A [d_X(a, x) \leq R]\}$$

for each $R > 0$. We define the Hausdorff distance between two sets

$$d_{\text{Haus}}(A, B) := \inf\{R \geq 0 : A \subseteq \mathcal{N}_R(B) \wedge B \subseteq \mathcal{N}_R(A)\}.$$

We will say that two sets $A, B \subseteq X$ are R -*equivalent* if they are within Hausdorff distance R .

For $x, y \in X$, we denote by $[x, y]$ an arbitrary geodesic between x and y . Note that such a geodesic may not be unique.

We now recall the notion of Gromov hyperbolicity due to M. Gromov [Gro87]. The version we present here is due to E. Rips. Comprehensive expositions can be found in [CDP90] and [BH99].

Definition 3.1. *Let (X, d_X) be a metric space. For a given $\delta > 0$, we say that (X, d_X) is δ -hyperbolic if every geodesic triangle is δ -slim, that is,*

$$\forall x, y, z \in X \ [[x, z] \subseteq \mathcal{N}_\delta([x, y]) \cup \mathcal{N}_\delta([y, z])].$$

We call (X, d_X) Gromov hyperbolic if it is δ -hyperbolic for some $\delta > 0$.

In Gromov hyperbolic spaces, geodesics with pairwise near endpoints follow travel.

Lemma 3.2. *Let x, y, x', y' be points on a δ -hyperbolic space X . Let $\gamma : [0, T] \rightarrow X$ and $\gamma' : [0, T'] \rightarrow X$ be the length parametrizations of $[x, y]$ and $[x', y']$, respectively. Then we have*

$$d_X(\gamma(t), \gamma'(t)) \leq d_X(x, x') + 4\delta. \quad \left(\forall 0 \leq t < \min(T', T - d_X(y, y') - \delta) \right)$$

This is a variant of [BH99, Lemma III.H.1.15]. Let us give a short proof.

Proof. As $\triangle xy y'$ is δ -slim, $\gamma(t)$ is δ -close to either $[x, y']$ or $[y, y']$. Note that

$$d_X(\gamma(t), [y, y']) \geq d_X(x, y) - d_X(y, y') - d_X(x, \gamma(t)) > \delta.$$

Hence, $\gamma(t)$ is δ -close to a point of $[x, y']$, which is δ -close to either $\gamma' = [x', y']$ or $[x, x']$. In the former case, we have $d_X(\gamma(t), \gamma'(t')) \leq 2\delta$ for some t' . Here, $t' = d_X(\gamma'(0), \gamma'(t')) \leq d_X(\gamma'(0), \gamma(0)) + t + d_X(\gamma(t), \gamma'(t')) \leq t + d_X(x, x') + 2\delta$, and similarly $t' \geq t - d_X(x, x') - 2\delta$. This implies that

$$d_X(\gamma(t), \gamma'(t)) \leq d_X(\gamma(t), \gamma'(t')) + |t' - t| \leq d_X(x, x') + 4\delta.$$

In the latter case, $\gamma(t)$ is 2δ -close to a point $p \in [x, x']$. Here, $d_X(p, x') = d_X(x, x') - d_X(x, p) \leq d_X(x, x') - t + 2\delta$. This implies that $d_X(p, \gamma'(t)) \leq d_X(x, x') + 2\delta$ and $d_X(\gamma(t), \gamma'(t)) \leq d_X(x, x') + 4\delta$ as desired. $\square$

Model examples of Gromov hyperbolic spaces are simplicial/ $\mathbb{R}$ -trees and real hyperbolic space $\mathbb{H}^d$. In these spaces, the following phenomenon happens: if you walk forward for some distance, and walk in another direction *without huge backtracking*, and walk in yet another direction without huge backtracking, and so on, then you will never come back to the original place. In order to formulate the property rigorously, let us define:

Definition 3.3. *Let (X, d_X) be a geodesic metric space. For $x, y, z \in X$, we define the Gromov product of y and z based at x by*

$$(y|z)_x := \frac{1}{2}[d_X(x, y) + d_X(x, z) - d_X(y, z)].$$

For example, in the standard Cayley graph of $F_2 = \langle a, b \rangle$, we have $(aaba|aab^{-1}ab)_{id} = 2$ since $aaba$ and $aab^{-1}ab$ share the first two letters.

The following property is called the *Gromov inequality*.

Lemma 3.4 ([BH99, Prop III.H.1.22]). *Let x, y, z, w be points on a δ -hyperbolic space X . Then we have*

$$(x|y)_w \geq \min((x|z)_w, (z|y)_w) - 4\delta.$$

We now formulate the local-to-global phenomenon mentioned above:

Lemma 3.5 ([Gou22, Lemma 3.7]). *Let $K > 0$, let $x_0, \dots, x_n$ be points in a δ -hyperbolic space X such that $d_X(x_{i-1}, x_i) \geq 2K + 3\delta + 1$ for each i and $(x_{i-1}|x_{i+1})_{x_i} \leq K$ for each i . Then $(x_i|x_k)_{x_j} \leq K + 2\delta$ for each $i < j < k$.*

Lemma 3.6 ([BH99, Proposition III.H.1.17]). *Let x, y, z be points in a δ -hyperbolic space, and let $p \in [x, y], q \in [x, z]$ be such that $d_X(x, p) = d_X(x, q) = (y|z)_x$. Then $[x, p]$ and $[x, q]$ are 4δ -fellow traveling. That means, for length parametrizations $\gamma, \gamma' : [0, (y|z)_x] \rightarrow X$ of $[x, p]$ and $[x, q]$, respectively, we have $d_X(\gamma(t), \gamma'(t)) \leq 4\delta$ for every $t \in [0, T]$.*

In particular, we have $d_X(y, [x, z]) \leq (x|z)_y + 4\delta$.

Let us now turn to isometries. Let (X, d_X) be a Gromov hyperbolic space and let g be its isometry. We say that g is *loxodromic* if there exists $\tau > 0$ such that $d_X(x, g^n x) \geq \tau n$ for each n and for each $x \in X$.

Prototypes of loxodromic isometries are the loxodromic isometries of $\mathbb{H}^d$. They act as a translation along an infinite geodesic. An isometry g of a

Gromov hyperbolic space X is called an *axial loxodromic* if there exist $\tau > 0$ and a geodesic $\gamma : \mathbb{R} \rightarrow X$ such that $g(\gamma(t)) = \gamma(t + \tau)$ for each $t \in \mathbb{R}$. In this case, we call γ an *axis* of g and denote it by $Ax(g)$. In general, given a group G acting on a Gromov hyperbolic space X and a loxodromic isometry $g \in G$, it is not hard to put another metric on X that is G -equivariantly quasi-isometric to the original one, so that g becomes a unital axial loxodromic isometry. See e.g. [BF02, Proposition 6.(2)].

Definition 3.7. Let (X, d_X) be a Gromov hyperbolic space. We say that two loxodromic isometries $g, h \in \text{Isom}(X)$ are independent if

$$\sup_{i,j \in \mathbb{Z}} (g^i x_0 | h^j x_0)_{x_0} < +\infty$$

for some (equivalently, every) $x_0 \in X$. We say that a group $G \leq \text{Isom}(X)$ is non-elementary if it contains two independent loxodromics.

We now introduce the nearest point projection.

Definition 3.8. Let (X, d_X) be a metric space and let $A \subseteq X$ be a locally compact subset. We define the nearest point projection $\pi_A(\cdot) : X \rightarrow 2^A$ as

$$\pi_A(x) := \left\{ a \in A : d_X(x, a) = \min_{y \in A} d_X(x, y) \right\}.$$

For $B, C \subseteq X$, we use the notation $\pi_A(B) := \bigcup_{x \in B} \pi_A(x)$, $\text{diam}_A(B) := \text{diam}_X(\pi_A(B))$ and $d_A(B, C) := \text{diam}_X(\pi_A(B \cup C))$.

Lemma 3.9. Let x, y, z be points on a δ -hyperbolic space X . Let $p \in [x, y]$ be such that $d_X(x, p) = (y|z)_x$. Then $\pi_{[x, y]}(z)$ is contained in $\mathcal{N}_{8\delta}(p)$.

In Gromov hyperbolic spaces, geodesics exhibit the so-called *contracting property*. The following is one formulation of the contracting property.

Lemma 3.10 ([CDP90, Proposition 10.2.1]). Let X be a δ -hyperbolic space, let $x, y \in X$, let γ be a geodesic and let $p \in \pi_\gamma(x)$, $q \in \pi_\gamma(y)$. Then we have

$$d_X(p, q) \leq \max(12\delta, 12\delta + d_X(x, y) - d_X(x, p) - d_X(q, y)).$$

This lemma has the following corollary.

Corollary 3.11. Let (X, d_X) be a δ -hyperbolic space, let $x, y \in X$ and let γ be a geodesic in X .

- (1) (coarse Lipschitzness) We have $\text{diam}(\pi_\gamma(x)) \leq 12\delta$ and $d_\gamma(x, y) \leq d_X(x, y) + 12\delta$.
- (2) (constriction) Let $p \in \pi_\gamma(x)$ and $q \in \pi_\gamma(y)$. Suppose that $d_X(p, q) > 12\delta$. Then $[p, q]$ is within Hausdorff distance 12δ from some subsegment $[x', y']$ of $[x, y]$, where $d_X(x', p), d_X(y', q) < 10\delta$.
- (3) (equivalent geodesics) Let γ' be a geodesic whose endpoints are pairwise D -near with the ones of γ . Then $\pi_\gamma(x)$ and $\pi_{\gamma'}(x)$ are $(2D + 28\delta)$ -equivalent.

We include the proof in Appendix B for completeness.

We now review the notion of weak proper discontinuity [BF02].

Definition 3.12. Let (X, d_X) be a Gromov hyperbolic space and let $G \leq \text{Isom}(X)$. We say that the action of G on X is proper if

$$\forall R > 0 \left[\#\{g \in G : d_X(x_0, gx_0) < R\} < +\infty \right].$$

Let $f \in G$ be a loxodromic isometry. We say that G has weak proper discontinuity (the WPD property) along f , or that f has the WPD property, if for each $R > 0$ there exists $L > 0$ such that

$$\#\{g \in G : d_X(x_0, gx_0) < R \text{ and } d_X(f^L x_0, gf^L x_0) < R\} < +\infty.$$

If f is axial in addition, we call it an axial WPD loxodromic.

A non-virtually cyclic finitely generated group is called acylindrically hyperbolic if it admits a WPD action on a Gromov hyperbolic space.

It is a fact that acylindrically hyperbolic groups are non-elementary [BF02]; any WPD action on a Gromov hyperbolic space by a non-virtually cyclic group entails infinitely many independent loxodromic isometries.

If G acts properly on a Gromov hyperbolic space, then every loxodromic element has the WPD property automatically. We record a theorem by M. Bestvina, K. Bromberg and K. Fujiwara.

Theorem 3.13 ([BBF15, Theorem H]). *Let G be an acylindrically hyperbolic group. Then G contains an element f and admits an isometric action on a Gromov hyperbolic space X , and there exists a constant $K > 0$ such that the following holds:*

- (1) f is a unital, axial WPD loxodromic isometry of X ;
- (2) for each $g \in G$, either
 - (a) (bounded projection) the nearest point projection of $Ax(f)$ onto $gAx(f)$ has diameter $\leq K$, or
 - (b) $g \cdot Ax(f) = Ax(f)$.

The collection of elements g in case (b), called the elementary closure $EC(f)$, is a virtually cyclic subgroup of G . Moreover, any loxodromic element independent of f lies outside $EC(f)$.

The following is called the Behrstock inequality [Beh06], which is a basic ingredient of the quasi-tree construction in [BBF15]. We sketch the proof for the reader's convenience.

Lemma 3.14. *Let X be a δ -hyperbolic space and let $\gamma_1, \gamma_2, \gamma_3$ be three geodesics with $\text{diam}_{\gamma_i}(\gamma_j) < K$ whenever $|i - j| = 1$. Then the following implication holds:*

$$d_{\gamma_2}(\gamma_1, \gamma_3) \geq 2K + 70\delta \Rightarrow d_{\gamma_1}(\gamma_2, \gamma_3) < 2K + 70\delta.$$

Proof. We will prove $d_{\gamma_1}(p, q) < 35\delta$ for each $q \in \gamma_3$ and $p \in \pi_{\gamma_2}(q)$. This would then imply $d_{\gamma_1}(\gamma_2, \gamma_3) < \max(\text{diam}_{\gamma_1}(\gamma_2) + 35\delta, 2 \cdot 35\delta) \leq K + 70\delta$.

Suppose to the contrary that $d_{\gamma_1}(p, q) \geq 35\delta$; let $u \in \pi_{\gamma_1}(p)$ and $v \in \pi_{\gamma_1}(q)$ be the diameter realizers. By Corollary 3.11(2), $[p, q]$ contains a subsegment

$$\begin{aligned}
[p', q'] \text{ with } d_X(p', u), d_X(q', v) &\leq 10\delta. \text{ Note that } d_X(p', q') \geq 15\delta. \text{ We have} \\
d_{\gamma_2}(q', q) &\geq d_{\gamma_2}(v, q) - 24\delta \quad (\because \text{Corollary 3.11(1)}) \\
&\geq d_{\gamma_2}(\gamma_1, \gamma_3) - \text{diam}_{\gamma_2}(\gamma_1) - \text{diam}_{\gamma_2}(\gamma_3) - 24\delta \\
&\geq (2K + 70\delta) - K - K - 24\delta \geq 46\delta.
\end{aligned}$$

By Corollary 3.11(2), $[q', q]$ meets $\mathcal{N}_{12\delta}(\gamma_2)$. This leads to the contradiction

$$d_X(q, \gamma_2) \leq d_X(q, q') + 12\delta < d_X(q, q') + d_X(q', p') \leq d_X(q, p). \quad \square$$

Corollary 3.15. *Let X be a δ -hyperbolic space and let $\gamma_0, \gamma_1, \dots, \gamma_n$ be geodesics with pairwise K -bounded projections such that $d_{\gamma_{i-1}}(\gamma_0, \gamma_i) \geq 6K + 210\delta$ for $1 < i \leq n$. Then $d_{\gamma_i}(\gamma_0, \gamma_n) =_{2K+70\delta} d_{\gamma_i}(\gamma_0, \gamma_{i+1})$ for each i .*

Proof. By Lemma 3.14, we have the implication

$$\begin{aligned}
&\left[\begin{array}{l} d_{\gamma_i}(\gamma_0, \gamma_n) \geq 4K + 140\delta \\ d_{\gamma_{i-1}}(\gamma_0, \gamma_i) \geq 4K + 140\delta \end{array} \right] \Rightarrow d_{\gamma_i}(\gamma_0, \gamma_{i-1}) < 2K + 70\delta \\
&\Rightarrow d_{\gamma_i}(\gamma_{i-1}, \gamma_n) \geq d_{\gamma_i}(\gamma_0, \gamma_n) - d_{\gamma_i}(\gamma_0, \gamma_{i-1}) \geq 2K + 70\delta \\
&\Rightarrow d_{\gamma_{i-1}}(\gamma_i, \gamma_n) < 2K + 70\delta \\
&\Rightarrow d_{\gamma_{i-1}}(\gamma_0, \gamma_n) =_{2K+70\delta} d_{\gamma_{i-1}}(\gamma_0, \gamma_i).
\end{aligned}$$

For the last implication, we used $|d_{\gamma_{i-1}}(\gamma_0, \gamma_n) - d_{\gamma_{i-1}}(\gamma_0, \gamma_i)| \leq d_{\gamma_{i-1}}(\gamma_i, \gamma_n)$. Now the induction from $i = n - 1$ to $i = 2$ yields the conclusion. $\square$

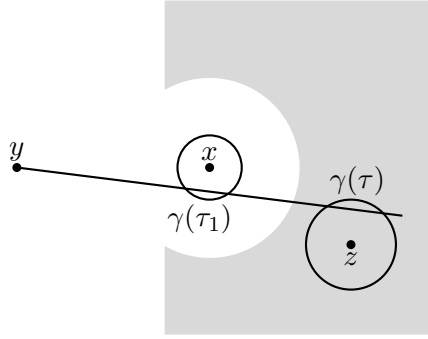
A similar induction yields the following. We omit the proof.

Corollary 3.16. *Let X be a δ -hyperbolic space and let $\mathcal{A}$ be a collection of unbounded geodesics such that $\text{diam}_\gamma(\gamma') \leq K$ for every pair of distinct elements $\gamma, \gamma' \in \mathcal{A}$. Let $\gamma_0, \dots, \gamma_n$ be elements of $\mathcal{A}$ such that $\gamma_{i-1} \neq \gamma_i$ for each $i \in \{1, \dots, n\}$ and $d_{\gamma_i}(\gamma_{i-1}, \gamma_{i+1}) \geq 10K + 350\delta$ for each $i \in \{1, \dots, n-1\}$. Then $d_{\gamma_j}(\gamma_i, \gamma_{j-1}) \leq 2K + 70\delta$ for each $0 \leq i < j \leq n$. In particular, γ_0 and γ_n are distinct.*

We now record another consequence of the WPD property. This intuitively tells us that the axis γ of a WPD element does not behave like a factor of a metric product. That is, there exists a uniform threshold D_0 such that, if an object is too far from γ compared to its size, then the object has D_0 -small projection on γ . Here, it does not matter if the diameter of the object is larger than D_0 .

Lemma 3.17 ([Sis16, Lemma 3.3], [Cho25, Lemma 3.2]). *Let G be a non-virtually cyclic group with a finite generating set $S \subseteq G$. Suppose that G acts on a Gromov hyperbolic space $X \ni x_0$ with a WPD loxodromic element $f \in G$. Then there exists $D_0 > 0$, and for each $k, M > 0$ there exists $R = R(k, M) > 0$, such that the following holds.*

Let $g, h \in G$ be such that $\|g\|_S \geq R$ and $\|h\|_S \leq M$. Then $\pi_{[x_0, f^k x_0]}(\{gx_0, ghx_0\})$ has diameter at most D_0 .

FIGURE 7. Definition of anti-halfspace $\mathfrak{A}_D(x, y)$.

4. HYPERBOLIC MAGIC LEMMA

In this section, we state and prove the so-called hyperbolic magic lemma [Hut19, Proposition 4.1]. Hutchcroft proved it for closed convex sets of $\mathbb{H}^d$, and our aim is to extend it to arbitrary Gromov hyperbolic spaces. Readers are invited to play with the animation¹.

Given $x, y \in X$ and $D > 0$, we define

$$\mathcal{H}_D(x, y) := \{z \in X : (z|y)_x \geq D\}.$$

Sets of this sort are called *halfspaces* rooted at x with radius parameter D .

For technical reasons, we introduce a variant of the notion of halfspaces. Given $x, y \in X$ and $D > 0$, let us define the *anti-halfspace*

$$\mathfrak{A}_D(x, y) := \left\{ z \in X \setminus \mathcal{N}_{6D}(x) : \begin{array}{l} \exists \text{ a geodesic } \gamma \text{ passing through} \\ y, \mathcal{N}_{D+100\delta}(x), \mathcal{N}_{D+200\delta}(z) \text{ in order} \end{array} \right\}.$$

This is almost, but not exactly, the complement of $\mathcal{H}_D(x, y)$. See Figure 7.

We record two elementary observations, whose proofs are omitted.

Observation 4.1. *Let $D \geq 1000\delta$. Then for every $x, y \in X$ we have $x \notin \mathfrak{A}_D(x, y)$. If moreover $d_X(x, y) \geq 100D$, then $y \notin \mathfrak{A}_D(x, y)$.*

Observation 4.2. *Let $D \geq 1000\delta$, let $x, y \in X$ and let $z \in \mathfrak{A}_D(x, y)$. Then $d_X(y, z) \geq d_X(x, y) + D$.*

Morally, $z \notin \mathcal{H}_D(x, y)$ or $z \in \mathfrak{A}_D(x, y)$ means that the sequence (y, x, z) is “aligned” along a geodesic in X . The linear order on geodesics motivates the following statement for points $a, a', b, b', c \in X$:

If (a, b, c) and (a, b', c) are aligned, then either (a, b, b') is aligned or (a, b', b) is aligned. If the former is the case and (a', b, c) is also aligned, then (a', b, b') is also aligned.

This statement does not hold as stated when we measure the alignment with respect to the Gromov products. This is why we introduce anti-halfspaces. Let us now formally prove the statement.

¹<https://inhyeokchoi48.github.io/research/binary>

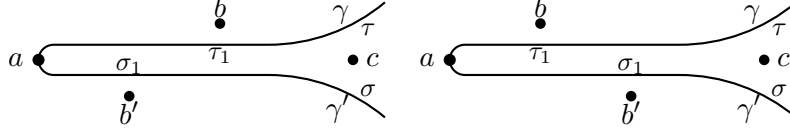


FIGURE 8. Two situations in the proof of Lemma 4.3. Left: $\tau_1 - \sigma_1 \geq 97D$. Right: $\sigma_1 - \tau_1 \geq 97D$.

Lemma 4.3. *Let X be δ -hyperbolic, let $D > 1000\delta$ and let $a, a', b, b', c \in X$ be such that $d_X(b, b'), d_X(b, c), d_X(b', c) \geq 100D$. Suppose that $c \in \mathfrak{A}_D(b, a') \cap \mathfrak{A}_D(b, a) \cap \mathfrak{A}_D(b', a)$. Then either $b \in \mathfrak{A}_D(b', a)$ or $b' \in \mathfrak{A}_D(b, a) \cap \mathfrak{A}_D(b, a')$.*

Proof. Let $\gamma(\cdot)$ be a geodesic starting at a and let $0 \leq \tau_1 \leq \tau$ be such that

$$d_X(\gamma(\tau_1), b) < D + 100\delta, \quad d_X(\gamma(\tau), c) < D + 200\delta.$$

Let $\gamma'(\cdot)$ be a geodesic starting at a and let $0 \leq \sigma_1 \leq \sigma$ be such that

$$d_X(\gamma'(\sigma_1), b') < D + 100\delta, \quad d_X(\gamma'(\sigma), c) < D + 200\delta.$$

See Figure 8 for schematics. Let $L := (\gamma(\tau)|\gamma'(\sigma))_a$. Lemma 3.6 tells us that $d_X(\gamma(t), \gamma'(t)) < 4\delta$ for $0 \leq t \leq L$. Note that

$$\tau - L = (a|\gamma'(\sigma))_{\gamma(\tau)} \leq d_X(\gamma(\tau), \gamma'(\sigma)) \leq 2D + 400\delta.$$

From this argument (and a symmetric one), we conclude $L \geq \max(\tau, \sigma) - 3D$.

Recall that $d_X(b', c) \geq 100D$ and $\sigma - \sigma_1 \geq 97D$. Hence, we have $\sigma_1 \leq \sigma - 97D \leq L - 94D \leq \tau - 94D$. We now have

$$(4.1) \quad d_X(\gamma(\sigma_1), b') \leq d_X(\gamma(\sigma_1), \gamma'(\sigma_1)) + d_X(\gamma'(\sigma_1), b') \leq D + 120\delta.$$

This implies that

$$\begin{aligned} 100D &\leq d_X(b, b') \leq d_X(b, \gamma(\tau_1)) + d_X(\gamma(\tau_1), \gamma(\sigma_1)) + d_X(\gamma(\sigma_1), b') \\ &\leq (D + 100\delta) + |\tau_1 - \sigma_1| + (D + 120\delta), \end{aligned}$$

i.e., $\tau_1 \leq \sigma_1 - 97D$ or $\tau_1 \geq \sigma_1 + 97D$. In the former case, the facts

$$d_X(\gamma(\tau_1), b) < D + 100\delta, \quad d_X(\gamma(\sigma_1), b') \leq D + 200\delta \quad (\text{by Inequality 4.1})$$

imply that $b' \in \mathfrak{A}_D(b, a)$. Similarly, $b \in \mathfrak{A}_D(b', a)$ holds in the latter case.

Let us now restrict to the former case, i.e., $\tau_1 \leq \sigma_1 - 97D$ and $b' \in \mathfrak{A}_D(b, a)$. Consider a geodesic η starting at a' and let $0 \leq \tau'_1 \leq \tau'$ be such that

$$d_X(\eta(\tau'_1), b) < D + 100\delta, \quad d_X(\eta(\tau'), c) < D + 200\delta.$$

Consider the geodesic triangle with vertices $\gamma(\tau_1), \gamma(\tau)$ and $\eta(\tau'_1)$. Since the triangle is δ -slim, $\gamma(\sigma_1) \in \gamma|_{[\tau_1, \tau]}$ is δ -close to either $[\gamma(\tau_1), \eta(\tau'_1)]$ or $[\eta(\tau'_1), \gamma(\tau)]$. Between them, the former lies in $\mathcal{N}_{3D}(\gamma(\tau_1))$, whereas $\gamma(\sigma_1)$ is $97D$ -far from $\gamma(\tau_1)$. Hence, $\gamma(\sigma_1)$ is δ -close to a point $p \in [\eta(\tau'_1), \gamma(\tau)]$.

Next, consider the geodesic triangle with vertices $\eta(\tau'_1), \eta(\tau')$ and $\gamma(\tau)$. This time, p is δ -close to either $[\eta(\tau'_1), \eta(\tau')]$ or $[\eta(\tau'), \gamma(\tau)]$. The latter one is contained in $\mathcal{N}_{3D}(\gamma(\tau))$. Meanwhile, $\gamma(\sigma_1)$ is $90D$ -far from $\gamma(\tau)$, so p is

$89D$ -far from $\gamma(\tau)$. Hence, p cannot be δ -close to $[\eta(\tau'), \gamma(\tau)]$, and is rather δ -close to $[\eta(\tau'_1), \eta(\tau')]$.

Summing up, $\gamma(\sigma_1)$ is 2δ -close to some point $q \in \eta|_{[\tau'_1, \tau']}$. This q is $(D + 122\delta)$ -close to b' . It follows that $b' \in \mathfrak{A}_D(b, a')$. $\square$

A subset $Y \subseteq X$ of a metric space is *uniformly locally finite* if

$$\sup_{y \in Y} \#(\mathcal{N}_R(y) \cap Y) < +\infty \quad (\forall R > 0).$$

The vertex set of a Cayley graph of a finitely generated group is uniformly locally finite. More generally, if a group G acts properly on a metric space $X \ni x_0$, then the G -orbit $G \cdot x_0$ is uniformly locally finite.

Proposition 4.4. *Let X be a δ -hyperbolic space and let Y be a uniformly locally finite subset of X . Then for each $\epsilon > 0$ and $D > 1000\delta + 1$ there exists a constant $N = N(\epsilon, D, Y)$ such that for every finite set $A \subseteq Y$ there exists a subset $A' \subseteq A$ satisfying:*

- (1) $\#A' \geq (1 - \epsilon)\#A$;
- (2) For each $a \in A'$ there exist halfspaces $\mathcal{H}_1, \mathcal{H}_2 \subseteq X$ rooted at a with radius parameter D such that $\#(A \setminus (\mathcal{H}_1 \cup \mathcal{H}_2)) \leq N$.

Proof. Because Y is uniformly locally finite, we have

$$\sup_{y \in Y} \#(\mathcal{N}_{100D}(y) \cap Y) =: M < +\infty.$$

We then define $N := 2M/\epsilon + M$. Note that N depends on ϵ, D, Y but not on the choice of $A \subseteq Y$.

Let $A \subseteq Y$ be a finite set, and let $\mathcal{A}_1$ be the collection of unwanted points:

$$\mathcal{A}_1 := \left\{ a \in A : \begin{array}{l} \#(A \setminus (\mathcal{H}_1 \cup \mathcal{H}_2)) > N \text{ for all pairs of halfspaces} \\ \mathcal{H}_1, \mathcal{H}_2 \text{ rooted at } a \text{ with radius parameter } D \end{array} \right\}.$$

Our goal is to prove that $\#\mathcal{A}_1 \leq \epsilon \cdot \#A$. We now pick a maximally $100D$ -separated subset $\mathcal{A}_2$ of $\mathcal{A}_1$, i.e., we have

- (1) $d_X(a, a') \geq 100D$ for each pair of distinct elements $a, a' \in \mathcal{A}_2$;
- (2) $\mathcal{A}_2$ is a maximal subset of $\mathcal{A}_1$ satisfying this property.

Then $\bigcup_{a \in \mathcal{A}_2} (\mathcal{N}_{100D}(a) \cap Y)$ covers the entire set $\mathcal{A}_1$ (if not, a missed element can be added to $\mathcal{A}_2$, breaking the maximality). Since $\mathcal{N}_{100D}(a) \cap Y$ has at most M elements for each $a \in \mathcal{A}_2$, we have $\#\mathcal{A}_2 \geq \frac{1}{M} \cdot \#\mathcal{A}_1$.

The proof will be complete once we show that $\#\mathcal{A}_2 \leq \frac{\epsilon}{M} \#A$. It can be established, for example, once we create disjoint complements of halfspaces rooted at each element of $\mathcal{A}_2$. However, the complements of halfspaces rooted at distinct elements might intersect. Our next goal is to extract some portion of $\mathcal{A}_2$ for which we can create disjoint complements of halfspaces.

Let us first prepare empty collections $\mathcal{B} = \mathcal{U} = \mathcal{G} = \emptyset$. They are meant to be collections of bad, undecided and good elements. Fix a basepoint $x_0 \in X$. Enumerate $\mathcal{A}_2$ by the distance from x_0 , i.e., let $\mathcal{A}_2 = \{a_1, a_2, \dots, a_{\#\mathcal{A}_2}\}$ be such that $d_X(x_0, a_i) \leq d_X(x_0, a_{i+1})$ for each i . At each step $i = 1, \dots, \#\mathcal{A}_2$,

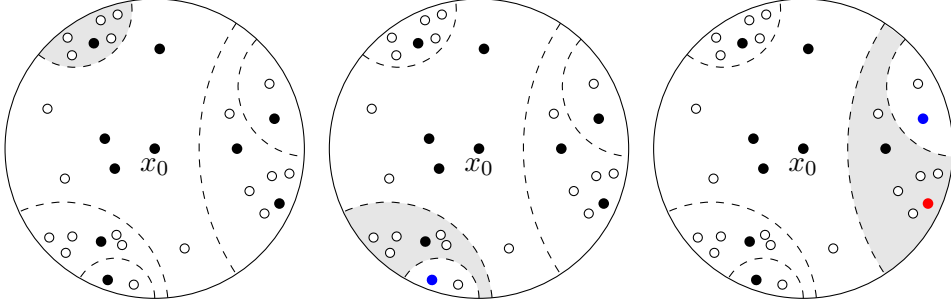


FIGURE 9. Three possible scenarios for the procedure in the proof of Proposition 4.4. The set A is the collection of all the dots, whereas $\mathcal{A}_2$ is the collection of colored dots. (Left) The anti-halfspace $\mathfrak{A}_i$ contains no colored dot. (Middle) The region $\mathfrak{A}_i \cap \mathfrak{A}'_i$ contains no colored dot. (Right) The region $\mathfrak{A}_i \cap \mathfrak{A}'_i$ contains a colored dot. In this case, the blue dot becomes b_i and the red dot becomes c_i .

we will define a point $b_i \in X$ and put a_i in either $\mathcal{B}$ or $\mathcal{G}$; this decision is final and shall not be modified further. We may put some other elements of $\mathcal{A}_2$ in $\mathcal{U}$, whose classification will change later. We will keep the balance $\#\mathcal{B} \leq \#\mathcal{U} + \#\mathcal{G}$ throughout. After the last step there will be no element of $\mathcal{U}$ left, so we will have $\#\mathcal{B} \leq \#\mathcal{G}$.

We now describe the procedure. At step i , we first declare $\mathfrak{A}_i := \mathfrak{A}_D(a_i, x_0)$.

- (1) If $\mathcal{A}_2 \cap \mathfrak{A}_i$ has no element, then we declare that $a_i \in \mathcal{G}$ and $b_i := x_0$.
- (2) If not, pick $b_i \in \mathcal{A}_2 \cap \mathfrak{A}_i$ closest to x_0 and declare $\mathfrak{A}'_i := \mathfrak{A}_D(a_i, b_i)$.
 - (a) If $\mathcal{A}_2 \cap \mathfrak{A}_i \cap \mathfrak{A}'_i$ has no element, then we declare that $a_i \in \mathcal{G}$.
 - (b) If not, we pick $c_i \in \mathcal{A}_2 \cap \mathfrak{A}_i \cap \mathfrak{A}'_i$ that is the closest to x_0 . We then declare $a_i \in \mathcal{B}$ and $b_i, c_i \in \mathcal{U}$. See Figure 9.

(If an element in $\mathcal{U}$ is declared good or bad, we remove it from $\mathcal{U}$.)

Till step i , $\mathcal{G} \cup \mathcal{B}$ consists of elements from $\{a_1, \dots, a_i\}$; it does not contain any of $a_{i+1}, a_{i+2}, \dots$. Let us observe what happens at step i .

In Case (1), $\mathcal{G}$ gains one more element that may or may not come from $\mathcal{U}$. $\mathcal{B}$ does not change. Overall, $\#\mathcal{B}$ stays the same and $\#\mathcal{U} + \#\mathcal{G}$ does not decrease. A similar situation occurs in Case (2-a).

In Case (2-b), $\mathcal{B}$ gains one element a_i , which might be from $\mathcal{U}$. In exchange, $\mathcal{U}$ gains elements b_i and c_i . Here, Observation 4.2 guarantees that

$$d_X(x_0, b_i), d_X(x_0, c_i) \geq d_X(x_0, a_i) + D.$$

Since $\mathcal{A}_2$ was labelled with respect to the distance from x_0 , we conclude that $b_i, c_i \in \{a_{i+1}, a_{i+2}, \dots\}$; neither b_i nor c_i come from $\mathcal{G} \cup \mathcal{B}$. We thus confirm that elements are never re-classified once they are put in $\mathcal{G} \cup \mathcal{B}$.

Furthermore, note that $b_i \in \mathfrak{A}_D(a_i, x_0) \cap \mathcal{A}_2$ and $c_i \in \mathfrak{A}_D(a_i, b_i) \cap \mathcal{A}_2$. By Observation 4.1, the former membership implies that $b_i \neq a_i$ and $d_X(b_i, a_i) \geq 100D$ (as $\mathcal{A}_2$ is $100D$ -separated), and the latter membership implies that

$c_i \notin \{a_i, b_i\}$. In particular, b_i, c_i are distinct. If b_i, c_i are not from $\mathcal{U}$ at step $i - 1$ and are genuinely new additions to $\mathcal{U}$, then we can conclude that $\#\mathcal{U}$ increases by at least 1 in Case (2-b). It remains to show

Claim 4.5. *For $i < j$ such that $a_i, a_j \in \mathcal{B}$, we have $\{b_i, c_i\} \cap \{b_j, c_j\} = \emptyset$.*

Proof of Claim 4.5. Suppose to the contrary that $b_i \in \{b_j, c_j\}$. By the construction of b_i, b_j and c_j , we have

$$b_i \in \mathfrak{A}_D(a_i, x_0) \cap \mathfrak{A}_D(a_j, x_0).$$

Note that $d_X(b_i, a_i), d_X(b_i, a_j) \geq 100D$. By applying Lemma 4.3 for $a = a' := x_0, b := a_i, b' := a_j$ and $c := b_i$, we conclude that either $a_i \in \mathfrak{A}_D(a_j, x_0)$ or $a_j \in \mathfrak{A}_D(a_i, x_0)$. The former implies that $d_X(a_i, x_0) > d_X(a_j, x_0)$ (Observation 4.2), contradicting the labelling scheme of $\{a_1, a_2, \dots\}$. The latter case contradicts the minimality of b_i in $\mathfrak{A}_D(a_i, x_0) \cap \mathcal{A}_2$, as $d_X(b_i, x_0) > d_X(a_j, x_0)$ (Observation 4.2). Hence, $b_i \in \{b_j, c_j\}$ is impossible.

Next, suppose to the contrary that $c_i \in \{b_j, c_j\}$. This implies that

$$c_i \in \mathfrak{A}_D(a_i, b_i) \cap \mathfrak{A}_D(a_i, x_0) \cap \mathfrak{A}_D(a_j, x_0).$$

Lemma 4.3 implies that either $a_i \in \mathfrak{A}_D(a_j, x_0)$ or $a_j \in \mathfrak{A}_D(a_i, b_i) \cap \mathfrak{A}_D(a_i, x_0)$. Again, the former contradicts the labelling scheme of $\mathcal{A}_2$, and the latter contradicts the minimality of c_i in $\mathfrak{A}_D(a_i, b_i) \cap \mathfrak{A}_D(a_i, x_0)$. Hence, $c_i \in \{b_j, c_j\}$ is also impossible, and this ends the proof. $\square$

Thanks to the claim, we conclude that $\#\mathcal{B} \leq \#\mathcal{U} + \#\mathcal{G}$ at each step. Moreover, after the last step, there is no element of $\mathcal{U}$ left. Hence $\#\mathcal{B} \leq \#\mathcal{G}$, and $\mathcal{G}$ takes up at least half of $\mathcal{A}_2$.

Now, with the final $\mathcal{G}$ in hand, for each $a_i \in \mathcal{G}$ we define

$$K_i := A \setminus (\mathcal{H}_D(a_i, x_0) \cup \mathcal{H}_D(a_i, b_i)).$$

Since $a_i \in \mathcal{G} \subseteq \mathcal{A}_2$, we have $\#K_i \geq N$ and $\#(K_i \setminus \mathcal{N}_{100D}(a_i)) \geq N - M$.

Claim 4.6. *For every pair of distinct elements $a_i, a_j \in \mathcal{G}$, $(K_i \setminus \mathcal{N}_{100D}(a_i))$ and $(K_j \setminus \mathcal{N}_{100D}(a_j))$ do not intersect.*

Proof of Claim 4.6. Suppose to the contrary that $K_i \setminus \mathcal{N}_{100D}(a_i)$ and $K_j \setminus \mathcal{N}_{100D}(a_j)$ have a common element $z \in A$ for some $i < j$ such that $a_i, a_j \in \mathcal{G}$. By Lemma 3.6, we deduce that $[x_0, z]$ and $[b_i, z]$ pass through the $(D + 4\delta)$ -neighborhood of a_i . We also know that $z \notin \mathcal{N}_{6D}(a_i)$. It follows that $z \in \mathfrak{A}_D(a_i, x_0) \cap \mathfrak{A}_D(a_i, b_i)$. Similarly, we have $z \in \mathfrak{A}_D(a_j, x_0) \cap \mathfrak{A}_D(a_j, b_j)$.

Now Lemma 4.3 implies that either $a_i \in \mathfrak{A}_D(a_j, x_0)$ or $a_j \in \mathfrak{A}_D(a_i, x_0) \cap \mathfrak{A}_D(a_i, b_i)$ holds. The former contradicts the labelling scheme of $\mathcal{A}_2$ (given Observation 4.2) and the latter contradicts the membership $a_i \in \mathcal{G}$. $\square$

With Claim 4.6 in hand, we have

$$\#A \geq \sum_{i: a_i \in \mathcal{G}} \#(K_i \setminus \mathcal{N}_{100D}(a_i)) \geq (N - M) \cdot \#\mathcal{G} \geq \frac{2M}{\epsilon} \cdot \frac{\#\mathcal{A}_2}{2} \geq \frac{1}{\epsilon} \#A_1.$$

This ends the proof of Proposition 4.4. $\square$

Now suppose that a group G is acting properly on $X \ni x_0$. Then the stabilizer of x_0 is finite, and the G -orbit of x_0 is uniformly locally finite. By Proposition 4.4, we conclude that:

Proposition 4.7. *Let X be a δ -hyperbolic space with a basepoint x_0 and let G be a group acting properly on X . Then for each $\epsilon, D > 0$ there exists a constant $N = N(\epsilon, D, G, X)$ such that for every finite set $A \subseteq G$ there exists a subset $A' \subseteq A$ satisfying:*

- (1) $\#A' \geq (1 - \epsilon)\#A$;
- (2) *For each $a \in A'$ there exist halfspaces $\mathcal{H}_1, \mathcal{H}_2 \subseteq X$ rooted at ax_0 with radius parameter D such that $\#\{g \in A : gx_0 \notin (\mathcal{H}_1 \cup \mathcal{H}_2)\} \leq N$.*

5. SUPPORTING HYPERPLANE LEMMA

Throughout this section, we fix a δ -hyperbolic space X with a basepoint x_0 and a group G acting on X . For proper and non-elementary actions, the hyperbolic magic lemma readily implies the supporting hyperplane lemma. Still, we will prove it for general WPD actions. Given $x, y \in X$, let

$$\begin{aligned}\mathcal{H}_{half}(x, y) &:= \{z \in X : z \text{ is closer to } x \text{ than to } y\}, \\ \mathcal{H}_G(a, b) &:= \{g \in G : gx_0 \in \mathcal{H}_{half}(ax_0, bx_0)\}.\end{aligned}$$

We say that $(f, l) \in G^2$ is a D -Schottky pair if the closed halfspaces

$$\overline{\mathcal{H}_{half}(fx_0, x_0)}, \overline{\mathcal{H}_{half}(f^{-1}x_0, x_0)}, \overline{\mathcal{H}_{half}(lx_0, x_0)}, \overline{\mathcal{H}_{half}(l^{-1}x_0, x_0)}$$

are pairwise disjoint, and every geodesic in X from one of these halfspaces to another one meets $\mathcal{N}_D(x_0)$. When the parameter D is not important, we will drop it and say that (f, l) is *Schottky*. For every pair of independent loxodromic isometries a and b , there exists $D > 0$ such that (a^m, b^n) is D -Schottky for every $m, n \geq D$ (cf. [Del91]). Note that:

Observation 5.1. *Let (f, l) be a D -Schottky pair. Then (f^k, l^m) is also a D -Schottky pair for every $k, m \neq 0$.*

Proof. We will prove the following by induction on k :

$$\mathcal{H}_{half}(a^{k+1}x_0, x_0) \subseteq \mathcal{H}_{half}(a^kx_0, x_0). \quad (\forall k > 0, \forall a \in \{f, f^{-1}, l, l^{-1}\})$$

Suppose to the contrary that there exists $a \in \{f, f^{-1}, l, l^{-1}\}$ and $y \in \mathcal{H}_{half}(a^{k+1}x_0, x_0) \setminus \mathcal{H}_{half}(a^kx_0, x_0)$. Then we have

$$d_X(y, a^{k+1}x_0) < d_X(y, x_0) \leq d_X(y, a^kx_0)$$

so $a^{-k}y \in \mathcal{H}_{half}(ax_0, x_0)$. Meanwhile, $a^{-k}y \in \overline{\mathcal{H}_{half}(a^{-k}x_0, x_0)}$ is contained in $\overline{\mathcal{H}_{half}(a^{-1}x_0, x_0)}$ by induction, which is disjoint from $\mathcal{H}_{half}(ax_0, x_0)$. This is a contradiction. $\square$

Observation 5.2. *Let (f, l) be a Schottky pair. Then for $a_1 \in \{f, f^{-1}, l, l^{-1}\}$, $\{a_1 \cdots a_{n-1} \overline{\mathcal{H}_{half}(a_nx_0, x_0)} : a_i \in \{f, f^{-1}, l, l^{-1}\}, a_{i-1} \neq a_i^{-1} \forall i \in \{2, \dots, n\}\}$ are disjoint subsets of $\mathcal{H}_{half}(a_1x_0, x_0)$ for each $n > 0$.*

Recall the definition of decoupled subsets (Definition 2.9).

Lemma 5.3. *Let X be a δ -hyperbolic space with a basepoint x_0 and let $G \leq \text{Isom}(X)$. Let $(f, l) \in G^2$ be a Schottky pair and let S be a finite generating set of G . Then the following holds for all sufficiently large k, m .*

Let $a_1, \dots, a_n \in \{f^k, f^{-k}, l^m, l^{-m}\}$ be such that $a_i \neq a_{i+1}^{-1}$ for each i . Let $s_i := a_1 \cdots a_i$ for each i . Then for each i , the following are decoupled:

$$(\mathcal{H}_G(s_{i-1}, s_i), id), (\mathcal{H}_G(s_{i+2}, s_{i+1}), s_n).$$

Proof. Let (f, l) be a D -Schottky pair; then (f^k, l^m) is also D -Schottky for $k, m > 0$. For $u \in \{f, f^{-1}, l, l^{-1}\}$ we fix a d_S -path P_u from id to u , and let

$$M := \max_{u \in \{f, f^{-1}, l, l^{-1}\}} \text{diam}_X(P_u \cdot x_0).$$

Let $K = \max_{s \in S} d_X(x_0, sx_0)$. We then claim the desired statement for every k, m such that $d_X(x_0, f^k x_0), d_X(x_0, l^m x_0) > 100(D + \delta + M + K)$.

Fixing such k and m , let us pick $a_1, \dots, a_n \in \{f^k, f^{-k}, l^m, l^{-m}\}$ such that $a_i \neq a_{i+1}^{-1}$ for each $1 \leq i < n$, and let $s_i := a_1 \cdots a_i$, for each $0 \leq i \leq n$. Let us now pick $g \in \mathcal{H}_G(s_{i-1}, s_i)$ and $g' \in \mathcal{H}_G(s_{i+2}, s_{i+1})$. Then we have

$$gx_0 \in s_i \mathcal{H}_{half}(a_i^{-1} x_0, x_0), \quad g'x_0 \in s_i a_{i+1} \mathcal{H}_{half}(a_{i+2} x_0, x_0) \subseteq s_i a_{i+1} \mathcal{H}_{half}(x_0, a_{i+1}^{-1} x_0).$$

By the D -Schottky property, $[gx_0, g'x_0]$ passes through $\mathcal{N}_D(s_i x_0)$. For a similar reason, it also passes through $\mathcal{N}_D(s_{i+1} x_0)$. This implies that

$$d_X(gx_0, g'x_0) \geq d_X(s_i x_0, s_{i+1} x_0) - 2D \geq 3K + \delta.$$

This implies that $d_S(g, g') \geq 3$ as desired.

For the second property for decoupling, let us define d_S -paths

$$P_{f^{\pm k}} := P_{f^{\pm 1}} \circ f^{\pm 1} P_{f^{\pm 1}} \circ \dots \circ f^{\pm(k-1)} P_{f^{\pm 1}},$$

$$P_{l^{\pm m}} := P_{l^{\pm 1}} \circ l^{\pm 1} P_{l^{\pm 1}} \circ \dots \circ l^{\pm(m-1)} P_{l^{\pm 1}}.$$

We claim that $P_{a_1} \circ s_1 P_{a_2} \circ \dots \circ s_{n-1} P_{a_n}$ satisfies the property. To show this, let $t > i$ such that $a_t = f^k$. Then $s_{t-1} P_{a_t} \cdot x_0$ is contained in the M -neighborhood of $\{s_{t-1} f^j x_0\}_{j \geq 0}$. Next, note that for each $j > 0$, we have

$$\begin{aligned} s_{t-1} f^j x_0 &\in \mathcal{H}_{half}(s_{t-1} f^j x_0, s_{t-1} f^{j-1} x_0) \subseteq \dots \subseteq \mathcal{H}_{half}(s_{t-1} f x_0, s_{t-1} x_0) \\ &\subseteq \mathcal{H}_{half}(s_{t-1} x_0, s_{t-2} x_0) \subseteq \dots \subseteq \mathcal{H}_{half}(s_{i+1} x_0, s_i x_0). \end{aligned}$$

By the D -Schottky property, $[s_{t-1} f^j x_0, s_{i-1} x_0]$ intersects $\mathcal{N}_D(s_i x_0)$, and

$$\begin{aligned} d_X(s_{t-1} f^j x_0, s_{i-1} x_0) &\geq d_X(s_{t-1} f^j x_0, s_i x_0) + d_X(s_i x_0, s_{i-1} x_0) - 2D \\ &\geq d_X(s_{t-1} f^j x_0, s_i x_0) + 98D + 100M. \quad (\forall j > 0) \end{aligned}$$

This implies $d_X(z, s_{i-1} x_0) > d_X(z, s_i x_0)$ for each $z \in s_{t-1} P_{a_t} \cdot x_0$. We can similarly handle the case $a_t \in \{f^{-k}, l^m, l^{-m}\}$ and conclude

$$s_{t-1} P_{a_t} \subseteq G \setminus \mathcal{H}_G(s_{i-1}, s_i). \quad (\forall t > i)$$

Similarly, we observe that

$$s_{t-1} P_{a_t} \subseteq G \setminus \mathcal{H}_G(s_{i+2}, s_{i+1}). \quad (\forall t \leq i)$$

Lastly, we have $id \in \mathcal{H}_G(s_0, s_1) \subseteq \mathcal{H}_G(s_1, s_2) \subseteq \dots \subseteq \mathcal{H}_G(s_{n-1}, s_n)$, and $s_n \in \mathcal{H}_G(s_n, s_{n-1}) \subseteq \dots \subseteq \mathcal{H}_G(s_1, s_0)$. This concludes the decoupling. $\square$

Given a Schottky pair $(f, l) \in G^2$, we can pick for each $g \in G$ an element $l_g \in \{l, l^{-1}\}$ such that $g^{-1}x_0 \notin \mathcal{H}_{half}(l_g x_0, x_0)$.

Example 5.4. Consider the free group $F_2 \simeq \langle a, b \rangle$ acting on its standard Cayley graph, and let $x_0 := id$. Then (a^k, b^l) is a 0-Schottky pair for any $k, l > 0$. For a 0-Schottky pair (a^{10}, b^{10}) , we can choose either element of $\{b^{10}, b^{-10}\}$ as l_a . However, b^{-10} cannot be chosen as $l_{a^2 b^{10}}$; it must be b^{10} .

Given $g \in G$ and $D > 0$, we define the following notion of anti-halfspace:

$$\mathfrak{A}_{D,f,l}(g) := \left\{ z \in X : \begin{array}{l} \exists \text{ geodesic } \gamma \text{ in } X \text{ that passes through} \\ x_0, \mathcal{N}_D(gl_g x_0), \mathcal{N}_D(gl_g f x_0), \mathcal{N}_{6D}(z) \text{ in order} \end{array} \right\}.$$

An immediate observation is:

Observation 5.5. Let $D > 100(d_X(x_0, lx_0) + \delta)$ and let (f, l) be a $0.1D$ -Schottky pair in G with $d_X(x_0, fx_0) \geq 20D$. Then we have

$$gl_g \mathcal{H}_{half}(f^2 x_0, fx_0) \subseteq \mathfrak{A}_{D,f,l}(g) \subseteq gl_g \mathcal{H}_{half}(fx_0, x_0) \quad \forall g \in G.$$

Moreover, $d_X(x_0, z) > d_X(x_0, gx_0) + D$ for all $z \in \mathfrak{A}_{D,f,l}(g)$.

Proof. Suppose that $y \in gl_g \mathcal{H}_{half}(f^2 x_0, fx_0)$. Then $[x_0, y]$ is a geodesic connecting $y \in gl_g \mathcal{H}_{half}(fx_0, x_0)$ to $x_0 \in gl_g \mathcal{H}_{half}(l_g^{-1} x_0, x_0)$. Since (f, l) is $0.1D$ -Schottky, we can find a point P in $[x_0, y] \cap \mathcal{N}_{0.1D}(gl_g x_0)$.

Similarly, there exists a point $Q \in [gl_g x_0, y] \cap \mathcal{N}_{0.1D}(gl_g f x_0)$. Since $[P, y]$ and $[gl_g x_0, y]$ are $(\delta + 0.1D)$ -equivalent, we can also find a point $Q' \in [P, y] \cap \mathcal{N}_{0.2D}(gl_g f x_0)$. We conclude that $y \in \mathfrak{A}_{D,f,l}(g)$.

Now, suppose that $z \in \mathfrak{A}_{D,f,l}(g)$, with a geodesic $\gamma(\cdot)$ starting from x_0 and times $0 \leq \tau_1 \leq \tau_2 \leq \tau$ such that

$$d_X(\gamma(\tau_1), gl_g x_0) < D, \quad d_X(\gamma(\tau_2), gl_g f x_0) < D, \quad d_X(\gamma(\tau), z) < 6D.$$

Then $d_X(\gamma(\tau), \gamma(\tau_2))$ is smaller than $d_X(\gamma(\tau), \gamma(\tau_1))$ by at least

$$\tau_2 - \tau_1 \geq d_X(gl_g x_0, gl_g f x_0) - 2D \geq 18D.$$

This forces that

$$d_X(\gamma(\tau), gl_g x_0) - d_X(\gamma(\tau), gl_g f x_0) \geq 18D - 2D = 16D.$$

As $d_X(\gamma(\tau), \cdot)$ and $d_X(z, \cdot)$ differ by at most $6D$, we get $z \in gl_g \mathcal{H}_{half}(fx_0, x_0)$. The ‘‘Moreover’’ part also follows immediately. $\square$

We now present an analogue of Lemma 4.3.

Lemma 5.6. Let X be a δ -hyperbolic space with basepoint x_0 and let $G \leq \text{Isom}(X)$. Let $(f, l) \in G^2$ be a Schottky pair, where f is a unital, axial WPD element whose axis $Ax(f)$ contains x_0 . Let S be a finite generating set of G . Then for each large enough D , there exists M such that the following holds.

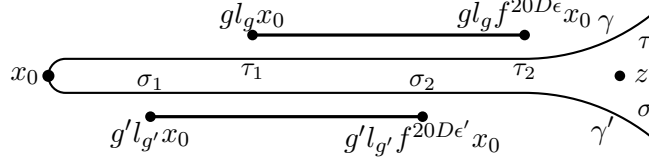


FIGURE 10. Case 2 in the proof of Lemma 5.6.

Let $g, g' \in G$ and let $\epsilon, \epsilon' \in \{+1, -1\}$ be such that $d_S(g, g') \geq M$ and

$$\mathfrak{A}_{D, f^{20D\epsilon}, l}(g) \cap \mathfrak{A}_{D, f^{20D\epsilon'}, l}(g') \neq \emptyset.$$

Then either $g'x_0 \in \mathfrak{A}_{D, f^{20D\epsilon}, l}(g)$ or $gx_0 \in \mathfrak{A}_{D, f^{20D\epsilon'}, l}(g')$.

Proof. Let D_0 be as in Lemma 3.17 for f and let

$$K_0 := 100(D_0 + d_X(x_0, lx_0) + \delta + 1).$$

We will prove the lemma for each $D > K_0$. Let $R = R(20D, 20D\|f\|_S)$ be as in Lemma 3.17 for f , and let $M := R + 20D\|f\|_S + 2\|l\|_S$.

Let $z \in \mathfrak{A}_{D, f^{20D\epsilon}, l}(g) \cap \mathfrak{A}_{D, f^{20D\epsilon'}, l}(g')$. Let $\gamma(\cdot)$ be a geodesic in X that starts at x_0 and let $0 \leq \tau_1 \leq \tau_2 \leq \tau$ be such that

$$d_X(\gamma(\tau_1), gl_g x_0) < D, \quad d_X(\gamma(\tau_2), gl_g f^{20D\epsilon} x_0) < D, \quad d_X(\gamma(\tau), z) < 6D.$$

Let $\gamma'(\cdot)$ be a geodesic starting at x_0 and let $0 \leq \sigma_1 \leq \sigma_2 \leq \sigma$ be such that

$$d_X(\gamma'(\sigma_1), g'l_{g'} x_0) < D, \quad d_X(\gamma'(\sigma_2), g'l_{g'} f^{20D\epsilon'} x_0) < D, \quad d_X(\gamma'(\sigma), z) < 6D.$$

Under the assumption $\sigma_1 \leq \tau_1$, we now aim to prove $gx_0 \in \mathfrak{A}_{D, f^{20D\epsilon'}, l}(g')$.

Let $L := (\gamma(\tau)|\gamma'(\sigma))_{x_0}$. Lemma 3.6 tells us that $d_X(\gamma(t), \gamma'(t)) < 4\delta$ for $0 \leq t \leq L$. Note that

$$\tau - L = (x_0|\gamma'(\sigma))_{\gamma(\tau)} \leq d_X(\gamma(\tau), \gamma'(\sigma)) \leq d_X(\gamma(\tau), z) + d_X(z, \gamma'(\sigma)) \leq 12D.$$

We can similarly conclude $L \geq \sigma - 12D$. Also,

$$\tau_2 - \tau_1 \geq d_X(gl_g x_0, gl_g f^{20D\epsilon} x_0) - 2D \geq 18D,$$

and similarly $\sigma_2 - \sigma_1 \geq 18D$.

There are now two subcases (of the case $\sigma_1 \leq \tau_1$).

Case 1: $\tau_1 \geq \sigma_2 - 4.5D$. In this case, we also have

$$\tau \geq \tau_2 \geq \tau_1 + 18D \geq \sigma_2 + 13.5D.$$

This implies that $L \geq \tau - 12D \geq \tau_1$ and

$$\begin{aligned} d_X(gx_0, \gamma'(\tau_1)) &\leq d_X(x_0, l_g x_0) + d_X(gl_g x_0, \gamma(\tau_1)) + d_X(\gamma(\tau_1), \gamma'(\tau_1)) \\ &\leq d_X(x_0, l_g x_0) + D + 4\delta. \end{aligned}$$

If $\sigma_1 \leq \sigma_2 \leq \tau_1$, this inequality serves as a certificate for $gx_0 \in \mathfrak{A}_{D, f^{20D\epsilon'}, l}(g')$.

If $\tau_1 < \sigma_2$, then $|\sigma_2 - \tau_1| \leq 4.5D$ and we can replace τ_1 with σ_2 as certificate.

We still have $gx_0 \in \mathfrak{A}_{D, f^{20D\epsilon'}, l}(g')$.

Case 2: $\sigma_1 \leq \tau_1 \leq \sigma_2 - 4.5D$. In this case, let $T := \min(\sigma_2, \tau_2 - 12D)$. Note that $(\tau_2 - 12D) - \tau_1 \geq 6D$ and $\sigma_2 - \tau_1 \geq 4.5D$; hence, $T - \tau_1 \geq 4.5D$.

Also, $T \leq \tau - 12D \leq L$, so $\gamma(t)$ and $\gamma'(t)$ are 4δ -close for times $t \in [0, T]$ by Lemma 3.2. In particular, $\gamma(\tau_1)$ and $\gamma(T)$ are contained in the sets

$$\mathcal{N}_{4\delta}([\gamma'(\tau_1), \gamma'(T)]) \subseteq \mathcal{N}_{4\delta}([\gamma'(\sigma_1), \gamma'(\sigma_2)]) \subseteq \mathcal{N}_{D+6\delta}(g'l_{g'}[x_0, f^{20D\epsilon'}x_0]).$$

This implies that

$$d_{g'l_{g'}[x_0, f^{20D\epsilon'}x_0]}(\gamma(\tau_1), \gamma(T)) \geq d_X(\gamma(T), \gamma(\tau_1)) - 2(D + 6\delta) \geq 2.5D - 12\delta.$$

Now note that $[\gamma(\tau_1), \gamma(\tau_2)]$ and $[gl_gx_0, gl_gf^{20D\epsilon}x_0]$ are $(2\delta + D)$ -equivalent, and $gl_g \cdot \{x_0, f^\epsilon x_0, f^{2\epsilon}x_0, \dots\}$ is 1-dense in the latter geodesic. (Recall that f is unital and $x_0 \in Ax(f)$.) Hence, there exists $0 < k < 20D$ such that $\gamma(T)$ is $(2\delta + D + 1)$ -close to $gl_gf^{k\epsilon}x_0$. For this k , we have

$$d_{g'l_{g'}[x_0, f^{20D\epsilon'}x_0]}(gl_gx_0, gl_gf^{k\epsilon}x_0) \geq 2.5D - 12\delta - (12\delta + D) - (12\delta + 2\delta + D + 1) \geq 0.1D.$$

Here, note that $d_S(gl_g, gl_gf^{k\epsilon}) \leq k \cdot \|f\|_S \leq 20D\|f\|_S$. By the definition of R in terms of Lemma 3.17, we conclude that $d_S(g'l_{g'}, gl_g) \leq R$. This implies that $d_S(g', g) \leq R + \|l_{g'}\|_S + \|l_g\|_S \leq M$, contradicting the assumption.

In summary, Case 2 is impossible; we have $gx_0 \in \mathfrak{A}_{D, f^{20D\epsilon'}, l}(g')$ whenever $\sigma_1 \leq \tau_1$. We can handle the remaining case by a symmetric argument. $\square$

The restriction that f is unital and axial is in fact unnecessary. We omit the proof of this more general version.

5.1. Main statement. Before stating the supporting hyperplane lemma, let us first prove the following analogue of Proposition 4.4.

Proposition 5.7. *Let X be a δ -hyperbolic space with a basepoint x_0 and let G be a group acting on X containing a Schottky pair (f, l) . Moreover, suppose that f is an axial WPD element whose axis $Ax(f)$ contains x_0 . Let S be a finite generating set.*

Then there exists D_0 such that, for each $\epsilon > 0$ and $D \geq D_0$ there exists a constant $N = N(\epsilon, D)$ such that for every finite set $A \subseteq G$ there exists a subset $A' \subseteq A$ satisfying:

- (1) $\#A' \geq (1 - \epsilon)\#A$;
- (2) For each $a \in A'$ there exists $\sigma \in \{+1, -1\}$ such that
$$\#(A \cdot x_0 \setminus \mathcal{H}_{half}(al_a f^{20D\sigma}x_0, al_a f^{40D\sigma}x_0)) \leq N.$$

As the statement suggests, the proof of Proposition 5.7 will follow the same line as that of Proposition 4.4.

Proof. The proposition is not affected by scaling up the metric on X . Hence, let us assume further that f is unital.

Let D_0 be the threshold for D as given in Lemma 5.6. By increasing D_0 if necessary, we can guarantee that (f, l) is $0.1D_0$ -Schottky and that $D_0 > 100(d_X(x_0, lx_0) + \delta)$. Now let $D \geq D_0$ and $\epsilon > 0$. Let $R = M(D)$ be as in Lemma 5.6, and let $N = 2 \cdot (2\#S)^R/\epsilon$.

For this proof, we will abbreviate $\mathfrak{A}_{D,f\pm 20D,l}(\cdot)$ by $\mathfrak{A}_{\pm}(\cdot)$. We will define

$$\mathcal{A}_1 := \left\{ a \in A : \#(A \cdot x_0 \setminus al_a \cdot \mathcal{H}_{half}(f^{20D\sigma}x_0, f^{40D\sigma}x_0)) \geq N \text{ for both } \sigma \in \{1, -1\} \right\}.$$

Let $\mathcal{A}_2 \subseteq \mathcal{A}_1$ be a maximally R -separated subset in the word metric d_S :

- (1) $d_S(a, a') \geq R$ for each pair of distinct elements $a, a' \in \mathcal{A}_2$;
- (2) $\mathcal{A}_2$ is a maximal subset of $\mathcal{A}_1$ satisfying this property.

Then $\bigcup_{a \in \mathcal{A}_2} (a \cdot B_S(R) \cap A)$ covers the entire set $\mathcal{A}_1$. We conclude

$$\#\mathcal{A}_2 \geq \frac{1}{\#B_S(R)} \cdot \#\mathcal{A}_1 \geq \frac{1}{(2\#S)^R} \#\mathcal{A}_1.$$

As before, we prepare $\mathcal{B} = \mathcal{U} = \mathcal{G} = \emptyset$. Enumerate $\mathcal{A}_2 = \{a_1, a_2, \dots, a_{\#\mathcal{A}_2}\}$ by the distance from x_0 , so that $d_X(x_0, a_i x_0) \leq d_X(x_0, a_{i+1} x_0)$ for each i . At each step $i = 1, \dots, \#\mathcal{A}_2$, we will put a_i in either $\mathcal{B}$ or $\mathcal{G}$. When a_i is declared good, we will also define its sign $\sigma(a_i) \in \{+1, -1\}$. This way, we will obtain a function $\sigma : \mathcal{G} \rightarrow \{+1, -1\}$ in the end.

We now describe the procedure. At step i ,

- (1) if $\mathcal{A}_2 \cdot x_0 \cap \mathfrak{A}_+(a_i) = \emptyset$, then we declare $a_i \in \mathcal{G}$ and $\sigma(a_i) = +1$;
- (2) if not (1) and if $\mathcal{A}_2 \cdot x_0 \cap \mathfrak{A}_-(a_i) = \emptyset$, then we declare $a_i \in \mathcal{G}$ and $\sigma(a_i) = -1$;
- (3) if neither (1) nor (2) applies, we pick $b_i, c_i \in \mathcal{A}_2$ such that $b_i x_0 \in \mathfrak{A}_+(a_i)$ and $c_i x_0 \in \mathfrak{A}_-(a_i)$, making sure that they are the closest possible choices to x_0 . We declare $a_i \in \mathcal{B}$ and $b_i, c_i \in \mathcal{U}$.

Till step i , $\mathcal{G} \cup \mathcal{B}$ consists of elements from $\{a_1, \dots, a_i\}$; it does not contain any of $a_{i+1}, a_{i+2}, \dots$ (*) We now describe what happens at step i .

In Case (1) or (2), $\mathcal{G}$ gains one more element that might be from $\mathcal{U}$ or not. Hence, $\#\mathcal{U} + \#\mathcal{G}$ does not decrease, while $\mathcal{B}$ does not change.

In Case (3), $\mathcal{B}$ gains one element a_i , which might be from $\mathcal{U}$. In exchange, $\mathcal{U}$ gains elements b_i and c_i . Here Observation 5.5 guarantees that $d_X(x_0, b_i x_0), d_X(x_0, c_i x_0) > d_X(x_0, a_i x_0)$ and the Schottky property of (f, l) guarantees that b_i and c_i are distinct. Since $\mathcal{A}_2$ was labelled with respect to the distance from x_0 , we conclude that $b_i, c_i \in \{a_{i+1}, a_{i+2}, \dots\}$; in other words, neither b_i nor c_i come from $\mathcal{G} \cup \mathcal{B}$. Hence, we conclude that elements in $\mathcal{G} \cup \mathcal{B}$ are never re-classified.

As before, we need to show that for each $i < j$ such that $a_i, a_j \in \mathcal{B}$, $\{b_i, c_i\}$ and $\{b_j, c_j\}$ are disjoint. Suppose to the contrary that $b_i = b_j$. Then Lemma 5.6 tells us that either (1) $a_i x_0 \in \mathfrak{A}_+(a_j)$ or (2) $a_j x_0 \in \mathfrak{A}_+(a_i)$. Case (1) means $d_X(x_0, a_i x_0) > d_X(x_0, a_j x_0)$, which is excluded by the labelling scheme. Case (2) violates the minimality of b_i , as we have

$$d_X(x_0, a_i x_0) < d_X(x_0, a_j x_0) < d_X(x_0, b_i x_0)$$

by Observation 5.5. Hence, $b_i = b_j$ cannot happen. Likewise, we can exclude the cases $b_i = c_j$, $c_i = b_j$ and $c_i = c_j$. Thus, $\{b_i, c_i\}$ and $\{b_j, c_j\}$ are disjoint.

By the same argument as in the previous proof, we have $\#\mathcal{G} \geq \#\mathcal{B}$ in the end. Moreover, for distinct $a, b \in \mathcal{G}$, $\mathfrak{A}_{\sigma(a)}(a)$ and $\mathfrak{A}_{\sigma(b)}(b)$ are disjoint; if not, Lemma 5.6 implies either $bx_0 \in \mathfrak{A}_{\sigma(a)}(a)$ or $ax_0 \in \mathfrak{A}_{\sigma(b)}(b)$, contradicting the goodness of a and b . Note that for each $a \in \mathcal{G} \subseteq \mathcal{A}_1$, $al_a\mathcal{H}_{half}(f^{40D\sigma(a)}x_0, f^{20D\sigma(a)}x_0)$ is a subset of $\mathfrak{A}_{\sigma(a)}(a)$ by Observation 5.5; each of them contains at least N elements of $A \cdot x_0$ by the definition of $\mathcal{A}_1$. We can thus conclude

$$\#A \geq \#A \cdot x_0 \geq N \cdot \#\mathcal{G} \geq N \cdot \frac{\#\mathcal{A}_2}{2} \geq \frac{N}{2 \cdot (2\#S)^R} \#\mathcal{A}_1 \geq \frac{1}{\epsilon} \#\mathcal{A}_1.$$

This ends the proof of Proposition 5.7. $\square$

We can now combine the ingredients and prove the following proposition.

Proposition 5.8. *Let X be a δ -hyperbolic space and let $G \leq \text{Isom}(X)$ be a non-virtually cyclic group with an axial WPD element $f \in G$. Let $x_0 \in Ax(f)$, and let S be a finite generating set of G . Then there exist an element $l \in G$ and constants $D, N > 0$ such that*

$$F := \left\{ u_1 \cdots u_{2N+3} : \begin{array}{l} u_1, \dots, u_{2N+3} \in \{f^D, f^{-D}, l, l^{-1}\}, \\ u_i \neq u_{i+1}^{-1} \ \forall i \in \{1, \dots, 2N+2\} \end{array} \right\},$$

$$\mathcal{H} := \left\{ u_1 \cdots u_N \mathcal{H}_G(id, f^D) : \begin{array}{l} u_1, \dots, u_N \in \{f^D, f^{-D}, l, l^{-1}\}, \\ u_i \neq u_{i+1}^{-1} \ \forall i \in \{1, \dots, N-1\}, \ u_N \in \{l, l^{-1}\} \end{array} \right\}$$

satisfy the following property.

For each finite set $A \subseteq G$ there exists $A' \subseteq A$ with $\#A' \geq \#A/2$ such that for each $a \in A'$, there exist $h \in F$ and $H \in \mathcal{H}$ such that $A \subseteq aH$ and $(H, id), (hH, h)$ are decoupled.

Proof. As G is non-elementary, we can take another loxodromic element $l \in G$ that is independent of f . By powering up f and l if necessary, we can assume that (f, l) is a Schottky pair and that the conclusion of Lemma 5.3 holds (for all k and m , not necessarily large).

Now, Proposition 5.7 provides constants D_0 and $N = N(1/2, D_0) > 6$ for $\epsilon = 1/2$. We will let $D := 20D_0$ and define F and $\mathcal{H}$ as in the statement. Now let $A \subseteq G$ be an arbitrary finite set and let $A' \subseteq A$ be the subset given by Proposition 5.7 for $\epsilon = 1/2$. Then for each $a \in A'$, there exists $\sigma \in \{+1, -1\}$ such that $A \cdot x_0$ has at most N elements outside $al_a \cdot \mathcal{H}_{half}(f^{\sigma D}x_0, f^{2\sigma D}x_0)$. Now, inside $a \cdot l_a \mathcal{H}_{half}(f^{2\sigma D}x_0, f^{\sigma D}x_0)$ there exist at least $2 \cdot 3^{N-5}$ disjoint subsets

$$\left\{ a \cdot l_a f^{2\sigma D} \cdot u_4 \cdots u_N \mathcal{H}_{half}(f^D x_0, x_0) : \begin{array}{l} u_5, \dots, u_{N-1} \in \{f^D, f^{-D}, l, l^{-1}\}, \\ u_4, u_N \in \{l, l^{-1}\}, \\ u_i \neq u_{i+1}^{-1} \ \forall i \in \{4, \dots, N-1\} \end{array} \right\}.$$

Among them, at least one misses $A \cdot x_0$ entirely. Call it $a \cdot u_1 \cdots u_N \mathcal{H}_{half}(f^D x_0, x_0)$. Then our choices will be

$$H_a := u_1 \cdots u_N \mathcal{H}_G(id, f^D]big), \quad h_a := u_1 \cdots u_N \cdot f^D \cdot l \cdot f^{-D} u_N^{-1} \cdots u_1^{-1}.$$

It is clear that $A \subseteq aH_a$. It remains to show that (H_a, id) and $(h_a H_a, h_a)$ are decoupled. This is done by applying Lemma 5.3 to

$$\begin{aligned} H_a &= \mathcal{H}_G(u_1 \cdots u_N, u_1 \cdots u_N f^D), \\ h_a H_a &= \mathcal{H}_G(u_1 \cdots u_N f^D l f^{-D}, u_1 \cdots u_N f^D l). \quad \square \end{aligned}$$

By Theorem 2.10, we have:

Corollary 5.9. *Let G be an acylindrically hyperbolic group and let Γ be a Cayley graph of G . Then Equation 2.7 holds: $\limsup_{p \nearrow p_c} (p_c - p) \chi_p < +\infty$.*

6. BRANCHING SET

Before we begin, recall the notions of barriers and roughly branching sets (Definition 2.12, 2.14). Recall also that $\mathcal{H}_R(x_0, y) := \{z \in X : (z|y)_{x_0} \geq R\}$. Our aim is to show:

Proposition 6.1. *Let X be a δ -hyperbolic space and let $G \leq \text{Isom}(X)$ be a non-virtually cyclic group with a unital, axial WPD element $f \in G$. Let $x_0 \in Ax(f)$. Let S be a finite generating set of G . Then there exists $r > 0$ such that the following holds.*

Let $R > 0$ and $y \in X$. Then there exists an r -roughly branching subset $B = B_1 \sqcup \dots \sqcup B_{R/r} \subseteq G$ such that, for every $g \in G$ with $gx_0 \in \mathcal{H}_R(x_0, y)$, every d_S -path connecting id to g passes through each of $B_1, \dots, B_{R/r}$.

Proof. Let $K_0 = K > 1000\delta$ be as in Theorem 3.13 for G and f . Recall that $EC(f)$ is a virtually cyclic subgroup of G . Now let

$$\mathcal{A} := \{gAx(f) : g \in G\}.$$

Note that $\text{diam}_\gamma(\gamma') \leq K_0$ for distinct axes $\gamma, \gamma' \in \mathcal{A}$ by Theorem 3.13.

Since G is non-virtually cyclic, there exist loxodromic elements $l, l' \in G$ such that f, l, l' are pairwise independent. By replacing l and l' with their suitable powers and by enlarging K_0 , we may assume that

- (1) $d_X(x_0, sx_0) \leq K_0$ for each $s \in S$,
 - (2) for every distinct $a, b \in \{f, f^{-1}, l, l^{-1}, l', l'^{-1}\}$ we have
- $$(6.1) \quad (a^i x_0 | b^j x_0)_{x_0} \leq 0.1K_0 \quad \forall i, j > 0,$$
- (3) $K_0 \leq d_X(x_0, lx_0), d_X(x_0, l'x_0) \leq 2K_0$.

Let $M := \|l\|_S + \|l'\|_S$.

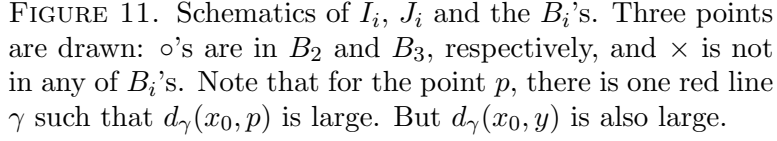
We now describe the roughly branching barrier. Let

$$I_i := [100K_0i - 25K_0, 100K_0i + 25K_0], \quad J_i := [100K_0i - K_0, 100K_0i + K_0],$$

$$B_i := \left\{ g \in G : \begin{array}{l} (gx_0 | y)_{x_0} \in I_i \text{ and} \\ \forall \gamma \in \mathcal{A} [d_\gamma(x_0, y) \geq 5K_0 \vee d_\gamma(x_0, gx_0) \leq 100K_0] \end{array} \right\}.$$

See Figure 11. We claim that:

Claim 6.2. *Let $P = (g_1, g_2, \dots, g_N)$ be a d_S -path such that $(g_1 x_0 | y)_{x_0} \in I_0$ and $(g_N x_0 | y)_{x_0} \in I_2$. Then there exists i such that $g_i \in B_1$.*


$$\mathcal{A}' := \{\gamma \in \mathcal{A} : d_\gamma(x_0, y) < 5K_0\}.$$

(1) Let $i(0)$ be the first step when $(g_{i(0)}x_0|y)_{x_0} \in J_1$. Let γ_0 be the closest element to x_0 of

(2) Having defined $i(0), \dots, i(k)$ and $\gamma_0, \dots, \gamma_k$, let $i(k+1)$ be the first step after $i(k)$ when $d_{\gamma_k}(x_0, g_{i(k+1)}x_0) < 80K_0$. Let γ_{k+1} be the closest element to x_0 of

Since P has finite length N , this will give the desired contradiction. In the rest of the proof, we will justify why the $i(k)$'s exist and why the $\mathcal{C}_k$'s are nonempty. Along the way, for each $k \geq 1$ we will also show that

- Let us now start the construction. Note that for each i , we have

Since $(g_i x_0 | y)_{x_0}$ changes from less than $30K_0$ to more than $170K_0$, there exists a step $i(0)$ for which $(g_{i(0)} x_0 | y)_{x_0}$ lies in J_1 .

Since P does not pass through B_1 , x_0 and $g_{i(0)}x_0$ are $100K_0$ -separated along some $\gamma \in \mathcal{A}'$. Hence, $\mathcal{C}_0$ is non-empty and we can pick γ_0 .

Now fix $l \geq 0$ and suppose that we have constructed $i(0) < \dots < i(l)$ and $\gamma_0, \dots, \gamma_l \in \mathcal{A}'$ satisfying properties (A) and (B) for $k \leq l$. We aim to show that $i(l+1)$ and γ_{l+1} are well defined and satisfy properties (A) and (B).

Before proceeding, fix some point $p \in \pi_{\gamma_0}(x_0)$, and let $z \in X$ be an arbitrary point satisfying $d_{\gamma_0}(x_0, z) \geq 70K_0$. As $\gamma_0 \in \mathcal{A}'$, we have $d_{\gamma_0}(x_0, y) \leq 5K_0$ and $d_{\gamma_0}(y, z) \geq 60K_0$. By Corollary 3.11(2), $[x_0, z]$ is 10δ -close to p , and $[y, z]$ is 10δ -close to $\pi_{\gamma_0}(y)$, which is again $5K_0$ -close to p . We conclude

$$d_X(x_0, z) =_{20\delta} d_X(x_0, p) + d_X(p, z), \quad d_X(y, z) =_{20\delta+10K_0} d_X(y, p) + d_X(p, z),$$

$$(y|z)_{x_0} =_{5.1K_0} (y|p)_{x_0}.$$

For example, $z = g_{i(0)}x_0$ satisfies the above requirement by definition, so we conclude $(y|p)_{x_0} =_{5.1K_0} (y|g_{i(0)}x_0)_{x_0}$.

Now suppose to the contrary that $d_{\gamma_l}(x_0, g_N x_0) \geq 80K_0$. By the assumption for this inductive step, we know that

$$d_{\gamma_k}(x_0, \gamma_{k+1}) \geq 75K_0 \quad (k = 0, \dots, l-1).$$

By applying Corollary 3.15, we deduce $d_{\gamma_0}(x_0, g_N x_0) \geq 72K_0$. Plugging in $z = g_N x_0$ to the above, we conclude that

$$(g_N x_0|y)_{x_0} =_{5.1K_0} (p|y)_{x_0} =_{5.1K_0} (g_{i(0)}x_0|y)_{x_0} \in J_1.$$

This implies that $(g_N x_0|y)_{x_0}$ lies in I_1 and not in I_2 , a contradiction. We conclude that $\{k > i(l) : d_{\gamma_l}(x_0, g_k x_0) < 80K_0\}$ is a nonempty set containing N , and we can pick its smallest element $i(l+1)$ as desired. Note that

$$d_{\gamma_l}(x_0, g_{i(l+1)-1}x_0) \geq 80K_0 \text{ and } d_{\gamma_l}(x_0, g_{i(l+1)}x_0) < 80K_0.$$

By Corollary 3.11(1), we have $d_{\gamma_l}(x_0, g_{i(l+1)}x_0) \geq 78K_0$. By applying Corollary 3.15 to $x_0, \gamma_0, \dots, \gamma_l, g_{i(l+1)}x_0$, we deduce $d_{\gamma_0}(x_0, g_{i(l+1)}x_0) \geq 76K_0$. Plugging in $z = g_{i(l+1)}x_0$ to the above, we conclude $(g_{i(l+1)}x_0|y)_{x_0} \in I_1$.

Next, we claim that $\mathcal{C}_{l+1}$ is nonempty. Recall that $g_{i(l+1)} \in P$ does not lie in B_1 , despite $(g_{i(l+1)}x_0|y)_{x_0} \in I_1$. This means that there exists $\gamma \in \mathcal{A}'$ with $d_\gamma(x_0, g_{i(l+1)}x_0) > 100K_0$. Meanwhile, Corollary 3.15 implies that

$$d_{\gamma_k}(x_0, g_{i(l+1)}x_0) \leq 87K_0 < d_\gamma(x_0, g_{i(l+1)}x_0) \quad (k = 0, \dots, l).$$

Hence, γ and $\gamma_0, \dots, \gamma_l$ are pairwise distinct elements of $\mathcal{A}'$, which have pairwise K_0 -bounded projections.

Suppose to the contrary that $d_X(x_0, \gamma) \leq d_X(x_0, \gamma_k)$ for some $k \in \{0, 1, \dots, l\}$. Let us pick the smallest possible k .

- (1) Case 1: $k = 0$. In this case, we have $d_{\gamma_0}(x_0, \pi_\gamma(x_0)) \leq 12\delta$ by Lemma 3.10. Meanwhile, by feeding $x_0, \gamma_0, \dots, \gamma_l, g_{i(l+1)}x_0$ into Corollary 3.15, we obtain $d_{\gamma_0}(x_0, g_{i(l+1)}x_0) \geq 72K_0$. Hence, we have

$$d_{\gamma_0}(x_0, \gamma) \leq 12\delta + \text{diam}_{\gamma_0}(\gamma) \leq 2K_0,$$

$$d_{\gamma_0}(\gamma, g_{i(l+1)}x_0) \geq d_{\gamma_0}(x_0, g_{i(l+1)}x_0) - 2K_0 \geq 70K_0.$$

Lemma 3.14 implies $d_\gamma(\gamma_0, g_{i(l+1)}x_0) \leq 3K_0$. As $d_\gamma(x_0, g_{i(l+1)}x_0) \geq 100K_0$, this implies that $d_\gamma(x_0, \gamma_0) \geq 97K_0$. Meanwhile, recall that $d_{\gamma_0}(x_0, g_{i(0)}x_0) \geq 78K_0$. Combining these two ingredients via Corollary 3.15, we deduce $d_\gamma(x_0, g_{i(0)}x_0) \geq 94K_0$, that is, $\gamma \in \mathcal{C}_0$. Moreover, γ is closer to x_0 than γ_0 is, because $d_\gamma(x_0, \gamma_0) \geq 97K_0$. This contradicts the minimality of $\gamma_0 \in \mathcal{C}_0$.

- (2) Case 2: $1 \leq k \leq l$ and $d_X(x_0, \gamma_{k-1}) < d_X(x_0, \gamma) \leq d_X(x_0, \gamma_k)$. Lemma 3.10 tells us that $d_\gamma(x_0, \pi_{\gamma_{k-1}}(x_0)) \leq 12\delta$, which implies

$$d_\gamma(x_0, \gamma_{k-1}) \leq 12\delta + \text{diam}_\gamma(\gamma_{k-1}) \leq 2K_0.$$

Similarly, we deduce $d_{\gamma_k}(x_0, \gamma) \leq 2K_0$. Recall from Corollary 3.15 that $d_{\gamma_k}(x_0, g_{i(l+1)}x_0)$ is larger than $72K_0$. Combining these two, we deduce $d_{\gamma_k}(\gamma, g_{i(l+1)}x_0) \geq 70K_0$ and $d_\gamma(\gamma_k, g_{i(l+1)}x_0) \leq 3K_0$. We now observe

$$\begin{aligned} d_\gamma(\gamma_{k-1}, \gamma_k) &\geq d_\gamma(x_0, g_{i(l+1)}x_0) - d_\gamma(x_0, \gamma_{k-1}) - d_\gamma(\gamma_k, g_{i(l+1)}x_0) \\ &\geq 100K_0 - 2K_0 - 3K_0 = 95K_0. \end{aligned}$$

This contradicts property (B) for k .

We conclude that $d_X(x_0, \gamma) > d_X(x_0, \gamma_l)$ and $\gamma \in \mathcal{C}_{l+1}$.

Having checked that $\mathcal{C}_{l+1}$ is nonempty indeed, let $\gamma_{l+1} \in \mathcal{C}_{l+1}$ be the distance minimizer. In order to check property (A) for $k = l + 1$, note that $d_{\gamma_{l+1}}(x_0, \pi_{\gamma_l}(x_0)) \leq 12\delta$ and $d_{\gamma_{l+1}}(x_0, \gamma_l) \leq 2K_0$ by Lemma 3.10 and the bounded projections. This implies that

$$(6.2) \quad d_{\gamma_{l+1}}(\gamma_l, g_{i(l+1)}x_0) \geq 80K_0 - 2K_0 \geq 78K_0,$$

and Lemma 3.14 tells us that $d_{\gamma_l}(\gamma_{l+1}, g_{i(l+1)}x_0) \leq 3K_0$. Since the choice of $g_{i(l+1)}$ forces $d_{\gamma_l}(x_0, g_{i(l+1)}x_0) = 2K_0 + 80K_0$, we can conclude property (A).

Now suppose to the contrary that there exists $\kappa \in \mathcal{A}'$ with $d_\kappa(\gamma_l, \gamma_{l+1}) \geq 90K_0$. For such a κ , we have

$$(6.3) \quad d_{\gamma_l}(\kappa, \gamma_{l+1}), d_{\gamma_{l+1}}(\kappa, \gamma_l) \leq 3K_0$$

by Lemma 3.14. Combining this with property (A) for $k = l + 1$, we conclude that $d_{\gamma_l}(x_0, \kappa) \geq 70K_0$. Consequently, κ is farther from x_0 than γ_l is. Next, by Lemma 3.14 we also know that $d_\kappa(x_0, \gamma_l) \leq 3K_0$. Then we have $d_\kappa(x_0, \gamma_{l+1}) \geq 87K_0$. In other words, κ is closer to x_0 than γ_{l+1} is. Lastly, by combining Inequality 6.2 and 6.3, we obtain $d_{\gamma_{l+1}}(\kappa, g_{i(l+1)}x_0) \geq 75K_0$. By using Lemma 3.14 again, we deduce $d_\kappa(\gamma_{l+1}, g_{i(l+1)}x_0) \leq 3K_0$. Now combining everything, we get

$$\begin{aligned} d_\kappa(x_0, g_{i(l+1)}x_0) &\geq d_\kappa(\gamma_l, \gamma_{l+1}) - d_\kappa(x_0, \gamma_l) - d_\kappa(\gamma_{l+1}, g_{i(l+1)}x_0) \\ &\geq 90K_0 - 3K_0 - 3K_0 \geq 80K_0. \end{aligned}$$

In conclusion, κ is an element of $\mathcal{C}_{l+1}$ that is closer to x_0 than γ_{l+1} is. This contradicts the definition of γ_{l+1} , and we conclude that property (B) holds for $k = l + 1$. This ends the induction step and the proof of Claim 6.2. $\square$

Similarly, for $g \in G$ such that $gx_0 \in \mathcal{H}_R(x_0, y)$, every d_S -path from id to g must pass through each of $B_1, B_2, \dots, B_{R/100K_0}$.

It remains to show that $B := \sqcup_{i \geq 1} B_i$ is roughly branching. Since $EC(f)$ is a finite extension of a quasi-isometrically embedded subgroup $\langle f \rangle$, the set

$$EC(f) \cap \{g \in G : d_X(x_0, gx_0) < 5K_0\}$$

is finite. Hence, it is contained in $\{g : \|g\|_S \leq R'\}$ for some R' . We claim that $\sqcup_{i \geq 1} B_i$ is $(R' + 4M + 200K_0\|f\|_S + 1)$ -roughly branching.

Let us take a subset B' of $\sqcup_{i \geq 1} B_i$ that is maximally $(R' + 2M + 1)$ -separated (in terms of the word metric d_S). We now construct a map $F : B' \rightarrow G$ as follows. Recall Condition 6.1. By Lemma 3.4, there exists $l_y \in \{l, l^{-1}\}$ such that

$$(l_y^{-1}x_0|y)_{x_0} \leq 0.1K_0 + 4\delta.$$

Moreover, for each $g \in G$ there exists $l_g \in \{l, l^{-1}, l'\}$ such that

$$(g^{-1}x_0|l_gx_0)_{x_0}, (g^{-1}y|l_gx_0)_{x_0} < 0.1K_0 + 4\delta.$$

We now set

$$F(g) := g \cdot l_g \cdot f^{200K_0} \cdot l_y. \quad (\forall g \in B')$$

We now claim that:

Claim 6.3. *If $a_1, a_2, \dots, a_k, b_1, \dots, b_k \in B'$ are such that*

$$F(a_1)F(a_2) \cdots F(a_k) = F(b_1)F(b_2) \cdots F(b_k),$$

then $a_1 = b_1$.

Proof of Claim 6.3. Let $U := F(a_1) \cdots F(a_k)$. Note that for each $i \in \{1, \dots, k\}$,

$$(x_0, a_ix_0, a_il_{a_i}x_0, a_il_{a_i}f^{200K_0}x_0, a_il_{a_i}f^{200K_0}l_yx_0)$$

is a sequence of 5 points with consecutive spacing $\geq K_0$ and consecutive Gromov products $\leq 0.1K_0 + 4\delta$. Here, we used the fact that

$$d_X(x_0, a_ix_0) \geq (a_ix_0|y)_{x_0} \geq 75K_0.$$

Moreover, for each $i \in \{2, \dots, k\}$,

$$(x_0, l_yx_0, l_ya_ix_0)$$

is a sequence of 3 points with consecutive spacing $\geq K_0$ and Gromov product $\leq 0.1K_0 + 8\delta$. By applying Lemma 3.5, we conclude that

$$(x_0, a_1l_{a_1}x_0, a_1l_{a_1}f^{200K_0}x_0, Ux_0)$$

has consecutive Gromov product $\leq 0.15K_0$. In particular, $d_\gamma(x_0, Ux_0) \geq 199K_0$ holds for $\gamma := a_1l_{a_1}Ax(f)$.

Similarly, we have $d_{\gamma'}(x_0, Ux_0) \geq 199K_0$ for $\gamma' := b_1l_{b_1}Ax(f)$. We now have three cases.

(1) $\gamma \neq \gamma'$ and $d_\gamma(x_0, \gamma') \geq 190K_0$. In this case, we have

$$d_\gamma(x_0, b_1x_0) \geq d_\gamma(x_0, \gamma') - \text{diam}_\gamma(\gamma') - d_\gamma(b_1x_0, b_1l_{b_1}x_0) \geq 180K_0.$$

Now let $m \in \mathbb{Z}$. Then by the construction, we have

$$(z, a_1x_0, a_1l_{a_1}x_0, a_1l_{a_1}f^m x_0)$$

has consecutive Gromov product less than $0.11K_0$ for $z \in \{x_0, y\}$. By Lemma 3.5 and Lemma 3.6, $[z, a_1l_{a_1}f^m x_0]$ intersects $\mathcal{N}_{0.12K_0}(a_1x_0)$ for $z \in \{x_0, y\}$. It follows that $(a_1l_{a_1}f^m x_0|y)_{x_0} =_{0.25K_0} (a_1x_0|y)_{x_0}$. Since this is true for all m , we have

$$\pi_{[x_0, y]}(a_1l_{a_1}Ax(f)) \subseteq \mathcal{N}_{0.4K_0}(\pi_{[x_0, y]}(a_1x_0)).$$

In particular, $\pi_{[x_0, y]}(a_1l_{a_1}Ax(f))$ has diameter $\leq 0.8K_0$, which implies $d_\gamma(x_0, y) \leq 0.9K_0 < 5K_0$. The presence of γ thus implies that $b_1 \notin \sqcup_i B_i$, which is a contradiction.

(2) $\gamma \neq \gamma'$ and $d_\gamma(x_0, \gamma') < 190K_0$. As $d_\gamma(x_0, Ux_0) > 199K_0$, we conclude $d_\gamma(\gamma', Ux_0) > 9K_0$. This implies that $d_{\gamma'}(\gamma, Ux_0) \leq 3K_0$ by Lemma 3.14. Meanwhile, we have $d_{\gamma'}(x_0, Ux_0) \geq 199K_0$. It follows that $d_{\gamma'}(x_0, \gamma) \geq 190K_0$. Now, a symmetric argument to the above yields that $a_1 \notin \sqcup_i B_i$, a contradiction.

(3) $\gamma = \gamma'$. Then we have $b_1l_{b_1}x_0 \in a_1l_{a_1}Ax(f)$ and $(b_1l_{b_1})^{-1}(a_1l_{a_1}) \in EC(f)$. Moreover, as we did in Case (1), we can apply Lemma 3.5 to the sequence $(x_0, a_1x_0, a_1l_{a_1}x_0, a_1l_{a_1}f^m x_0)$ for each $m \in \mathbb{Z}$. As a result, we obtain $(x_0|z)_{a_1l_{a_1}x_0} \leq 0.115K_0$ for all $z \in a_1l_{a_1}Ax(f)$, including $z = b_1l_{b_1}x_0$. Hence, we have

$$d_X(x_0, b_1l_{b_1}x_0) \geq d_X(x_0, a_1l_{a_1}x_0) + d_X(a_1l_{a_1}x_0, b_1l_{b_1}x_0) - 0.3K_0.$$

Similarly, we have

$$d_X(x_0, a_1l_{a_1}x_0) \geq d_X(x_0, b_1l_{b_1}x_0) + d_X(a_1l_{a_1}x_0, b_1l_{b_1}x_0) - 0.3K_0.$$

Combining these two, we conclude that $(a_1l_{a_1})^{-1}(b_1l_{b_1})$ lies in $EC(f) \cap \{g \in G : d_X(x_0, gx_0) < 5K_0\}$. Hence, it has d_S -norm at most R' , and $d_S(a_1, b_1) \leq R' + 2M$ holds. Since B' is $(R' + 2M + 1)$ -separated, we conclude $a_1 = b_1$. $\square$

Now an inductive argument leads to:

Claim 6.4. *If $a_1, a_2, \dots, a_k, b_1, \dots, b_k \in B'$ are such that*

$$F(a_1)F(a_2) \cdots F(a_k) = F(b_1)F(b_2) \cdots F(b_k),$$

then $a_i = b_i$ for each $i = 1, \dots, k$.

It remains to check that $\sqcup_i B_i$ is contained in a bounded neighborhood of $F(B')$. Given any $a \in \sqcup_i B_i$, it is $(R' + 2M + 1)$ -close to some $a' \in B'$, as B' is a maximal $(R' + 2M + 1)$ -separated subset of $B = \sqcup_i B_i$. Now, $F(a')$ and a' are $(2M + 200K_0\|f\|_S)$ -close. In summary, a is $(R' + 4M + 1 + 200K_0\|f\|_S)$ -close to $F(B')$ as desired. $\square$

Corollary 6.5. *Let G be a finitely generated group admitting a proper and non-elementary action on a Gromov hyperbolic space and let Γ be a Cayley graph of G . Then Equation 2.8 holds:*

$$\limsup_{p \nearrow p_c} \left\{ \frac{\sum_{g,h \in A} \tau_p(g,h)}{\chi_p \cdot \#A} : A \subseteq G \text{ finite} \right\} = 0.$$

Proof. Let G be as above, with a finite generating set S , and let $\Gamma = \text{Cay}(G, S)$. Combining Proposition 4.7 and Proposition 6.1, this group satisfies the assumption of Theorem 2.18 with

$$\mathcal{H}_D := \{ \{g \in G : gx_0 \in \mathcal{H}_D(x_0, y)\} : y \in X \}$$

and $S_D = \mathcal{G}_{D,E} := \emptyset$. By Theorem 2.18, we conclude Equation 2.8 for Γ . $\square$

By combining Corollary 5.9 and 6.5, we conclude:

Corollary 6.6. *Let G be a finitely generated group admitting a proper and non-elementary action on a Gromov hyperbolic space and let Γ be a Cayley graph of G . Then $p_c(\Gamma) < p_u(\Gamma)$ and $\nabla_{p_c}(\Gamma) < +\infty$.*

7. BARRIERS IN ACYLINDRICALLY HYPERBOLIC GROUPS

Let G be an acylindrically hyperbolic group with a finite generating set S . Let $f \in G$, let (X, d_X) be a δ -hyperbolic space and let K be the constant given by Theorem 3.13. More explicitly, G acts on X and contains a unital, axial WPD loxodromic isometry f . Let $x_0 \in Ax(f)$. By increasing K if necessary, we may also suppose that $d_X(x_0, sx_0) < K$ for each $s \in S$. Also, let D_0 be the constant given by Lemma 3.17. We fix these choices throughout the section. Recall the notations

$$\begin{aligned} \mathcal{N}_R(A) &:= \{x \in X : \exists a \in A [d_X(x, a) \leq R]\}, \\ B_S(R) &:= \{g \in G : \|g\|_S \leq R\}. \end{aligned} \quad (\forall R > 0, \forall A \subseteq X)$$

For each $D, E, \rho > 0$ and $g \in G$, we define a version of anti-halfspace:

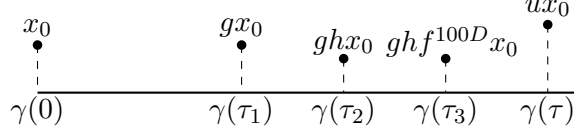
$$\mathfrak{A}_{D,E;\rho}^{\pm}(g) := \left\{ u \in G : \begin{array}{l} \exists h \in B_S(E), \exists \text{ geodesic } \gamma \text{ passing through} \\ \mathcal{N}_{\rho D}(x_0), \mathcal{N}_D(gx_0), \mathcal{N}_{20\delta}(ghx_0), \\ \mathcal{N}_{20\delta}(ghf^{\pm 100D}x_0) \text{ and } \mathcal{N}_{18D}(ux_0) \text{ in order} \end{array} \right\}.$$

Explicitly, we say $u \in \mathfrak{A}_{D,E;\rho}^{\pm}(g)$ if there exist $h \in B_S(E)$, a geodesic γ starting from $\mathcal{N}_{\rho D}(x_0)$ at time 0, and $0 \leq \tau_1 \leq \tau_2 \leq \tau_3 \leq \tau$ such that

$$\begin{aligned} \gamma(\tau_1) &\in \mathcal{N}_D(gx_0), & \gamma(\tau_2) &\in \mathcal{N}_{20\delta}(ghx_0), \\ \gamma(\tau_3) &\in \mathcal{N}_{20\delta}(ghf^{\pm 100D}x_0), & \gamma(\tau) &\in \mathcal{N}_{18D}(ux_0). \end{aligned}$$

We say that the tuple $(h, \gamma, \tau_1, \tau_2, \tau_3, \tau)$ is a *certificate* for the membership $u \in \mathfrak{A}_{D,E;\rho}^{\pm}(g)$. See Figure 12 for schematics. We also define

$$\begin{aligned} \mathfrak{A}_{D,E;\rho}(g) &:= \mathfrak{A}_{D,E;\rho}^{+}(g) \cup \mathfrak{A}_{D,E;\rho}^{-}(g), \\ \mathfrak{A}_{D,E}(g) &:= \mathfrak{A}_{D,E;0}(g). \end{aligned}$$

FIGURE 12. Schematics for $\mathfrak{A}_{D,E;\rho}^+(g)$.

The following analogue of Observation 4.1 and Observation 4.2 is immediate from the definition of anti-halfspaces.

Observation 7.1. *Let $0 \leq \rho \leq 2$, let $D \geq 10\delta$, let $E > 0$, let $g \in G$ and let $u \in \mathfrak{A}_{D,E;\rho}(g)$. Then $u \notin \{id, g\}$ and $d_X(x_0, ux_0) \geq d_X(x_0, gx_0) + 50D$.*

We present an analogue of Lemma 4.3, which was essential for constructing the “lineage” between points.

Lemma 7.2. *Let $D > 100(K + D_0 + \delta)$ and $E > 0$. Then there exists $R > 0$ such that the following holds.*

Let $0 \leq \rho, \rho' \leq 2$ and let $g, g', u \in G$ be such that $d_S(g, g') \geq R$. Suppose that $u \in \mathfrak{A}_{D,E;\rho}(g)$ with a certificate $(h, \gamma, \tau_1, \tau_2, \tau_3, \tau)$ and $u \in \mathfrak{A}_{D,E;\rho'}(g')$ with a certificate $(h', \gamma', \tau'_1, \tau'_2, \tau'_3, \tau')$. Then either $\gamma(t) \in \mathcal{N}_{18D}(g'x_0)$ for some $t \geq \tau_3$ or $\gamma'(t) \in \mathcal{N}_{18D}(gx_0)$ for some $t \geq \tau'_3$.

Proof. Let $R' := R(100D, 2E + 100D\|f\|_S)$ be as in Lemma 3.17 for the WPD element f . We will prove the statement for $R := R' + 2E$.

Let us prove the lemma for the case $u \in \mathfrak{A}_{D,E;\rho}^+(g) \cap \mathfrak{A}_{D,E;\rho'}^+(g')$ only; the other cases can be handled similarly. Here are some facts:

- (1) $d_X(\gamma(0), \gamma'(0)) \leq 4D$ and $d_X(\gamma(\tau), \gamma'(\tau')) \leq 36D$.
- (2) $|\tau - \tau'| \leq 40D$.
- (3) $\tau_3 - \tau_2 =_{40\delta} d_X(x_0, f^{100D}x_0) = 100D$.
- (4) $\tau'_1 + 59D \leq \tau'_2 + 59D \leq \tau'_3 - 40D \leq \tau' - 40D \leq \tau$.
- (5) $\gamma(\tau_2 + kD)$ and $ghf^{kD}x_0$ are 80δ -close for $k = 50, 59$.

Analogous facts hold after swapping γ, τ_i, g, h , with γ', τ'_i, g', h' .

We will prove that if $\tau_2 \leq \tau'_2$, then $\gamma(t) \in \mathcal{N}_{18D}(g'x_0)$ for $t := \tau'_1 + 11D$.

A symmetric argument will deal with the other situation.

Case 1. $\tau_3 \leq \tau'_1 + 11D$. Note that $\tau'_1 + 11D \leq \tau'_3 - 40D \leq \tau' - 40D \leq \tau$ by Fact (4). By Lemma 3.2 we have

$$\begin{aligned} d_X(\gamma(\tau'_1 + 11D), g'x_0) &\leq d_X(\gamma(\tau'_1 + 11D), \gamma'(\tau'_1 + 11D)) \\ &\quad + d_X(\gamma'(\tau'_1 + 11D), \gamma'(\tau'_1)) + d_X(\gamma'(\tau'_1), g'x_0) \\ &\leq (4D + 4\delta) + 11D + D \leq 18D. \end{aligned}$$

Case 2. $\tau_3 \in [\tau'_1 + 11D, \tau'_2 + 50D]$. Note that $\tau_2, \tau'_1 \leq \tau_3 - 11D$ and $\tau_2, \tau'_1 \leq (\tau'_2 + 50D) - 10D$. This implies

$$\text{diam}([\tau_2, \tau_3] \cap [\tau'_1, \tau'_2 + 50D]) \geq 11D.$$

Meanwhile, Lemma 3.2 implies that $\gamma(t)$ and $\gamma'(t)$ are $(4D + 40\delta)$ -close for $t = \tau'_1, \tau'_2 + 50D$, as $\tau'_1, \tau'_2 + 50D \leq \tau' - 40D \leq \tau$. We now have an implication chain

$$\begin{aligned} d_{\gamma([\tau_2, \tau_3])}(\gamma(\tau'_1), \gamma(\tau'_2 + 50D)) &\geq 11D \\ \Rightarrow d_{\gamma([\tau_2, \tau_3])}(\gamma'(\tau'_1), \gamma'(\tau'_2 + 50D)) &\geq 11D - 2(4D + 52\delta) \quad (\because \text{Corollary 3.11(1)}) \\ \Rightarrow d_{gh[x_0, f^{100D}x_0]}(\gamma'(\tau'_1), \gamma'(\tau'_2 + 50D)) &\geq 3D - 200\delta \quad (\because \text{Corollary 3.11(3)}) \\ \Rightarrow d_{gh[x_0, f^{100D}x_0]}(g'x_0, g'h'f^{50D}x_0) &\geq 3D - 2D - 300\delta. \quad (\because \text{Corollary 3.11(1)}) \end{aligned}$$

This contradicts Lemma 3.17, as $\|(gh)^{-1}g'\|_S \geq \|g^{-1}g'\|_S - \|h\|_S \geq R - E \geq R'$, and $\|h'f^{50D}\|_S \leq 50D\|f\|_S + E$.

Case 3. $\tau_3 \geq \tau'_2 + 50D$. In this case, we have the inclusion

$$[\tau'_2, \tau'_2 + 9D] \subseteq [\tau_2, \tau_2 + 59D] \cap [\tau'_2, \tau'_3].$$

This implies $d_{\gamma'([\tau'_2, \tau'_3])}(\gamma'(\tau_2), \gamma'(\tau_2 + 59D)) \geq 9D$.

Meanwhile, as $\tau_2 + 59D \leq \tau - 40D \leq \tau'$, Lemma 3.2 tells us that

$$\begin{aligned} d_X(\gamma'(\tau_2), ghx_0) &\leq (4D + 4\delta) + d_X(\gamma(\tau_2), ghx_0) \leq 4D + 90\delta, \\ d_X(\gamma'(\tau_2 + 59D), ghf^{59D}x_0) &\leq (4D + 4\delta) + d_X(\gamma(\tau_2 + 59D), ghf^{59D}x_0) \\ &\leq 4D + 90\delta. \end{aligned}$$

Then Corollary 3.11(1), (3) tells us that

$$d_{g'h'[x_0, f^{100D}x_0]}(ghx_0, ghf^{59D}x_0) \geq D - 300\delta.$$

As $\|(gh)^{-1}g'h'\|_S \geq R - 2E \geq R'$ and $\|f^{59D}\|_S \leq 59D\|f\|_S$, this inequality contradicts Lemma 3.17.

Therefore, only Case 1 is possible and $\gamma(t) \in \mathcal{N}_{18D}(g'x_0)$ indeed holds. $\square$

Lemma 7.3. *Let $D > 100(K + D_0 + \delta)$, let $E > 0$ and let $R = R(D, E)$ be as in Lemma 7.2. Let $g, g', u, v \in G$ be such that*

$$d_S(g, g') \geq R \text{ and } u \in \mathfrak{A}_{D,E}(g) \cap \mathfrak{A}_{D,E}(g') \cap gv^{-1}\mathfrak{A}_{D,E}(v).$$

Then either $g \in \mathfrak{A}_{D,E}(g')$ or $g' \in \mathfrak{A}_{D,E}(g) \cap gv^{-1}\mathfrak{A}_{D,E}(v)$ holds.

Proof. By Lemma 7.2, we already know that either $g \in \mathfrak{A}_{D,E}(g')$ or $g' \in \mathfrak{A}_{D,E}(g)$. We will focus on the latter case.

Let $(h, \gamma, \tau_1, \tau_2, \tau_3, \tau)$ be a certificate for the membership $vg^{-1}u \in \mathfrak{A}_{D,E}(v)$, and let $(h', \gamma', \tau'_1, \tau'_2, \tau'_3, \tau')$ be a certificate for $u \in \mathfrak{A}_{D,E}(g')$. Since we are in the case $g' \in \mathfrak{A}_{D,E}(g)$, there exists $z \in \mathcal{N}_{18D}(g'x_0) \subseteq \mathcal{N}_{19D}(\gamma'(\tau'_1))$ such that $(x_0|z)_{gx_0} \leq D$ holds. Meanwhile, Observation 7.1 says $d_X(\gamma'(\tau'_1), gx_0) \geq 50D$ and $(\gamma'(\tau'_1)|z)_{gx_0} \geq 30D$. By the Gromov inequality (Lemma 3.4), we have $(x_0|\gamma'(\tau'_1))_{gx_0} \leq D + 4\delta$. Hence, there exists some $\tau'_0 \in [0, \tau'_1]$ such that $\gamma'(\tau'_0) \in \mathcal{N}_{2D}(gx_0)$. This means that

$$(g^{-1}\gamma'(\cdot + \tau'_0), h', \tau'_1 - \tau'_0, \tau'_2 - \tau'_0, \tau'_3 - \tau'_0, \tau' - \tau'_0)$$

serves as a certificate for $g^{-1}u \in \mathfrak{A}_{D,E;2}(g^{-1}g')$. Meanwhile,

$$(v^{-1}\gamma(\cdot + \tau_1), h, 0, \tau_2 - \tau_1, \tau_3 - \tau_1, \tau - \tau_1)$$

serves as a certificate for $g^{-1}u \in \mathfrak{A}_{D,E;1}(id)$.

We can now apply Lemma 7.2 to $g^{-1}u \in \mathfrak{A}_{D,E;1}(id) \cap \mathfrak{A}_{D,E;2}(g^{-1}g')$. Then either $id \in \mathfrak{A}_{D,E;2}(g^{-1}g')$ or $g^{-1}g' \in \mathfrak{A}_{D,E;1}(id)$. Since the former case violates Observation 7.1, we are led to the latter case. In particular, we obtain a timing $T \geq \tau_3$ such that $d_X(v^{-1}\gamma(T), g^{-1}g'x_0) \leq 18D$. This implies a stronger conclusion that $g' \in gv^{-1}\mathfrak{A}_{D,E}(v)$, as desired. $\square$

Now for each $g \in G$ we define

$$\mathcal{H}_{D,E,f}(g) := \left\{ u \in G : \begin{array}{l} (gx_0|ux_0)_{x_0} > 0.5D \text{ or} \\ \nexists h \in B_S(E) \left[d_{hAx(f)}(x_0, ux_0) \geq 140D \right] \end{array} \right\}.$$

We call it an f -halfspace with radius parameters (D, E) . This is related to anti-halfspaces $\mathfrak{A}_{D,E}$ because:

Lemma 7.4. *For each $D > 100(\delta + 1)$, $E \geq 0$ there exists $F > E$ such that*

$$[u \notin \mathcal{H}_{D,E,f}(g^{-1})] \Rightarrow [gu \in \mathfrak{A}_{D,F}(g)] \quad (\forall g, u \in G).$$

Proof. Let $u \notin \mathcal{H}_{D,E,f}(g^{-1})$, with some $h \in B_S(E)$ such that $d_{hAx(f)}(x_0, ux_0) \geq 140D$. Let $h' \in \{hf^i\}_{i \in \mathbb{Z}}$ be such that $[\pi_{hAx(f)}(x_0), \pi_{hAx(f)}(ux_0)]$ contains $[h'x_0, h'f^{140D}x_0]$ and such that $\pi_{hAx(f)}(x_0)$ is 1-close to $\{h'x_0, h'f^{140D}x_0\}$. Since the projection is large, Corollary 3.11(2) tells us that $[h'x_0, h'f^{140D}x_0]$ is contained in $\mathcal{N}_{12\delta}([x_0, ux_0])$. Meanwhile, $[h'f^{20D}x_0, h'f^{120D}x_0]$ is a subsegment of $[h'x_0, h'f^{140D}x_0]$ that is at least $10D$ -far from $\pi_{hAx(f)}(x_0)$. By the triangle inequality, x_0 cannot be $5D$ -close to $[h'f^{20D}x_0, h'f^{120D}x_0]$.

Now draw the geodesic $\gamma : [0, T] \rightarrow X$ from x_0 to gux_0 and let $\tau_1 := (gux_0|gx_0)_{x_0}$. Let also $p \in [gx_0, gux_0]$ be such that $d_X(p, gx_0) = (x_0|gux_0)_{gx_0}$. Then by Lemma 3.6, $\gamma|_{[\tau_1, T]}$ and $[p, gux_0]$ are 4δ -fellow traveling and $\gamma(\tau_1) \in \mathcal{N}_{0.5D+4\delta}(gx_0)$. Here, $[gx_0, p]$ and $[p, gux_0]$ cover the geodesic $[gx_0, gux_0]$, and $gh'f^{20D}[x_0, f^{100D}x_0]$ does not intersect $\mathcal{N}_{5D}(gx_0)$. Hence, we have

$$\begin{aligned} gh'f^{20D}[x_0, f^{100D}x_0] &\subseteq \mathcal{N}_{12\delta}([gx_0, gux_0]) \setminus \mathcal{N}_{5D}(gx_0) \\ &\subseteq \mathcal{N}_{12\delta}([p, gux_0]) \subseteq \mathcal{N}_{16\delta}(\gamma|_{[\tau_1, T]}). \end{aligned}$$

Furthermore, recall that

$$\begin{aligned} d_X(hx_0, h'x_0) &\leq d_X(hx_0, \pi_{hAx(f)}(x_0)) + (140D + 1) \\ &\leq 2d_X(x_0, hx_0) + 140D + 1 \leq 2EK + 140D + 1. \end{aligned}$$

This implies that

$$\begin{aligned} \|h'f^{20D}\|_S, \|h'f^{120D}\|_S &\leq \|h\|_S + \|f\|_S \cdot (d_X(hx_0, h'x_0) + 120D) \\ &\leq E + (2EK + 260D + 1)\|f\|_S. \end{aligned}$$

We conclude that $gu \in \mathfrak{A}_{D,F}(g)$ for $F := E + (2EK + 260D + 1)\|f\|_S$. $\square$

We can now state:

Proposition 7.5. *Let X be a δ -hyperbolic space and let G be a non-virtually cyclic group acting on X with an axial, unital WPD loxodromic element f . Let S be a finite generating set of G . Let $x_0 \in Ax(f)$. Then for each $D > 100(K + D_0 + \delta)$ and $\epsilon, E, E' > 0$, there exists a constant $N = N(\epsilon, D, E, E')$ such that every finite set $A \subseteq G$ has a subset $A' \subseteq A$ satisfying:*

- (1) $\#A' \geq (1 - \epsilon)\#A$;
- (2) For each $a \in A'$ there exist f -halfspaces $\mathcal{H}_1, \mathcal{H}_2 \subseteq G$ with radius parameters (D, E) such that

$$\#(A \setminus a \cdot ((\mathcal{H}_1 \cup \mathcal{H}_2) \setminus \{g \in G : \|g\|_S \leq E'\})) \leq N.$$

Proof. Note that $\{g \in G : \|g\|_S \leq E'\}$ has at most $(2\#S)^{E'}$ elements. Hence, the statement for general E' will follow once we prove it for $E' = 0$. For this reason we set $E' = 0$. Given $E > 0$, we let $F = F(D, E)$ be as in Lemma 7.4 and let $R = R(D, F)$ be as in Lemma 7.3. We set

$$N = N(\epsilon, D, E) := \frac{2 \cdot (2\#S)^R}{\epsilon}.$$

Let us begin the proof by collecting unwanted elements, i.e.,

$$\mathcal{A}_1 := A \setminus A' = \left\{ a \in A : \#(A \setminus a(\mathcal{H}_1 \cup \mathcal{H}_2)) \geq N \text{ for every pair of } f\text{-halfspaces } \mathcal{H}_1, \mathcal{H}_2 \text{ with radius parameters } (D, E) \right\}.$$

Let $\mathcal{A}_2$ be a maximally R -separated subset of $\mathcal{A}_1$, i.e., we have

- (1) $d_S(a, a') \geq R$ for each pair of distinct elements $a, a' \in \mathcal{A}_2$;
- (2) $\mathcal{A}_2$ is a maximal subset of $\mathcal{A}_1$ satisfying this property.

Then $\bigcup_{a \in \mathcal{A}_2} a \cdot \{g \in G : \|g\|_S \leq R\}$ covers entire $\mathcal{A}_1$. Hence, we have

$$\#\mathcal{A}_2 \geq \frac{1}{(2\#S)^R} \cdot \#\mathcal{A}_1.$$

As before, we first prepare empty collections $\mathcal{B} = \mathcal{U} = \mathcal{G} = \emptyset$. Enumerate $\mathcal{A}_2$ by the distance from x_0 , i.e., let $\mathcal{A}_2 = \{a_1, a_2, \dots, a_{\#\mathcal{A}_2}\}$ be such that $d_X(x_0, a_i x_0) \leq d_X(x_0, a_{i+1} x_0)$ for each i .

We now describe a procedure that takes place throughout $\#\mathcal{A}_2$ steps. At step i , we first declare $\mathfrak{A}_i := \mathfrak{A}_{D,F}(a_i)$.

- (1) If $\mathcal{A}_2 \cap \mathfrak{A}_i$ has no element, then we declare that $a_i \in \mathcal{G}$ and $b_i := id$.
- (2) If not, pick $b_i \in \mathcal{A}_2 \cap \mathfrak{A}_i$ whose orbit point $b_i x_0$ is the *closest* to x_0 .

We then declare $\mathfrak{A}'_i := b_i \mathfrak{A}_{D,F}(b_i^{-1} a_i)$.

- (a) If $\mathcal{A}_2 \cap \mathfrak{A}_i \cap \mathfrak{A}'_i$ has no element, then we declare that $a_i \in \mathcal{G}$.
- (b) If not, we pick $c_i \in \mathcal{A}_2 \cap \mathfrak{A}_i \cap \mathfrak{A}'_i$ whose orbit point $c_i x_0$ is the *closest* to x_0 . We then declare $a_i \in \mathcal{B}$ and $b_i, c_i \in \mathcal{U}$.

(If an element in $\mathcal{U}$ is declared good or bad, it is not undecided anymore; we remove it from $\mathcal{U}$.)

In Case (1), $\mathcal{G}$ gains one more element that might be from $\mathcal{U}$ or not. $\mathcal{B}$ does not change. Overall, $\#\mathcal{B}$ stays the same and $\#\mathcal{U} + \#\mathcal{G}$ does not decrease. A similar situation occurs in Case (2-a).

In Case (2-b), $\mathcal{B}$ gains one element a_i , which might be from $\mathcal{U}$. In exchange, $\mathcal{U}$ gains elements b_i and c_i . Here Observation 7.1 guarantees that $d_X(x_0, b_i x_0), d_X(x_0, c_i x_0) > d_X(x_0, a_i x_0)$. Since $\mathcal{A}_2$ was labelled with respect to the distance from x_0 , we conclude that $b_i, c_i \in \{a_{i+1}, a_{i+2}, \dots\}$; in other words, neither b_i nor c_i come from $\mathcal{G} \cup \mathcal{B}$. We thus confirm that elements are never re-classified once they are put in $\mathcal{G} \cup \mathcal{B}$.

Furthermore, Observation 7.1 guarantees that $d_X(b_i x_0, c_i x_0) > 50D$. Hence b_i and c_i are distinct elements. If b_i, c_i are genuinely new additions to $\mathcal{U}$ and are not re-used from $\mathcal{U}$ at step $i-1$, then we can conclude that $\#\mathcal{U}$ increases by at least 1 in Case (2-b). It remains to show

Claim 7.6. *For $i < j$ such that $a_i, a_j \in \mathcal{B}$, we have $\{b_i, c_i\} \cap \{b_j, c_j\} = \emptyset$.*

Proof of Claim 7.6. Suppose to the contrary that $b_i \in \{b_j, c_j\}$. That is,

$$b_i \in \mathfrak{A}_{D,F}(a_i) \cap \mathfrak{A}_{D,F}(a_j).$$

Here, recall that $d_S(a_i, a_j) \geq R$. Lemma 7.3 tells us that $a_i \in \mathfrak{A}_{D,F}(a_j)$ or $a_j \in \mathfrak{A}_{D,F}(a_i)$. The former contradicts the labelling scheme of $\{a_1, a_2, \dots\}$, while the latter contradicts the minimality of $b_i \in \mathfrak{A}_{D,F}(a_i)$.

Next, suppose to the contrary that $c_i \in \{b_j, c_j\}$. That is,

$$c_i \in \mathfrak{A}_{D,F}(a_i) \cap b_i \mathfrak{A}_{D,F}(b_i^{-1} a_i) \cap \mathfrak{A}_{D,F}(a_j).$$

Lemma 7.3 implies that either $a_i \in \mathfrak{A}_{D,F}(a_j)$ or $a_j \in \mathfrak{A}_{D,F}(a_i) \cap b_i \mathfrak{A}_{D,F}(b_i^{-1} a_i)$. The former contradicts $d_X(x_0, a_i x_0) < d_X(x_0, a_j x_0)$, and the latter contradicts the minimality of $c_i \in b_i \mathfrak{A}_{D,F}(b_i^{-1} a_i) \cap \mathfrak{A}_{D,F}(a_i)$. $\square$

Thanks to the claim, we conclude that $\#\mathcal{B} \leq \#\mathcal{U} + \#\mathcal{G}$ at each step. But recall also that $a_i \in \mathcal{A}_2$ is declared good or bad at step i and is not affected thereafter. Hence, after the last step, there is no element of $\mathcal{U}$ left. Hence, we have $\#\mathcal{B} \leq \#\mathcal{G}$, and $\mathcal{G}$ takes up at least half of $\mathcal{A}_2$.

After the procedure, for each $i \in \{1, \dots, \#\mathcal{A}_2\}$ with $a_i \in \mathcal{G}$, we define

$$K_i := A \setminus a_i(\mathcal{H}_{D,E,f}(a_i^{-1}) \cup \mathcal{H}_{D,E,f}(a_i^{-1} b_i)).$$

Since $a_i \in \mathcal{G} \subseteq \mathcal{A}_2$, we have $\#K_i \geq N$. The remaining claim is:

Claim 7.7. *For every pair of distinct elements $a_i, a_j \in \mathcal{G}$, K_i and K_j do not intersect.*

Suppose to the contrary that K_i and K_j have a common element u for some $i < j$ such that $a_i, a_j \in \mathcal{G}$. By Lemma 7.4, we have

$$u \in \mathfrak{A}_{D,F}(a_i) \cap b_i \mathfrak{A}_{D,F}(b_i^{-1} a_i) \cap \mathfrak{A}_{D,F}(a_j) \cap b_j \mathfrak{A}_{D,F}(b_j^{-1} a_j).$$

Then we have $a_j \in \mathfrak{A}_{D,F}(a_i) \cap b_i \mathfrak{A}_{D,F}(b_i^{-1} a_i)$ or $a_i \in \mathfrak{A}_{D,F}(a_j) \cap b_j \mathfrak{A}_{D,F}(b_j^{-1} a_j)$ by Lemma 7.3. This contradicts the goodness of a_i or a_j . Hence, such a common element u cannot exist and K_i and K_j are disjoint.

With Claim 7.7 in hand, we have

$$\#A \geq \sum_{i: a_i \in \mathcal{G}} \#K_i \geq N \cdot \#\mathcal{G} \geq N \cdot \frac{\#\mathcal{A}_2}{2} \geq N \cdot \frac{\#\mathcal{A}_1}{2 \cdot (2\#S)^R} \geq \frac{1}{\epsilon} \#\mathcal{A}_1. \quad \square$$

We now have to check the branching property. Recall that $\mathcal{A}$ is the collection of all translates of $Ax(f)$. Let

$$\begin{aligned}\mathcal{NF}_D &:= \{g \in G : \forall \gamma \in \mathcal{A} [d_\gamma(x_0, gx_0) < D]\}, \\ \mathcal{NF}_D^i &:= \mathcal{NF}_D \cap \{g : d_S(id, g) = i\}, \\ \mathcal{NF}_D^{\geq i} &:= \mathcal{NF}_D \cap \{g : d_S(id, g) \geq i\}.\end{aligned}$$

We then observe that:

Proposition 7.8. *For each D , $\mathcal{NF}_D$ is r -roughly branching for some r .*

Proof. It suffices to prove that $\mathcal{NF}_D \setminus EC(f)$ is roughly branching for each D . To justify this, take an element $h \in G \setminus EC(f)$; this is possible as G is not virtually cyclic. Then for each $g \in G$, either g or gh lies outside $EC(f)$. Furthermore, Corollary 3.11(1) tells us that

$$\sup_{\gamma \in \mathcal{A}} d_\gamma(x_0, gx_0) - \sup_{\gamma \in \mathcal{A}} d_\gamma(x_0, ghx_0) < d_X(x_0, hx_0) + 12\delta < +\infty. \quad (\forall g \in G)$$

Hence, for each D there exists D' such that $\mathcal{NF}_D$ is contained in the D' -neighborhood of $(\mathcal{NF}_{D'} \setminus EC(f))$, which is roughly branching.

We will henceforth prove that $\mathcal{NF}_D \setminus EC(f)$ is roughly branching for a given D . Let us define $E := \lceil 10D + 100K + 1000\delta \rceil$. Recall that the set

$$EC(f) \cap \{g \in G : d_{Ax(f)}(x_0, gx_0) < E\}$$

is a finite set. Namely, it is contained in $B_S(R')$ for some R' . Let $R = R' + 2$.

Consider a subset A of $\mathcal{NF}_D \setminus EC(f)$ that is maximally R -separated in the word metric d_S . Let $a_1, \dots, a_k, b_1, \dots, b_k \in A$ be such that

$$a_1 f^E a_2 f^E \dots a_k f^E = b_1 f^E b_2 f^E \dots b_k f^E =: U.$$

Let $\gamma_i := a_1 f^E \dots a_i Ax(f)$ for each $i = 0, \dots, k$. First, note that $\gamma_i \neq \gamma_{i+1}$ for each i , as each a_i lies outside $EC(f)$. We then observe that

$$\begin{aligned}d_{\gamma_i}(\gamma_{i-1}, \gamma_{i+1}) &\geq d_{\gamma_i}(a_1 f^E \dots a_{i-1} f^E x_0, a_1 f^E \dots a_i f^E a_{i+1} x_0) \\ &= d_{Ax(f)}(a_i^{-1} x_0, f^E a_{i+1} x_0) \\ &\geq E - d_{Ax(f)}(x_0, a_i^{-1} x_0) - d_{Ax(f)}(x_0, a_{i+1} x_0) \\ &\geq E - 2 \cdot D \geq 8D + 100K + 1000\delta.\end{aligned}$$

By applying Corollary 3.16 inductively, we conclude that γ_i 's are pairwise distinct axes in $\mathcal{A}$, and $d_{\gamma_j}(\gamma_i, \gamma_l) \geq 8D + 90K + 900\delta$ for each $i < j < l$. This in particular implies that

$$d_{\gamma_1}(x_0, Ux_0) \geq 8D + 90K + 900\delta - \text{diam}_{\gamma_1}(\gamma_0) - \text{diam}_{\gamma_1}(\gamma_k) \geq 8D + 80K + 900\delta.$$

The same discussion holds for $\gamma'_i := b_1 f^E \dots b_i Ax(f)$. We now claim that $\gamma_1 = \gamma'_1$. If not, they have pairwise K -bounded projections. This means

$$d_{\gamma_1}(\gamma'_1, \gamma_2) \geq d_{\gamma_1}(x_0, \gamma_2) - d_{\gamma_1}(x_0, b_1 x_0) - \text{diam}_{\gamma_1}(\gamma'_1) \geq (E - 2D) - D - K.$$

Similarly, $d_{\gamma'_1}(\gamma_1, \gamma'_2) \geq E - 3D - K$. Then, the sequence $(\eta_1, \dots, \eta_{2k}) := (\gamma_k, \dots, \gamma_1, \gamma'_1, \dots, \gamma'_k)$ satisfies $\eta_i \neq \eta_{i+1}$ and $d_{\eta_i}(\eta_{i-1}, \eta_{i+1}) \geq E - 3D - K$ for

each i . Again, by applying Corollary 3.16, we deduce that η_i 's are pairwise distinct. But this contradicts the fact that $\gamma_k = UA_x(f) = \gamma'_k$.

Hence, $\gamma_1 = \gamma'_1$ and $a_1^{-1}b_1 \in EC(f)$. Moreover, we have

$$d_{Ax(f)}(x_0, a_1^{-1}b_1x_0) \leq d_{a_1Ax(f)}(x_0, a_1x_0) + d_{a_1Ax(f)}(x_0, b_1x_0) \leq 2D.$$

Hence, $a_1^{-1}b_1 \in B_S(R')$. Since $a_1, b_1 \in A$ are chosen from an R -separated set, we conclude $a_1 = b_1$.

Inductively, we deduce $a_i f^E = b_i f^E$ for each i . Hence, A is $E\|f\|_S$ -roughly branching, and $\mathcal{NF}_D \setminus EC(f)$ is $(R + E\|f\|_S)$ -roughly branching. $\square$

We now observe that $\mathcal{NF}_D^{\geq i}$ indeed serves as a “barrier”.

Proposition 7.9. *For every $D > 100(D_0 + K + \delta + 1)$ and $E > 0$, there exists $E' > 0$ such that the following holds. Let $g \in G$ be such that*

(1) *there does not exist $h \in G$ such that*

$$d_S(id, h) \leq E' \text{ and } d_{hAx(f)}(x_0, gx_0) \geq 70D.$$

(2) *$d_S(id, g) \geq E$.*

Let $(id = g_0, g_1, \dots, g_N := g)$ be a d_S -path between id and g . Then there exists t such that $g_t \in \mathcal{NF}_{100D}^{\geq E}$.

Proof. Let $E' = R(100D, E)$ be as in Lemma 3.17.

Suppose to the contrary that a d_S -path $P = (g_0, \dots, g_N)$ never intersects $\mathcal{NF}_{100D}^{\geq E}$. We will then construct an infinite sequence $i(0) < i(1) < \dots$ in $\{1, \dots, N\}$ and pick $\gamma_0, \gamma_1, \dots \in \mathcal{A}$ as follows.

(1) Let $i(0)$ be the last step within $B_S(E)$. Let γ_0 be the closest element to x_0 of

$$\mathcal{C}_0 := \{\gamma \in \mathcal{A} : d_\gamma(x_0, g_{i(0)}x_0) \geq 80D\}.$$

(2) Having defined $i(0), \dots, i(k)$ and $\gamma_0, \dots, \gamma_k$, let $i(k+1)$ be the first step after $i(k)$ when $d_{\gamma_k}(x_0, g_{i(k+1)}x_0) < 80D$. Let γ_{k+1} be the closest element to x_0 of

$$\mathcal{C}_{k+1} := \{\gamma \in \mathcal{A} : d_X(x_0, \gamma) > d_X(x_0, \gamma_k) \wedge d_\gamma(x_0, g_{i(k+1)}x_0) \geq 80D\}.$$

Since P has finite length N , this will give the desired contradiction. Note that this is very similar to the proof of Claim 6.2! From now on, we will justify why the $i(k)$'s exist and why the $\mathcal{C}_k$'s are nonempty. Along the way, we will also ensure for each $k \geq 1$ that

(A) $d_{\gamma_{k-1}}(x_0, \gamma_k) =_{5D} 80D$.

(B) there is no $\gamma \in \mathcal{A} \setminus \{\gamma_{k-1}, \gamma_k\}$ such that $d_\gamma(\gamma_{k-1}, \gamma_k) \geq 90D$.

Let us start the construction. As mentioned earlier, we set

$$i(0) := \max\{i : d_S(id, g_i) \leq E\}.$$

One very important fact is that $g_t \notin B_S(E)$ for all $t > i(0)$.

If $g_{i(0)} \in \mathcal{NF}_{100D}$, then it is in $\mathcal{NF}_{100D}^{\geq E}$, which is against the assumption. Thus, there exists $h \in G$ such that $d_{hAx(f)}(x_0, g_{i(0)}x_0) \geq 100D \geq$

$80D$. By replacing h with an element of $\{hf^i\}_{i \in \mathbb{Z}}$, we may suppose that $[\pi_{hAx(f)}(x_0), \pi_{hAx(f)}(g_{i(0)}x_0)]$ contains $[hx_0, hf^{100D}x_0]$. Corollary 3.11(2) tells us that $[x_0, g_{i(0)}x_0]$ has a subsegment that is 12δ -equivalent to $[hx_0, hf^{100D}x_0]$. In other words, $d_{[hx_0, hf^{100D}x_0]}(x_0, g_{i(0)}x_0) \geq 99D > D_0$. Note also that $\|g_{i(0)}\|_S = E$ by definition. Lemma 3.17 then tells us that $\|h\|_S \leq E'$.

In summary, $\mathcal{C}_0$ is a nonempty collection, and each element of $\mathcal{C}_0$ is realized as $hAx(f)$ for some $\|h\|_S \leq E'$. Let us take $\gamma_0 = hAx(f) \in \mathcal{C}_0$ closest to x_0 , i.e., the one with the smallest $d_X(x_0, hAx(f))$.

Now fix $l \geq 0$ and suppose that we have constructed $i(0) < \dots < i(l)$ and $\gamma_0, \dots, \gamma_l \in \mathcal{A}$ satisfying properties (A) and (B) for $k \leq l$. We aim to show that $i(l+1)$ and γ_{l+1} are well defined and satisfy properties (A) and (B).

Suppose to the contrary that $d_{\gamma_l}(x_0, g_N x_0) \geq 80D$. By the assumption for this inductive step, we know that

$$d_{\gamma_k}(x_0, \gamma_{k+1}) \geq 75D \quad (k = 0, \dots, l-1).$$

By applying Corollary 3.15, we deduce $d_{\gamma_0}(x_0, g_N x_0) \geq 72D$. Since $\|h\|_S \leq E'$, this violates Condition (1) for g_N . We conclude that $\{k > i(l) : d_{\gamma_l}(x_0, g_k x_0) < 80D\}$ is a nonempty set containing N , and we can pick its smallest element $i(l+1)$ as desired. Note that

$$d_{\gamma_l}(x_0, g_{i(l+1)-1}x_0) \geq 80D \text{ and } d_{\gamma_l}(x_0, g_{i(l+1)}x_0) < 80D.$$

By Corollary 3.11(1), we have $d_{\gamma_l}(x_0, g_{i(l+1)}x_0) \geq 78D$.

Next, we claim that $\mathcal{C}_{l+1}$ is nonempty. Since $g_{i(l+1)} \in P$ lies outside $B_S(E)$ but does not lie in $\mathcal{NF}_{100D}^{\geq E}$, there exists $\gamma \in \mathcal{A}$ with $d_\gamma(x_0, g_{i(l+1)}x_0) \geq 100D$. Meanwhile, Corollary 3.15 implies that

$$d_{\gamma_k}(x_0, g_{i(l+1)}x_0) \leq 87D < d_\gamma(x_0, g_{i(l+1)}x_0) \quad (k = 0, \dots, l).$$

Hence, γ and $\gamma_0, \dots, \gamma_l$ are pairwise distinct elements of $\mathcal{A}$, which have pairwise K -bounded projections.

Suppose to the contrary that $d_X(x_0, \gamma) \leq d_X(x_0, \gamma_k)$ for some $k \in \{0, 1, \dots, l\}$. Let us pick the smallest possible k .

- (1) Case 1: $k = 0$. In this case, we have $d_{\gamma_0}(x_0, \pi_\gamma(x_0)) \leq 12\delta$ by Lemma 3.10. Meanwhile, by feeding $x_0, \gamma_0, \dots, \gamma_l, g_{i(l+1)}x_0$ into Corollary 3.15, we obtain $d_{\gamma_0}(x_0, g_{i(l+1)}x_0) \geq 72D$. Hence, we have

$$\begin{aligned} d_{\gamma_0}(x_0, \gamma) &\leq 12\delta + \text{diam}_{\gamma_0}(\gamma) \leq 2D, \\ d_{\gamma_0}(\gamma, g_{i(l+1)}x_0) &\geq d_{\gamma_0}(x_0, g_{i(l+1)}x_0) - 2D \geq 70D. \end{aligned}$$

Lemma 3.14 implies that $d_\gamma(\gamma_0, g_{i(l+1)}x_0) \leq 3D$. As $d_\gamma(x_0, g_{i(l+1)}x_0) \geq 100D$, this implies that $d_\gamma(x_0, \gamma_0) \geq 97D$. Meanwhile, recall that $d_{\gamma_0}(x_0, g_{i(0)}x_0) \geq 80D$. Combining these two ingredients via Corollary 3.15, we deduce $d_\gamma(x_0, g_{i(0)}x_0) \geq 94D$, that is, $\gamma \in \mathcal{C}_0$. Moreover, γ is closer to x_0 than γ_0 is, because $d_\gamma(x_0, \gamma_0) \geq 97D$. This contradicts the minimality of $\gamma_0 \in \mathcal{C}_0$.

- (2) Case 2: $1 \leq k \leq l$ and $d_X(x_0, \gamma_{k-1}) < d_X(x_0, \gamma) \leq d_X(x_0, \gamma_k)$. Lemma 3.10 tells us that $d_\gamma(x_0, \pi_{\gamma_{k-1}}(x_0)) \leq 12\delta$, which implies

$$d_\gamma(x_0, \gamma_{k-1}) \leq 12\delta + \text{diam}_\gamma(\gamma_{k-1}) \leq 2D.$$

Similarly, we have $d_{\gamma_k}(x_0, \gamma) \leq 2D$. Recall from Corollary 3.15 that $d_{\gamma_k}(x_0, g_{i(l+1)}x_0) > 72D$. From these we deduce $d_{\gamma_k}(\gamma, g_{i(l+1)}x_0) \geq 70D$ and $d_\gamma(\gamma_k, g_{i(l+1)}x_0) \leq 3D$. We now observe

$$\begin{aligned} d_\gamma(\gamma_{k-1}, \gamma_k) &\geq d_\gamma(x_0, g_{i(l+1)}x_0) - d_\gamma(x_0, \gamma_{k-1}) - d_\gamma(\gamma_k, g_{i(l+1)}x_0) \\ &\geq 100D - 2D - 3D = 95D. \end{aligned}$$

This contradicts property (B) for k .

We conclude that $d_X(x_0, \gamma) > d_X(x_0, \gamma_l)$ and $\gamma \in \mathcal{C}_{l+1}$.

Having checked that $\mathcal{C}_{l+1}$ is nonempty indeed, let $\gamma_{l+1} \in \mathcal{C}_{l+1}$ be the distance minimizer. In order to check property (A) for $k = l + 1$, note that $d_{\gamma_{l+1}}(x_0, \pi_{\gamma_l}(x_0)) \leq 12\delta$ and $d_{\gamma_{l+1}}(x_0, \gamma_l) \leq 2D$ by Lemma 3.10 and the bounded projections. This implies that

$$(7.1) \quad d_{\gamma_{l+1}}(\gamma_l, g_{i(l+1)}x_0) \geq 80D - 2D \geq 78D,$$

and Lemma 3.14 implies $d_{\gamma_l}(\gamma_{l+1}, g_{i(l+1)}x_0) \leq 3D$. Since $d_{\gamma_l}(x_0, g_{i(l+1)}x_0) = 2D \cdot 80D$ by definition, we can deduce property (A) as desired.

Now suppose to the contrary that there exists some $\kappa \in \mathcal{A} \setminus \{\gamma_l, \gamma_{l+1}\}$ with $d_\kappa(\gamma_l, \gamma_{l+1}) \geq 90D$. For such κ , we have

$$(7.2) \quad d_{\gamma_l}(\kappa, \gamma_{l+1}), d_{\gamma_{l+1}}(\kappa, \gamma_l) \leq 3D$$

by Lemma 3.14. Combining this with property (A) for $k = l + 1$, we conclude that $d_{\gamma_l}(x_0, \kappa) \geq 70D$. Consequently, κ is farther from x_0 than γ_l is. Next, by Lemma 3.14 we also know that $d_\kappa(x_0, \gamma_l) \leq 3D$. Then we have $d_\kappa(x_0, \gamma_{l+1}) \geq 87D$. In other words, κ is closer to x_0 than γ_{l+1} is. Lastly, by combining Inequality 7.1 and 7.2, we obtain $d_{\gamma_{l+1}}(\kappa, g_{i(l+1)}x_0) \geq 75D$. By using Lemma 3.14 again, we deduce $d_\kappa(\gamma_{l+1}, g_{i(l+1)}x_0) \leq 3D$. Now combining everything, we get

$$\begin{aligned} d_\kappa(x_0, g_{i(l+1)}x_0) &\geq d_\kappa(\gamma_l, \gamma_{l+1}) - d_\kappa(x_0, \gamma_l) - d_\kappa(\gamma_{l+1}, g_{i(l+1)}x_0) \\ &\geq 90D - 3D - 3D \geq 80D. \end{aligned}$$

In conclusion, κ is an element of $\mathcal{C}_{l+1}$ that is closer to x_0 than γ_{l+1} is. This contradicts the definition of γ_{l+1} , and we conclude that property (B) holds for $k = l + 1$. This ends the induction step. $\square$

Corollary 7.10. *Let Γ be a Cayley graph of an acylindrically hyperbolic group G with respect to a finite generating set. Then Equation 2.8 holds.*

Proof. We can fix an action of G on a δ -hyperbolic space X with a unital, axial WPD element $f \in G$. Let S be a finite generating set for G that gives rise to $\Gamma = \text{Cay}(G, S)$. Let $x_0 \in Ax(f)$.

Given $D, E > 0$, we define

$$\begin{aligned}\mathcal{H}_D &:= \left\{ \{g \in G : gx_0 \in \mathcal{H}_{0.5D}(x_0, ux_0)\} : u \in G \right\}, \\ S_D &:= \mathcal{NF}_{200D} = \sqcup_{i=1}^{\infty} \mathcal{NF}_{200D}^i.\end{aligned}$$

Let $E' = E'(2D, E)$ be as in Proposition 7.9 for $2D$ and E . We define

$$\mathcal{G}_{D,E} := \left\{ g \in G : \|g\|_S \geq E \wedge \nexists h \in G [\|h\|_S \leq E' \wedge d_{hAx(f)}(x_0, gx_0) \geq 140D] \right\}.$$

Now given $\epsilon > 0$, we let $N = N(\epsilon, D, E'(2D, E), E)$ be as in Proposition 7.5.

First, Proposition 7.8 tells us that S_D is roughly branching for each D . Second, Proposition 6.1 provides a constant $r > 1$ such that for each $\mathcal{H} \in \mathcal{H}_D$ there exists an r -roughly branching subset $B = B_1 \sqcup \dots \sqcup B_{D/r}$ that is a disjoint union of D/r d_S -barriers $B_1, \dots, B_{D/r}$ between id and $\mathcal{H}$.

Finally, for each $D, E > 0$ we observe that $\mathcal{NF}_{200D}^{\geq E}$ is a d_S -barrier between id and $\mathcal{G}_{D,E}$ by Proposition 7.9. Lastly, Proposition 7.5 guarantees that for each finite $A \subseteq G$ there exists $A' \subseteq A$ with $\#A' \geq (1 - \epsilon)\#A$ such that for each $a \in A'$, there exist $\mathcal{H}_1, \mathcal{H}_2 \in \mathcal{H}_D$ such that

$$\#(A \setminus a(\mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{G}_{D,E})) \leq N.$$

We can thus apply Theorem 2.18 and conclude Equation 2.8. $\square$

Combining this with Corollary 5.9 and Theorem 2.5, we conclude:

Corollary 7.11. *Let G be an acylindrically hyperbolic group and let Γ be a Cayley graph of G with respect to a finite generating set. Then $p_c(\Gamma) < p_u(\Gamma)$ and $\nabla_{p_c}(\Gamma) < +\infty$.*

APPENDIX A. PROOF OF THEOREM 2.10

We summarize Hutchcroft's proof of Theorem 2.10. Let G, S, F and $\mathcal{H}$ be as in the assumption. Our goal is to find $\epsilon > 0$ such that $\left(\frac{d}{dp}\right)_+ \chi_p \geq \epsilon \chi_p^2$ holds for all $p_c/2 \leq p < p_c$. Then by integration we will have $\chi_p \leq \epsilon^{-1}(p_c - p)^{-1}$ for each $p_c/2 < p < p_c$, which leads to Equation 2.7.

We first define a $\{0, 1\}$ -valued function $I = I(a, A) \subseteq \Omega$ for inputs $a \in G$ and $A \subseteq G$:

$$I(a, A) := 1_{\{\exists h \in F, H \in \mathcal{H} [A \subseteq aH \text{ and } (H, id), (hH, h) \text{ are decoupled}]\}}.$$

By our assumption, we have

$$\sum_{a \in A} I(a, A) \geq \frac{1}{2} \#A$$

for each $A \subseteq G$. We now define a function $Y : \Omega \times G \rightarrow \mathbb{R}$:

$$Y(\omega, a) := I(a, C_\omega(id)) 1_{a \in C_\omega(id)}.$$

We will now fix $p_c/2 < p < p_c$. We have

$$\sum_{a \in G} \mathbb{E}_p Y(\omega, a) = \mathbb{E}_p \sum_{a \in C_\omega(id)} I(a, C_\omega(id)) \geq \mathbb{E}_p \left(\frac{1}{2} \#C_\omega(id) \right) = \frac{1}{2} \chi_p.$$

Meanwhile, we have

$$\begin{aligned}
\sum_{a \in G} \mathbb{E}_p Y(\omega, a) &= \sum_{a \in G} \mathbb{E}_p Y(\omega, a^{-1}) = \mathbb{E}_p \sum_{a: a^{-1} \in C_\omega(id)} I(a^{-1}, C_\omega(id)) \\
&= \mathbb{E}_p \sum_{a: id \in C_\omega(a)} I(id, C_\omega(a)) \\
&= \mathbb{E}_p \sum_{a \in C_\omega(id)} I(id, C_\omega(id)) = \mathbb{E}_p [\#C(id) \cdot I(id, C_\omega(id))].
\end{aligned}$$

Let $N = (\#F) \cdot (\#\mathcal{H})$. Among at most N candidates, there exists a deterministic pair (H, h) such that $(H, id), (hH, h)$ are decoupled and such that

$$\mathbb{E}_p [\#C(id) \cdot 1_{\{C(id) \subseteq H\}}] \geq \frac{1}{2N} \chi_p.$$

By applying the action of h , we also have

$$\mathbb{E}_p [\#C(h) \cdot 1_{\{C(h) \subseteq hH\}}] \geq \frac{1}{2N} \chi_p.$$

Now, note that the event $\{C(id) \subseteq H\}$ is determined solely by edges incident to H , and $\{C(h) \subseteq hH\}$ is determined solely by edges incident to hH . Since the two sets are disjoint, the two events are independent. We conclude that

$$\mathbb{E}_p [(\#C(id)) \cdot (\#C(h)) \cdot 1_{\{C(id) \subseteq H \text{ and } C(h) \subseteq hH\}}] \geq \frac{1}{4N^2} \chi_p^2.$$

Since (H, id) and (hH, h) are decoupled, there exists a d_S -path $P_1 \circ P_2$ connecting id to h such that P_1 is not adjacent to hH and P_2 is not adjacent to H . Let $e = \overrightarrow{g_n g_{n+1}}$ be the first (oriented) edge of P_2 . As described in Hutchcroft's proof, a standard conversion of finitely many states guarantees a constant c_p , which is a linear combination of finitely many products and ratios of p and $1 - p$, hence bounded on compact subsets of $(0, 1)$, such that

$$\mathbb{E}_p [(\#C(g_n)) \cdot (\#C(g_{n+1})) \cdot 1_{\{g_n \not\leftrightarrow g_{n+1}\}}] \geq c_p \mathbb{E}_p [(\#C(id)) \cdot (\#C(h)) \cdot 1_{\{C(id) \subseteq H, C(h) \subseteq hH\}}].$$

By applying the action by g_n^{-1} , we conclude that

$$\mathbb{E}_p [(\#C(id)) \cdot (\#C(s)) \cdot 1_{id \not\leftrightarrow s}] \geq c \chi_p^2$$

for some c uniform on $p \in [0.5p_c, p_c]$ and for some element $s = s(p)$ of $S \cup S^{-1}$.

We now recall Russo's formula. For each $g \in G$, a closed edge $e = \overrightarrow{e^- e^+} \in \mathcal{E}^\rightarrow$ is pivotal for the event $\{id \leftrightarrow g\}$ if and only if $id \leftrightarrow e^-$, $e^+ \leftrightarrow g$ and $e^- \not\leftrightarrow e^+$. Hence, we have

$$\left(\frac{d}{dp}\right)_+ \tau_p(g) \geq \frac{1}{1-p} \sum_{e \in \mathcal{E}^\rightarrow} \mathbb{P}_p(\{id \leftrightarrow e^-\} \cap \{e^- \not\leftrightarrow e^+\} \cap \{e^+ \leftrightarrow g\}).$$

Since $\tau_p(g)$ is a monotone function in p for each $g \in G$, summing over finitely many g 's and taking limits implies

$$\begin{aligned}
\left(\frac{d}{dp}\right)_+ \chi_p &\geq \frac{1}{1-p} \sum_{g \in G} \sum_{e \in \mathcal{E} \rightarrow} \mathbb{P}_p(\{id \leftrightarrow e^-\} \cap \{e^- \not\leftrightarrow e^+\} \cap \{e^+ \leftrightarrow g\}) \\
&= \frac{1}{1-p} \sum_{s \in S \cup S^{-1}} \sum_{h \in G} \sum_{g \in G} \mathbb{P}_p(\{id \leftrightarrow h\} \cap \{h \not\leftrightarrow hs\} \cap \{hs \leftrightarrow g\}) \\
&= \frac{1}{1-p} \sum_{s \in S \cup S^{-1}} \sum_{h \in G} \sum_{g \in G} \mathbb{P}_p(\{id \leftrightarrow h\} \cap \{h \not\leftrightarrow hs\} \cap \{hs \leftrightarrow hsg\}) \\
&= \frac{1}{1-p} \sum_{g, h \in G, s \in S \cup S^{-1}} \mathbb{P}_p(\{h^{-1} \leftrightarrow id\} \cap \{id \not\leftrightarrow s\} \cap \{s \leftrightarrow g\}).
\end{aligned}$$

The last term is bounded from below by $\frac{1}{1-p} \mathbb{E}_p[(\#C(id)) \cdot (\#C(s)) \cdot 1_{id \not\leftrightarrow s}]$ for our choice $s = s(p)$. Hence, we conclude that $(d/dp)_+ \chi_p$ is uniformly coarsely bounded from below by χ_p^2 for $p \in (p_c/2, p_c)$. This ends the proof.

APPENDIX B. PROOF OF COROLLARY 3.11

We sketch the proof of Corollary 3.11.

- (1) It is a direct consequence of Lemma 3.10.
- (2) We have

$$d_X(x, p) + d_X(p, y) - 12\delta \leq d_X(x, p) + d_X(p, q) + d_X(q, y) - 12\delta \leq d_X(x, y).$$

Hence, $(x|y)_p \leq 6\delta$. Pick $x' \in [x, y]$ such that $d_X(x, x') = (y|p)_x$. Then by Lemma 3.6, x' is 10δ -close to p . Similarly, the point $y' \in [x, y]$ such that $d_X(x, y') = (y|q)_x$ is 10δ -close to q . Note that

$$d_X(x, p) \leq d_X(x, y) - d_X(q, y) - d_X(p, q) + 12\delta < d_X(x, q) - 12\delta + 12\delta = d_X(x, q).$$

For a similar reason, we have $d_X(y, q) < d_X(y, p)$. This implies that

$$\begin{aligned}
d_X(x, x') &= \frac{1}{2}[d_X(x, y) + d_X(x, p) - d_X(y, p)] \\
&\leq \frac{1}{2}[d_X(x, y) + d_X(x, q) - d_X(y, q)] = d_X(x, y').
\end{aligned}$$

Hence, x' comes earlier than y' along $[x, y]$. As $\triangle x'py'$ and $\triangle py'q$ are δ -slim, we know that $[x', y'] \subseteq \mathcal{N}_\delta([x', p]) \cup \mathcal{N}_{2\delta}([p, q]) \cup \mathcal{N}_{2\delta}([y', q]) \subseteq \mathcal{N}_{12\delta}([p, q])$. Similarly, $[p, q] \subset \mathcal{N}_{12\delta}([x', y'])$ holds.

- (3) Let $\gamma = [y, z]$ and $\gamma' = [y', z']$. Let $a \in [y, z]$ be such that $d_X(y, a) = (x|z)_y$, let $b \in [y, x]$ be such that $d_X(y, b) = (x|z)_y$, let $c \in [y, x]$ be such that $d_X(y, c) = (x|z')_y$, let $d \in [x, z']$ be such that $d_X(z', d) = (x|y)_{z'}$, let $e \in [x, z']$ be such that $d_X(z', e) = (x|y')_{z'}$, and let $f \in [y', z']$ be such that $d_X(z', f) = (x|y')_{z'}$.

By Lemma 3.9, a and f are 8δ -equivalent to $\pi_\gamma(x)$ and $\pi_{\gamma'}(x)$, respectively. Moreover, Lemma 3.6 tells us that $d_X(a, b), d_X(c, d), d_X(e, f) \leq 4\delta$, and the triangle inequality tells us that $d_X(b, c) \leq d_X(z, z') \leq D$,

$d_X(d, e) \leq d_X(y, y') \leq D$. In conclusion, $\pi_\gamma(x)$ and $\pi_{\gamma'}(x)$ are $(2D + 28\delta)$ -equivalent as desired.

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